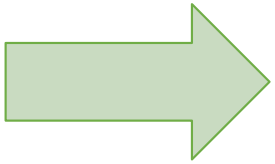
Chapter 2
Symmetric Encryption

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Plan



1. Computational definitions of security
2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
6. Pseudorandom functions and permutations
7. Block ciphers
8. Practical considerations

How to change the security definition?

we will require that m_0, m_1 are chosen by a **poly-time adversary**

An encryption scheme is **perfectly secret** if for every $m_0, m_1 \in \mathcal{M}$

$\mathbf{Enc}(K, m_0)$ **and** $\mathbf{Enc}(K, m_1)$ **are identically distributed**

we will require that no **poly-time adversary** can distinguish $\mathbf{Enc}(K, m_0)$ from $\mathbf{Enc}(K, m_1)$

A game

(Enc,Dec) – an encryption scheme



security parameter
1^n



adversary
(polynomial-time probabilistic Turing machine)

oracle

chooses m_0, m_1 such that
$|m_0|=|m_1|$

m_0, m_1

1. selects k randomly from $\{0,1\}^n$
2. chooses a random $b = 0,1$
3. calculates
   $c := \mathrm{Enc}(k, m_b)$

has to guess b

c

Alternative name: **has indistinguishable encryptions in the presence of an eavesdropper**

Security definition:
We say that **(Enc,Dec)** is **semantically-secure** if any **polynomial time** adversary guesses b correctly with probability at most $\frac{1}{2} + \varepsilon(n)$, where ε is negligible.

Testing the definition

Suppose the adversary can compute ***k*** from **Enc(*k*,*m*)**.
Can he win the game?

YES!

Suppose the adversary can compute **some bit of *m*** from **Enc(*k*,*m*)**. Can he win the game?

YES!

Multiple messages

In real-life applications we need to encrypt **multiple messages with one key**.

The adversary may learn something about the key by looking at

ciphertexts $c_1,\ldots,c_t$ of

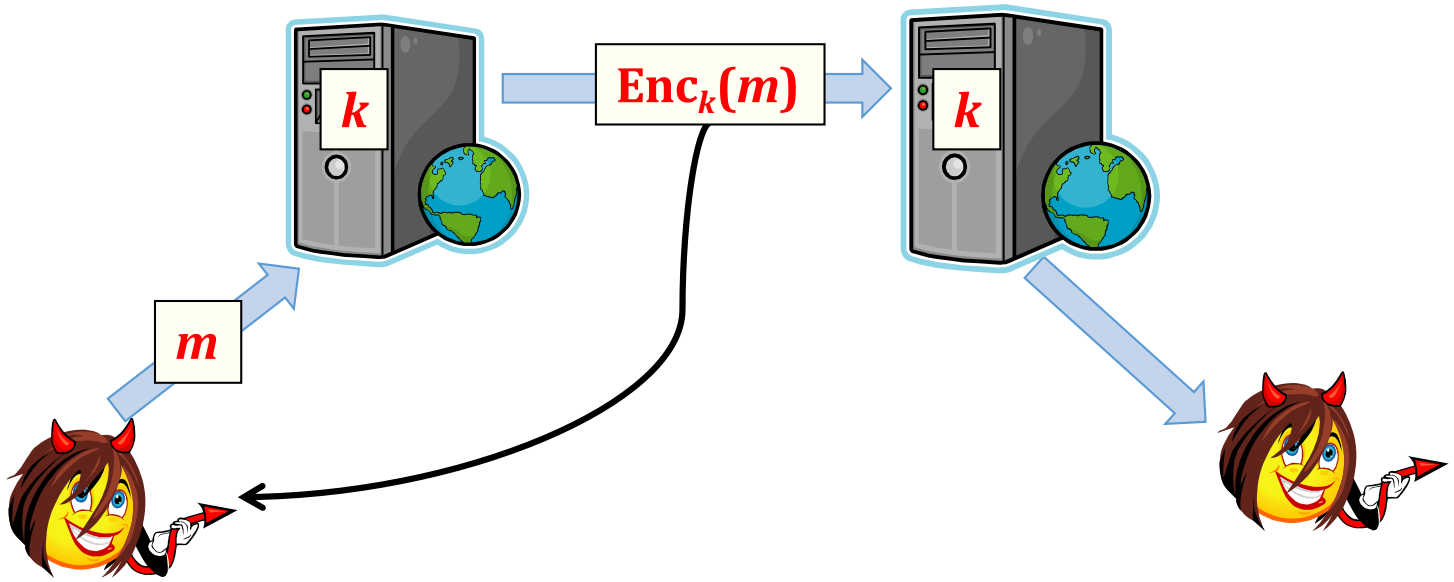
some messages $m_1,\ldots,m_t$.

How are these messages chosen?
let's say: the adversary can choose them!

(good tradition: be as pessimistic as possible)

A real-life example

Example: routing

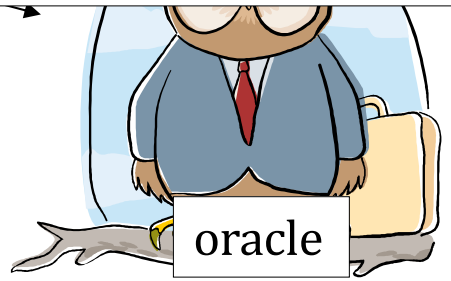


A chosen-plaintext attack (CPA)



security parameter 1^n

1. chooses random $k \leftarrow \{0,1\}^n$
2. chooses random $b \leftarrow \{0,1\}$



oracle

chooses m'_1

m'_1

$c_1 = \mathrm{Enc}(k, m'_1)$

chooses m'_t

m'_t

$c_t = \mathrm{Enc}(k, m'_t)$

challenge phase:

chooses m_0, m_1

m_0, m_1

$c = \mathrm{Enc}(k, m_b)$

the interaction continues . . .

has to guess b

CPA-security

We will sometimes say: IND-CPA-secure

Security definition

We say that **(Enc,Dec) has indistinguishable encryptions under a chosen-plaintext attack (CPA)** if

every **randomized polynomial time** adversary
guesses $\boldsymbol{b}$ correctly
with probability at most $\frac{1}{2} + \boldsymbol{\varepsilon}(\boldsymbol{n})$, where $\boldsymbol{\varepsilon}$ is negligible.

Observation

A **CPA-secure** encryption scheme cannot be deterministic.

Typical options:

- **Enc** has a "state" (e.g. a counter)
- **Enc** is randomized, i.e., it takes as additional input:
  - some perfect randomness ***R***, or
  - takes as an **nonce** ***R***

weaker requirement

nonce = "**n**umber used **once**"

Other attacks known in the literature

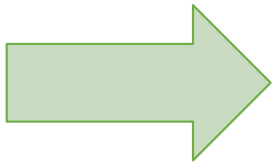


- **ciphertext-only attack** – the adversary has no information about the plaintext
- **known plaintext attack** – the plaintext are drawn from some distribution that the adversary does not control
- **batch chosen-plaintext attack** – like the **CPA** attack, but the adversary has to choose $m'_1,\ldots, m'_t$ at once.

("our" **CPA**-attack is also called the "**adaptive CPA-attack**")

- **chosen ciphertext attack** – we will discuss it later…

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Is it possible to prove security?

Bad news:

Theorem

If semantically-secure encryption exists (with $|k| < |m|$)



P ≠ NP

Intuition: if **P = NP** then the adversary can guess the key...

(formal proof – exercises)

Moral:
"If **P=NP**, then the semantically-secure encryption is broken"

Is it 100% true?

Not really...

This is because even if **P=NP** we do not know what are the constants.

Maybe **P=NP** in a very "inefficient way"...

To prove security of a cryptographic scheme we need to show **a lower bound on the computational complexity of some problem.**

In the "asymptotic setting" that would mean that at least we show that **P ≠ NP**.

Does the implication in the other direction hold? (that is: does **P ≠ NP** imply anything for cryptography?)

No! (at least as far as we know)

Therefore

proving that an encryption scheme is secure is probably much harder than proving that **P ≠ NP**.

What can we prove?



We can prove conditional results.

That is, we can show theorems of a type:

Suppose that some "computational assumption **A**" holds

⇓

then scheme **X** is secure.

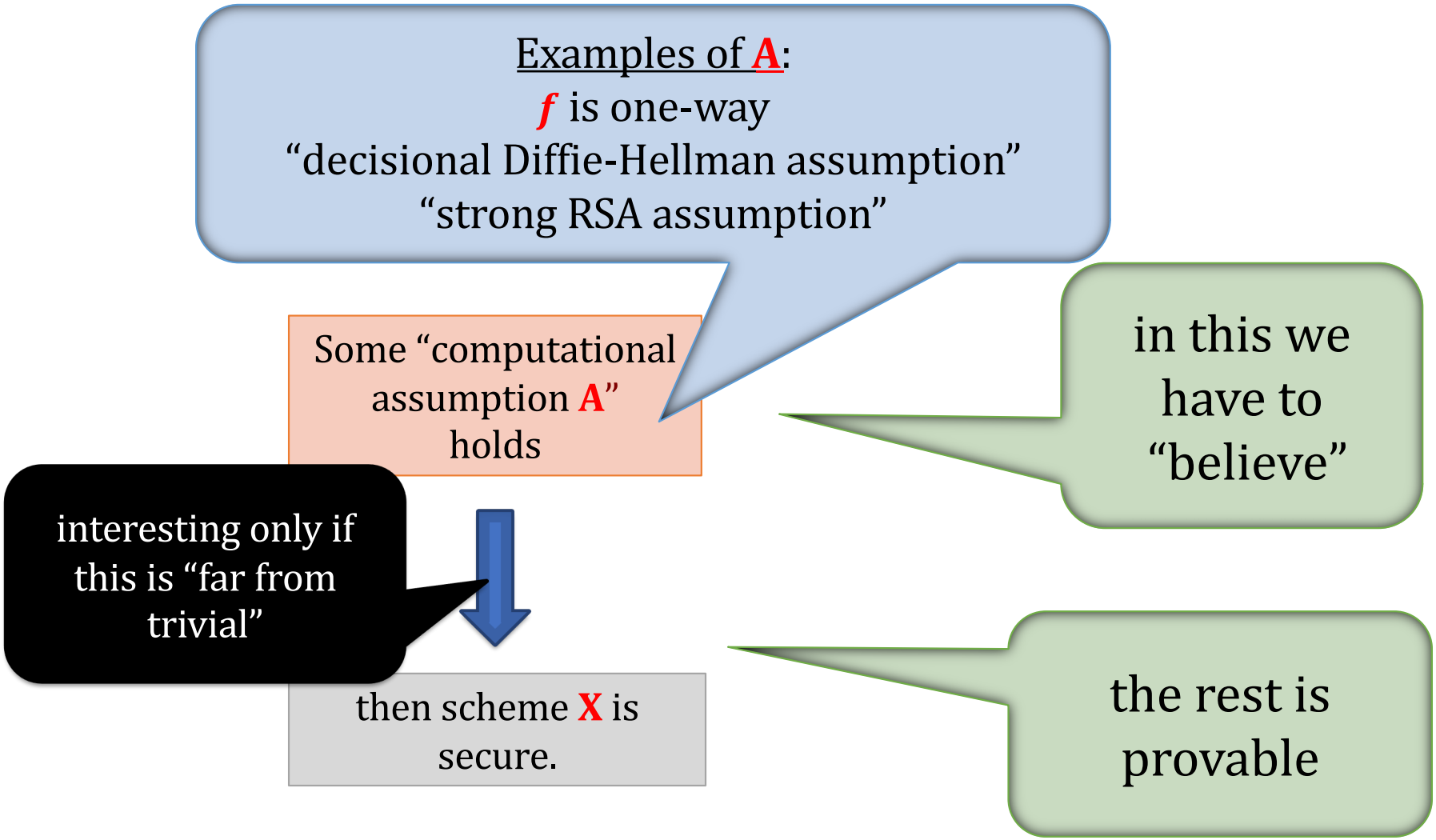
Suppose that some scheme **Y** is secure

⇓

then scheme **X** is secure.

Research program in cryptography

Base the security of cryptographic schemes on a small number of well-specified "computational assumptions".



Example

Suppose that G is a "cryptographic pseudorandom generator"



we can construct a secure encryption scheme based on G

A natural question

What is "**the minimal assumption**" needed for cryptography?

Answer: existence **one-way functions**.

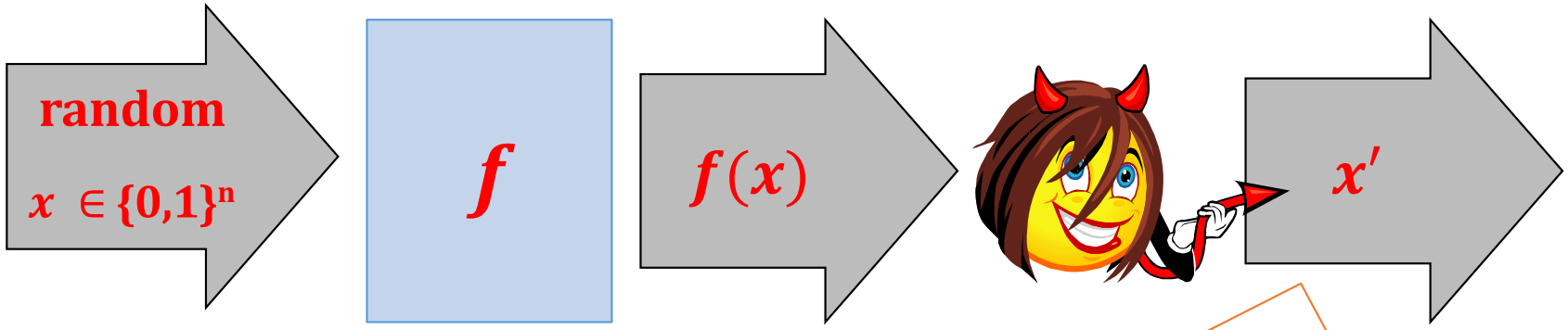
We introduce them now.

One-way functions

A function

$$f : \{0,1\}^* \rightarrow \{0,1\}^*$$

is **one-way** if it is: **(1)** poly-time computable, and **(2)** "hard to invert it".



probability that any poly-time adversary outputs x' such that

$$f(x) = f(x')$$

is negligible in n.

A real-life analogue: phone book



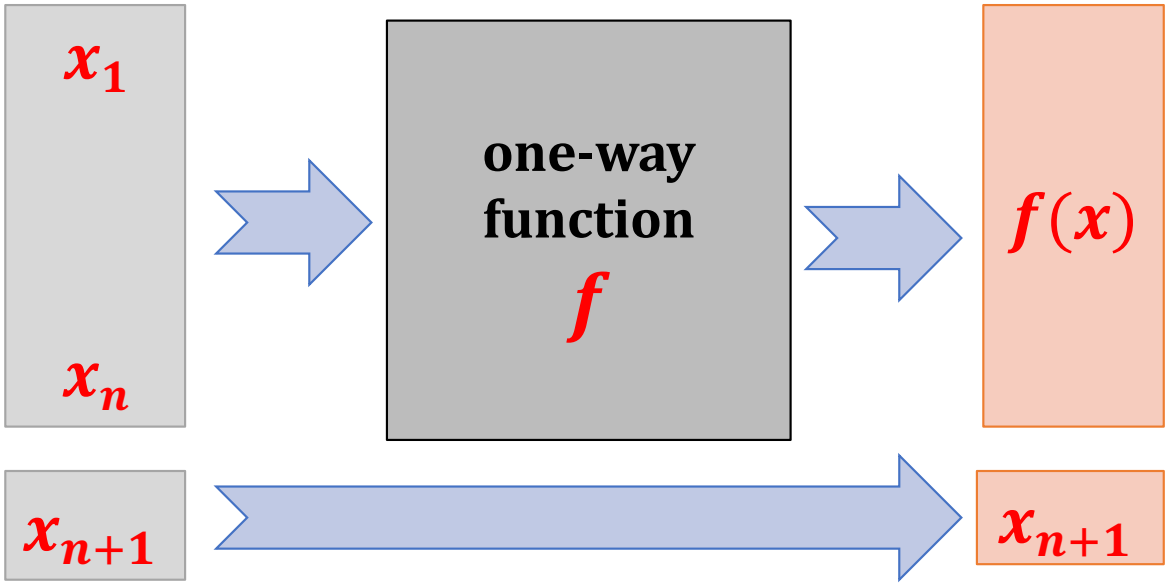
A function:

people → numbers

is “one way”.

One way functions **do not** "hide all the input"

Example:



$f'(x_1, \ldots, x_{n+1}) := f(x_1, \ldots, xn) \,||\, x_{n+1}$ is also a one-way function

One way-functions – main properties

They are "**hard to invert**" on random input x, but:

- "**inverting**" can also mean finding some other x' such that

$$f(x) = f(x')$$

- it's ok if **partial information** on x can be computed from $f(x)$.

Their existence is the "**minimal assumption**" needed for cryptography.

Why "minimal assumption?

Recall the following idea from the previous lecture:

(m – a message)

1. the key K is chosen uniformly at random
2. $C := \mathrm{Enc}_K(m)$ is given to the adversary

An idea

"The adversary should not be able to compute K."

Observation: it's not a sufficient condition for security.

But it is **necessary**.

if the adversary can compute K from $C := \mathrm{Enc}_K(m)$ from some m (and $|K| < |m|$) then **(Enc,Dec)** is **not** secure

Moral: f defined as $f(K) := \mathrm{Enc}_K(m)$ has to be **one-way**.

One-way functions more formally

experiment **(machine M, function f)**

1. pick a random element $x \leftarrow \{0,1\}^n$
2. let $y := f(x)$,
3. let x' be the output of M on y
4. we say that M **won** if $f(x') = y$.

We will say that a poly-time computable $f: \{0,1\}^* \to \{0,1\}^*$ is **one-way** if

$\forall$ **polynomial-time Turing Machine M**

P (M wins) is negligible

Example of a (candidate for) a one-way function

If **P=NP** then **one-way functions don't exist**.

Therefore currently no function can be proven to be one-way.

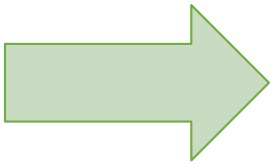
But there exist candidates.

Example:

$f(p,q) = pq$, where p and q are primes such that $|p| = |q|$.

this function is defined on
primes × primes,
not on
{0,1}*
but it's just a technicality

Plan

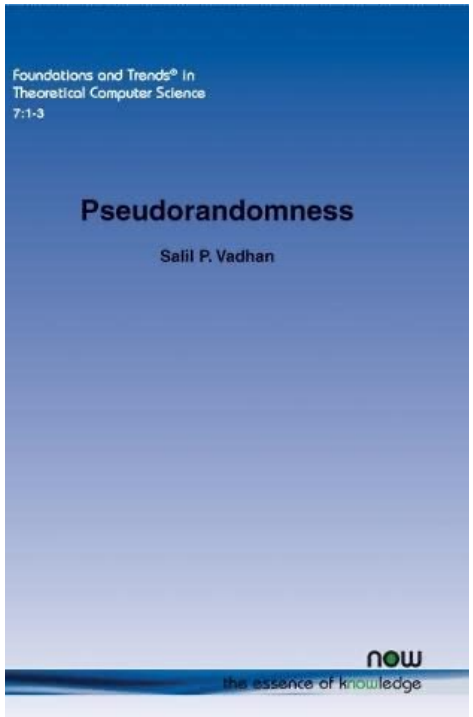


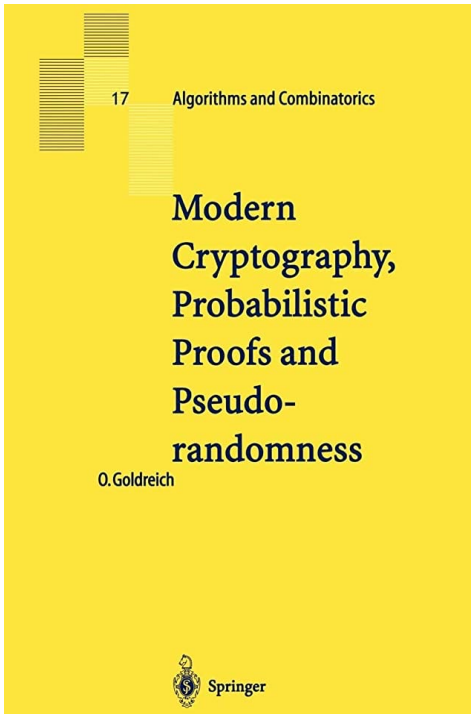
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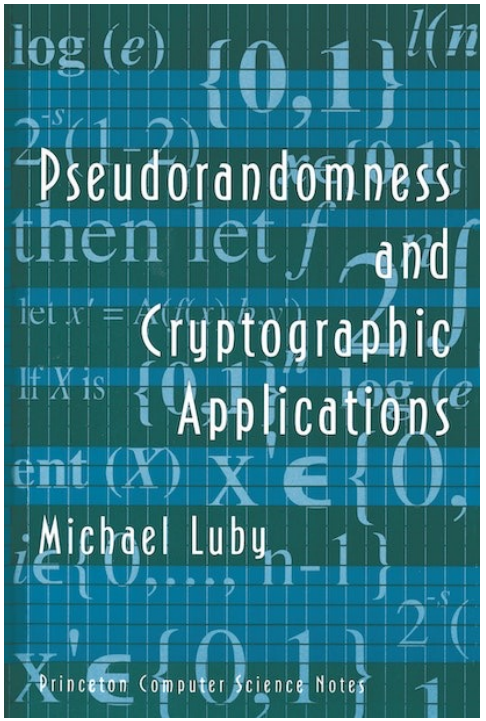
Our tool: pseudorandomness

An important topic on its own.

Applications outside of cryptography (e.g.: **derandomization** of randomized algorithms).





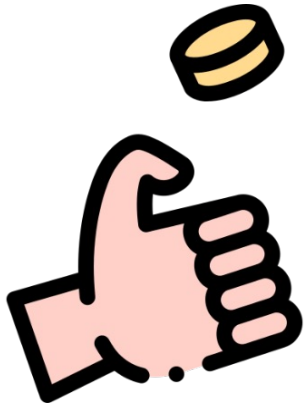


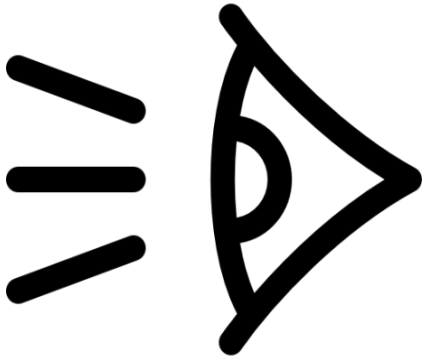
Intution

Randomness depends on the "**power of an observer**".

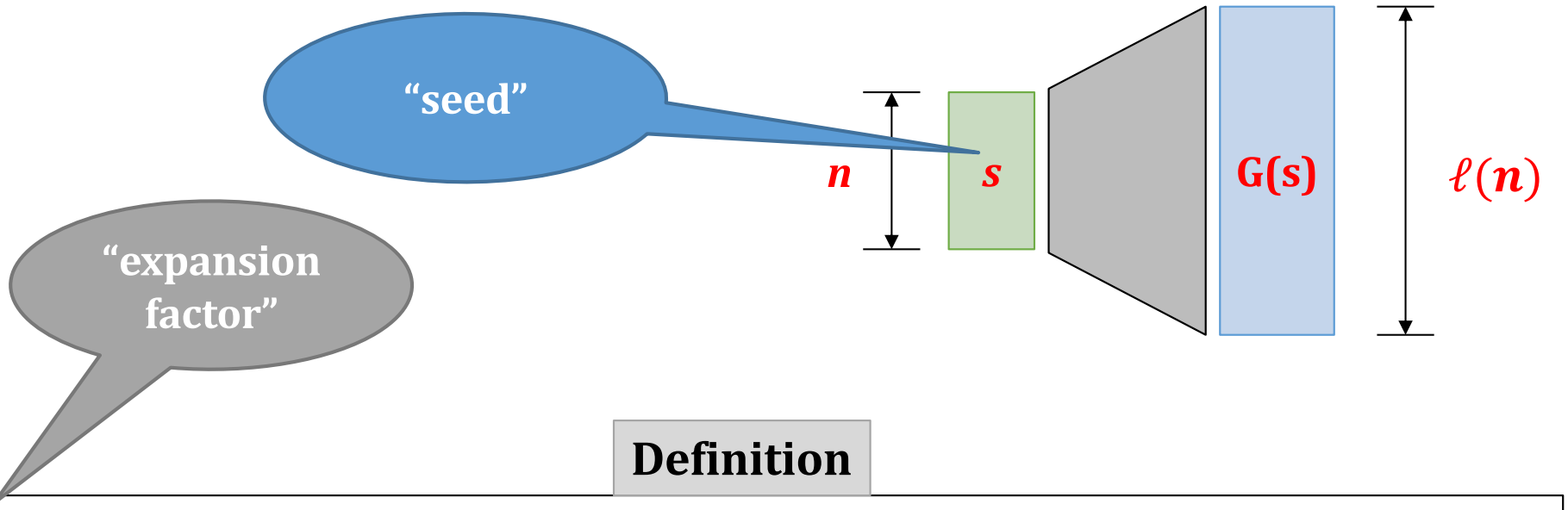
Think of a **coin-toss**:

An observer that can record and process coin behavior **can predict the outcome**





Pseudorandom generators



Definition

ℓ – polynomial such that always $\ell(n) > n$

An algorithm $G : \{0,1\}^* \to \{0,1\}^*$ is called a **pseudorandom generator (PRG)** if

for every n

and for every s such that $|s| = n$

we have

$|G(s)| = \ell(n)$.

and for a random s the value $G(s)$ "**looks random**".

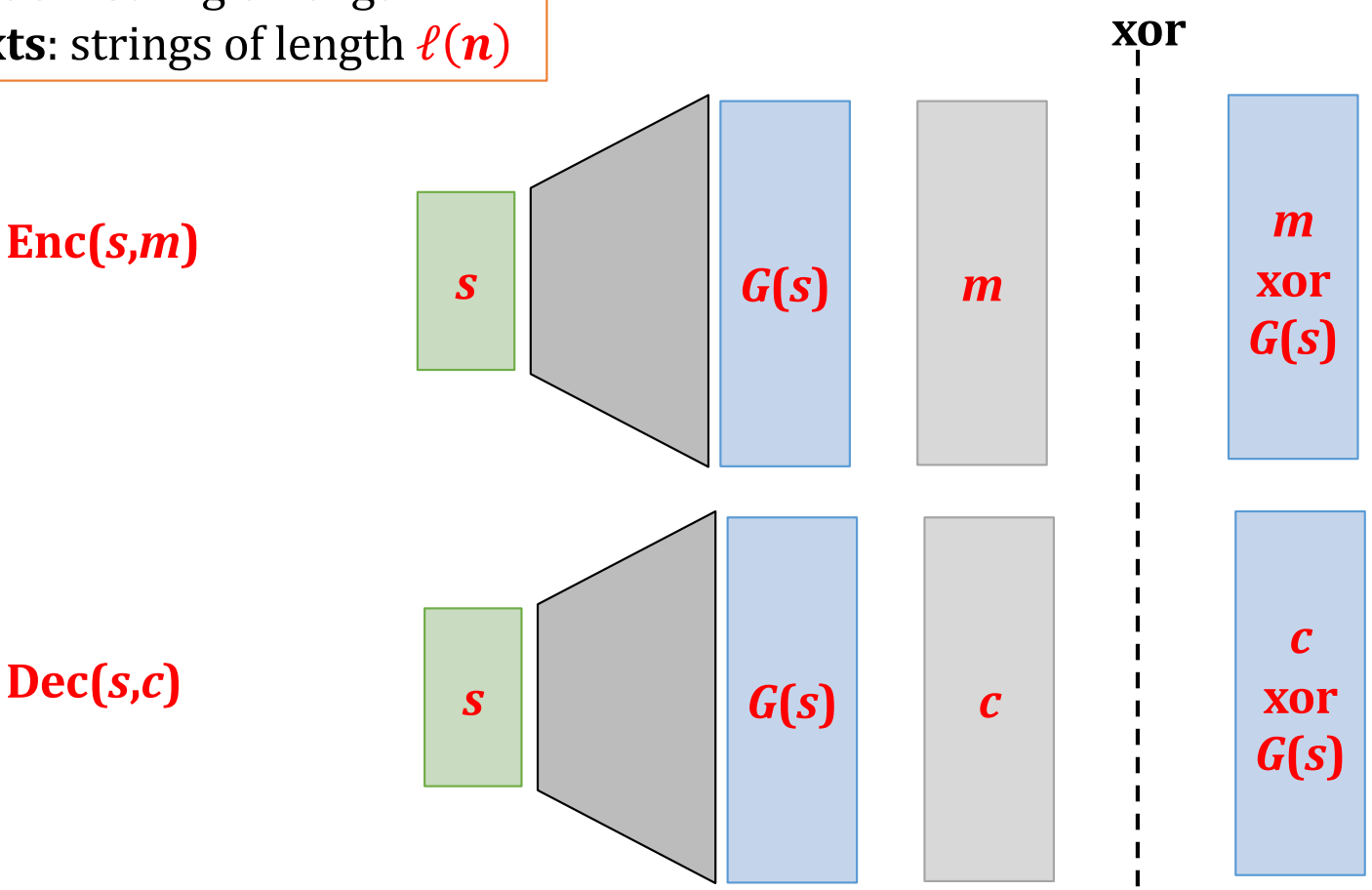
this has to be formalized

Idea

Use PRGs to "shorten" the key in the one time pad

for a moment just consider a **single message case**

Key: random string of length n
Plaintexts: strings of length $\ell(n)$



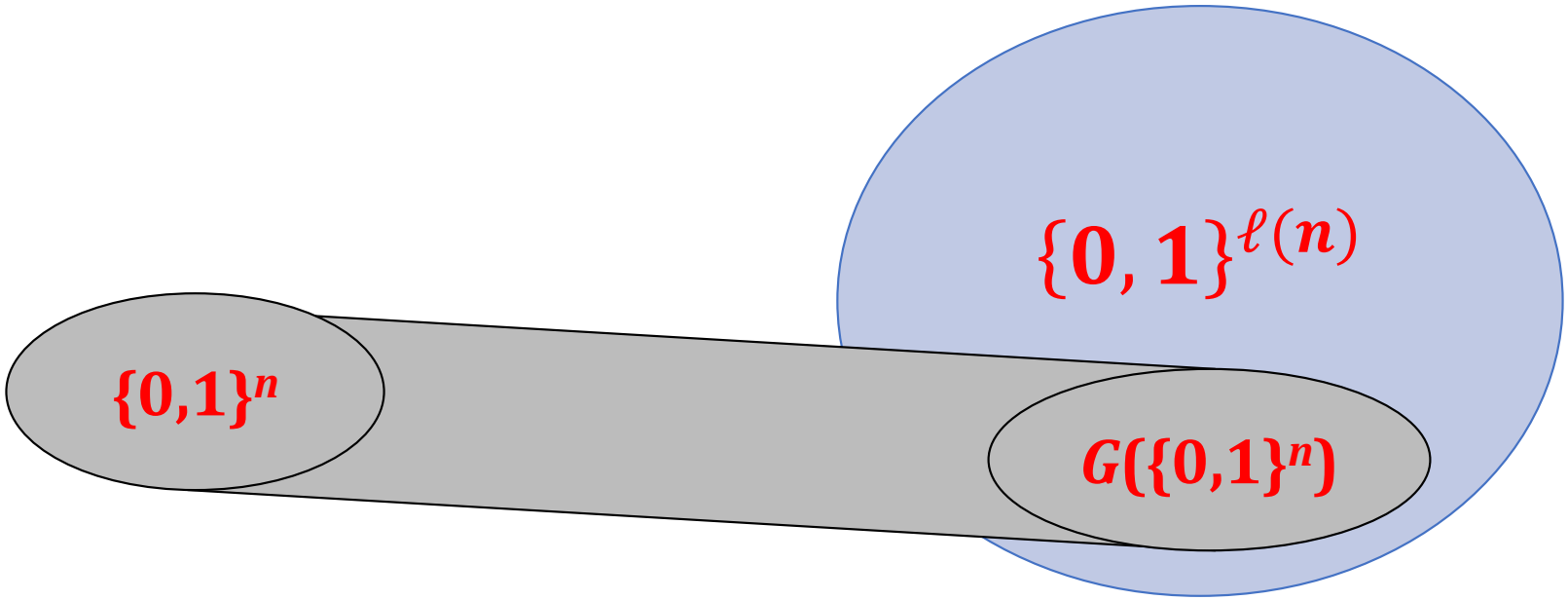
If we use a "normal PRG" – this idea doesn't work
We have to use the **cryptographic PRGs.**

“Looks random”

Suppose $s \in \{0,1\}^n$ is chosen randomly.

Can $G(s) \in \{0,1\}^{\ell(n)}$ be uniformly random?

No!



“Looks random”

What does it mean?

Non-cryptographic applications:

should pass **some statistical tests**.

Cryptography:

should pass **all polynomial-time tests**.

Non-cryptographic PRGs

Example: **Linear Congruential Generators (LCG)** defined recursively

- $X_0 \in Z_m$ – the key
- for $n = 1, 2, \ldots$ let

$$X_{n+1} := (a \cdot X_n + c) \bmod m$$

output: $Y_1, Y_2, \ldots$ where

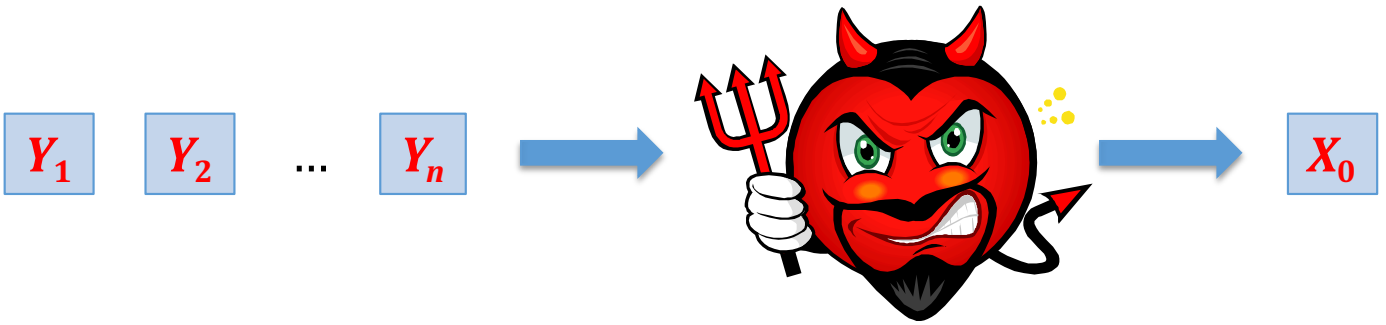
Y_i = first t bits of each X_i

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rand() function in Windows – an **LCG** with

$$a = 214013,\ c = 2531011,\ m = 2^{32},\ t = 15$$

How to break an LRS



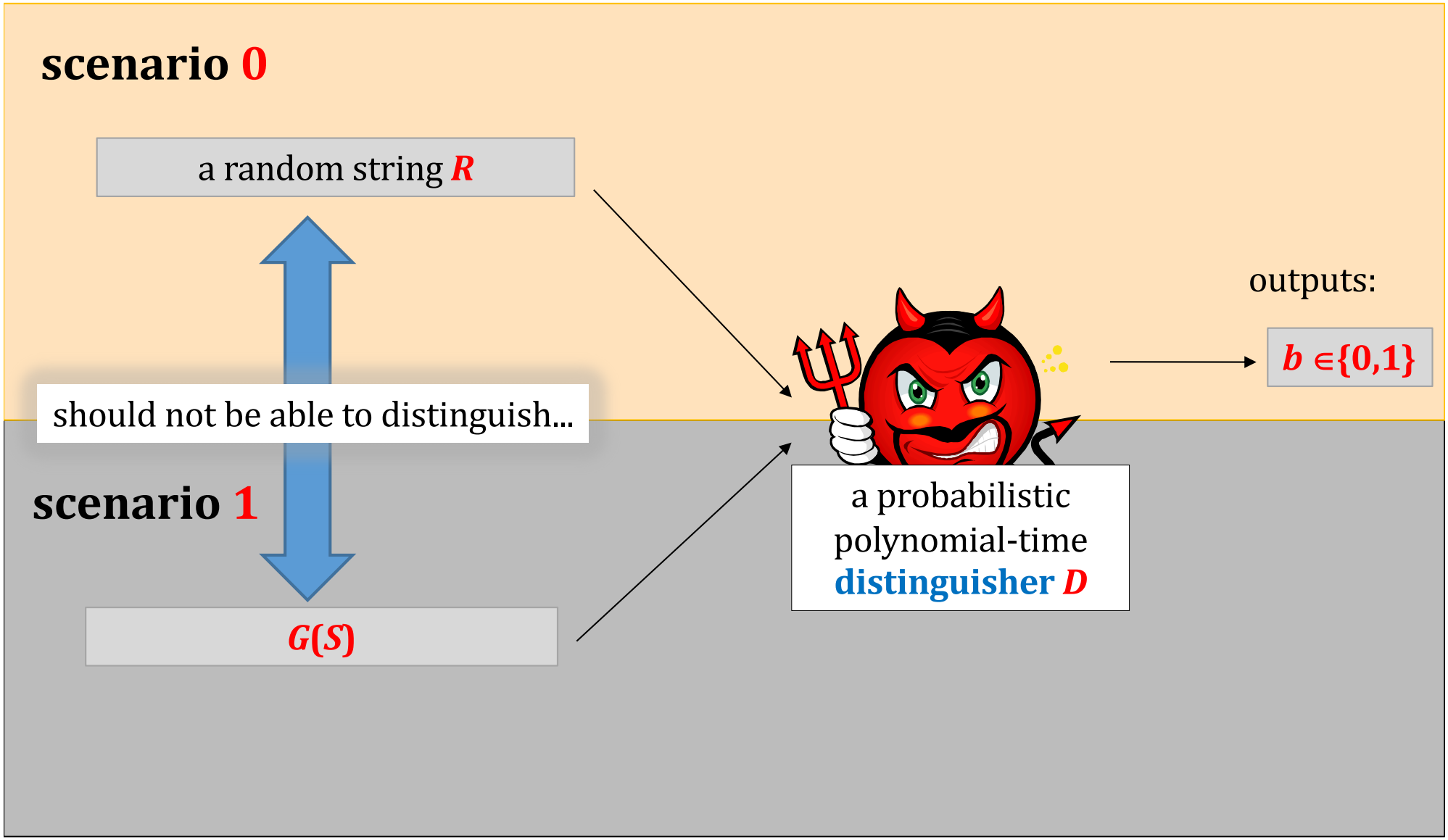
Solve linear equations with "partial knowledge" (because you only know first t bits)

See:

G. Argyros and A. Kiayias: **I Forgot Your Password: Randomness Attacks Against PHP Applications**, *USENIX Security '12*

(successful attacks on password-recovery mechanisms in PHP)

PRG – main idea of the definition



Cryptographic PRG

a random string R

or

$G(S)$ (where S random)



outputs:

0 if he thinks it's R

1 if he thinks it's $G(S)$

Should not be able to distinguish...

Definition

n – a parameter

S – a variable distributed uniformly over $\{0,1\}^n$

R – a variable distributed uniformly over $\{0,1\}^{\ell(n)}$

G is a **cryptographic PRG** if

for every polynomial-time Turing Machine D

we have that

$$\left|P(D(R)=1)-P\left(D\left(G(S)\right)=1\right)\right|$$

is negligible in n.

Constructions

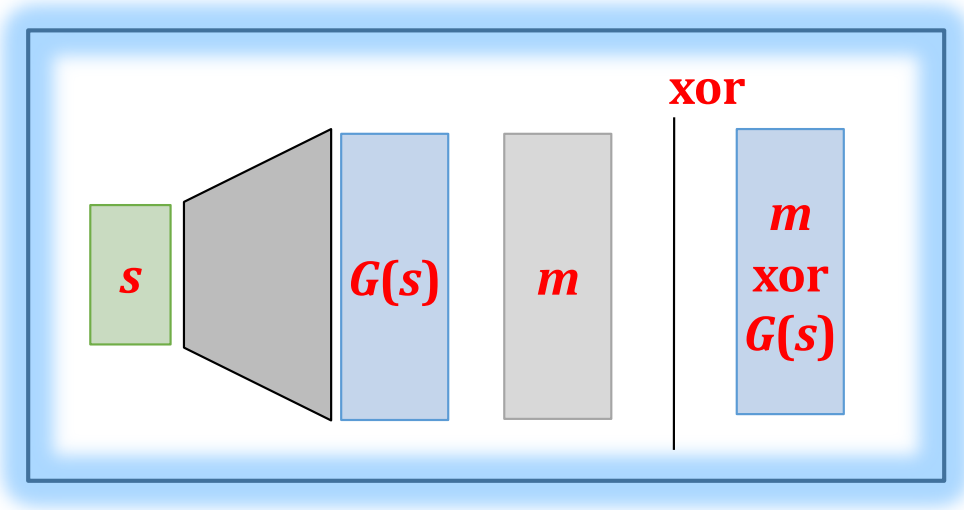
There exists constructions of cryptographic pseudorandom-generators, that are **conjectured** to be secure.

We will discuss them later...

Theorem

(for simplicity consider only the single message case)

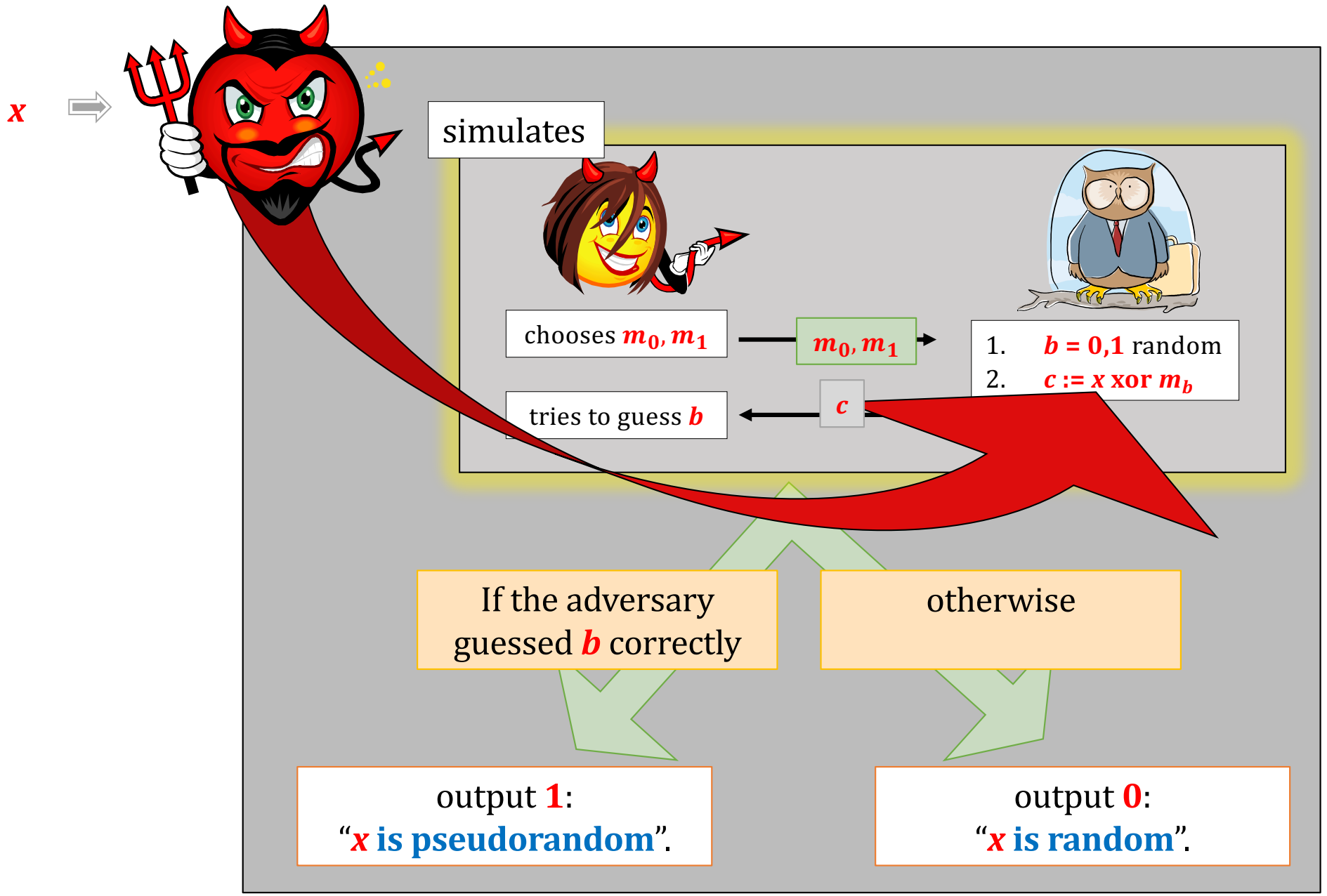
If **G** is a **cryptographic PRG** then the encryption scheme constructed before is semantically secure.



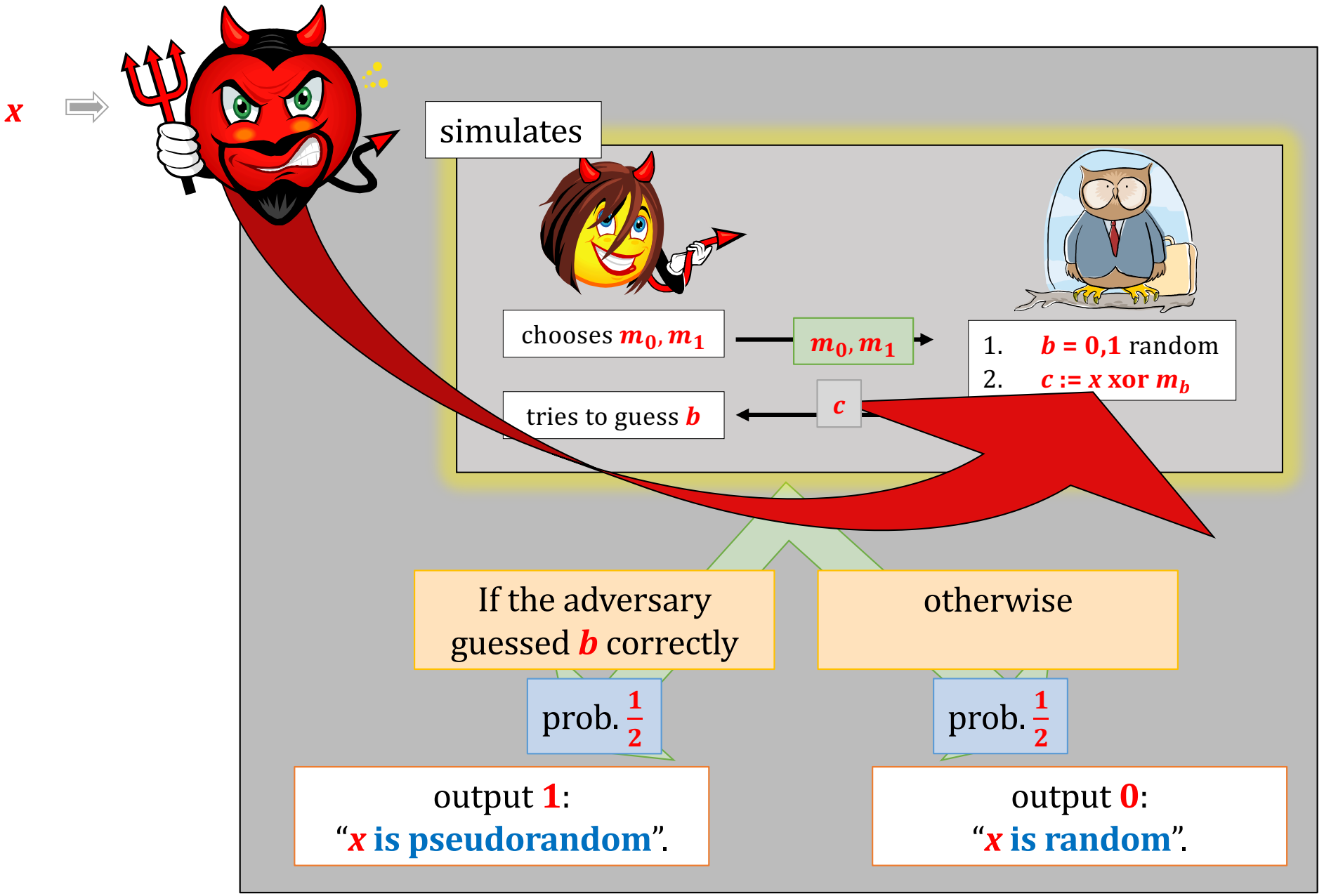
cryptographic PRGs exist → CPA-secure encryption exists

Proof (sketch)

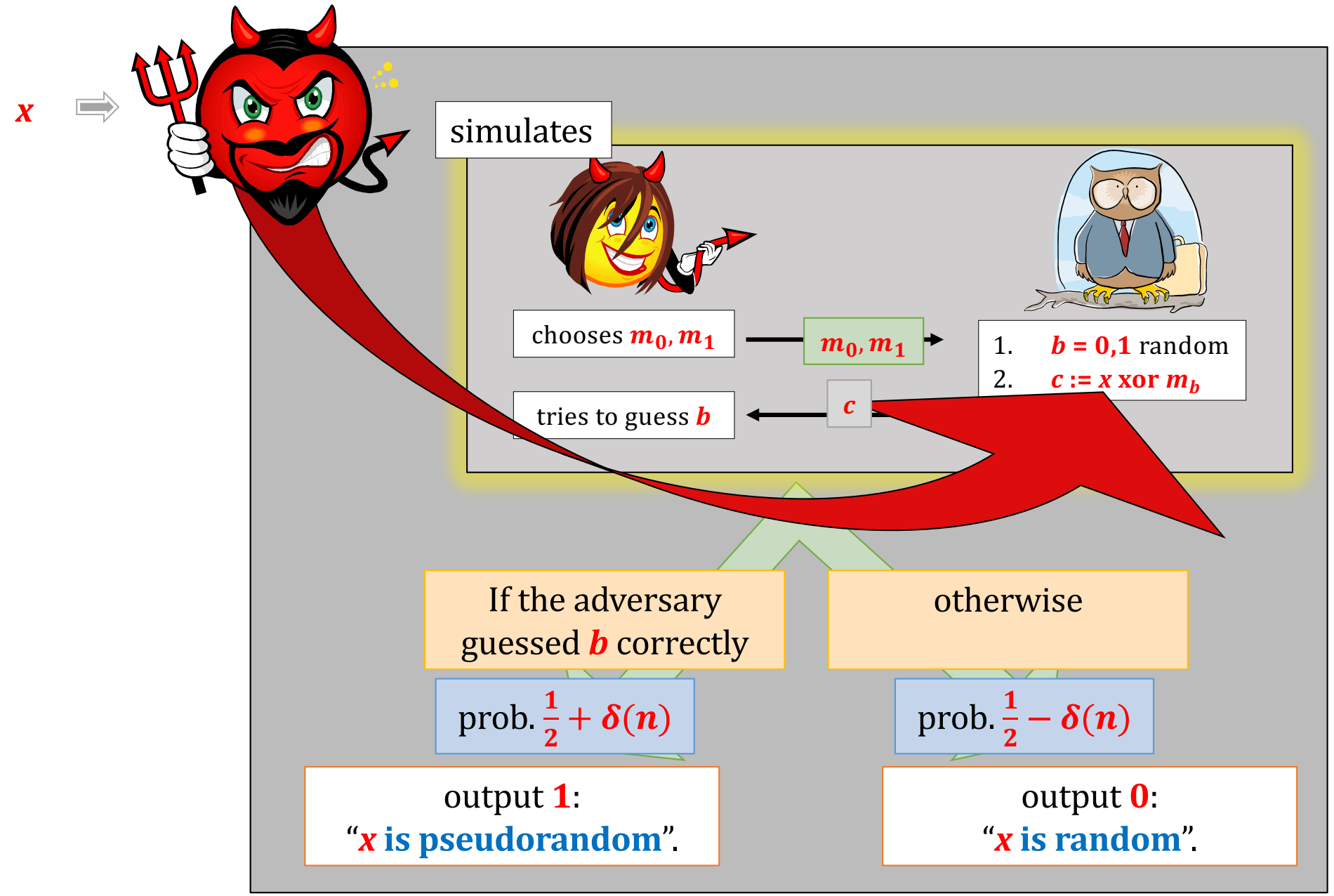
Suppose that it is **not** secure.
Therefore, there exists a poly-time adversary that wins the "guessing game" with probability $\frac{1}{2} + \delta(n)$ where $\delta(n)$ is not negligible.


x
simulates
chooses m_0, m_1
m_0, m_1
1. $b = 0,1$ random
2. $c := x$ xor m_b
tries to guess b
c
If the adversary guessed b correctly
otherwise
output 1: "x is pseudorandom".
output 0: "x is random".

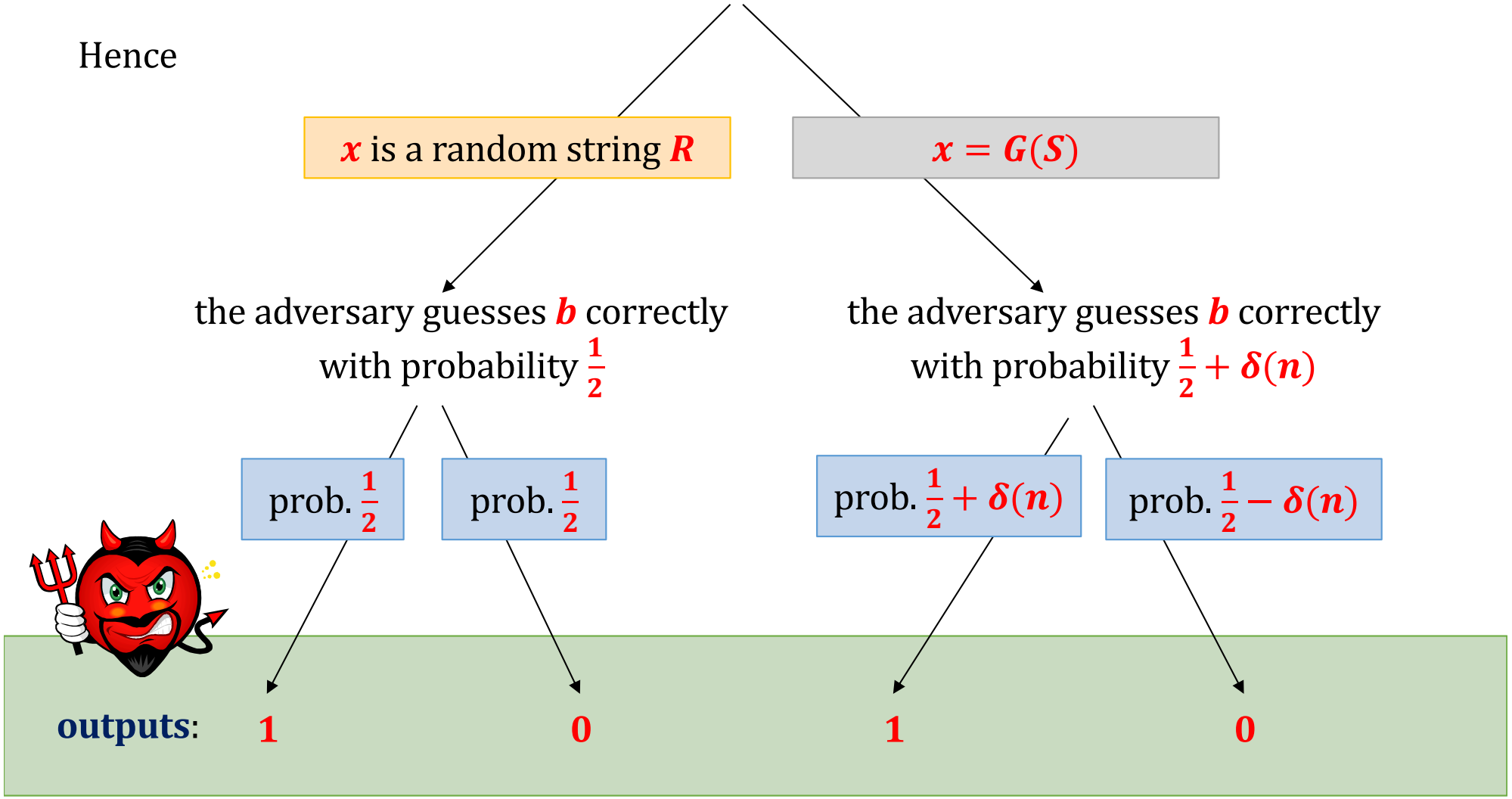
"scenario 0": x is a random string



"scenario 1": $x = G(S)$



Hence



$$|P(D(R) = 1) - P(D(G(S)) = 1| = \left|\frac{1}{2} - \left(\frac{1}{2} + \delta(n)\right)\right| = |\delta(n)|$$

Since δ is not negligible G cannot be a **cryptographic PRG**.

The complexity

The distinguisher  simply simulated

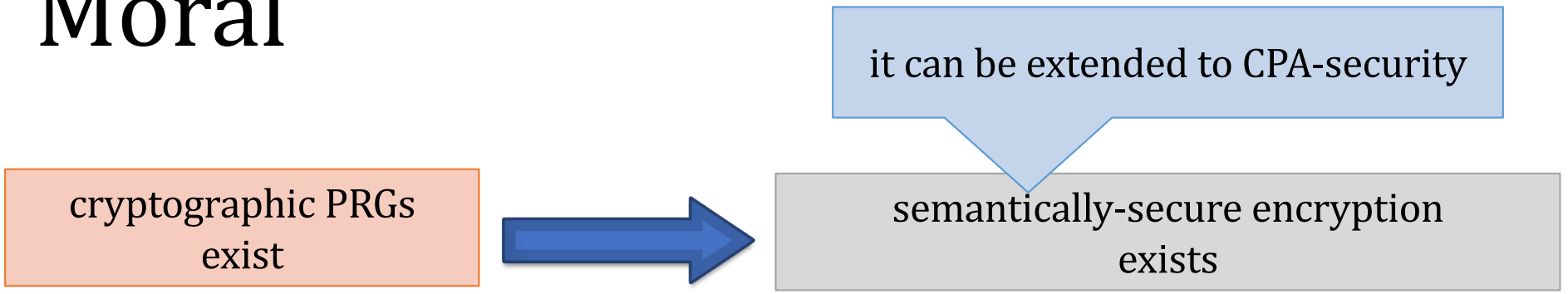
one execution of the adversary 

against the oracle  .

Hence he works in polynomial time.

QED

Moral



To construct secure encryption it suffices to construct a secure PRG.

Moreover, we can also state the following:

Informal remark. The reduction is tight.

A question

What if the distinguisher  needed to simulate

1000 executions of the adversary  ?

An (informal) answer

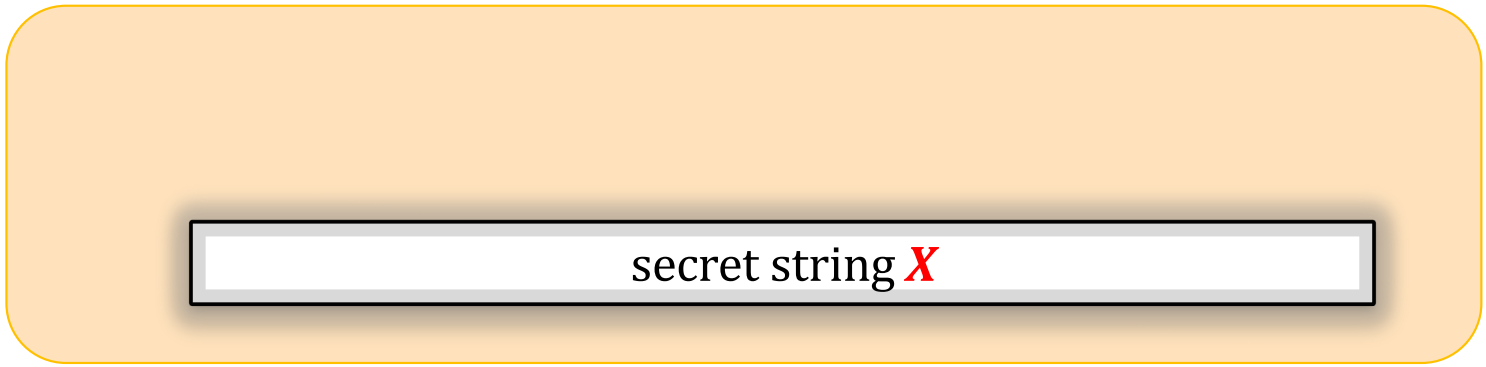
Then, the encryption scheme would be "**1000** times less secure" than the pseudorandom generator.

Why?

To achieve the same result  needs to work **1000** times more than  .

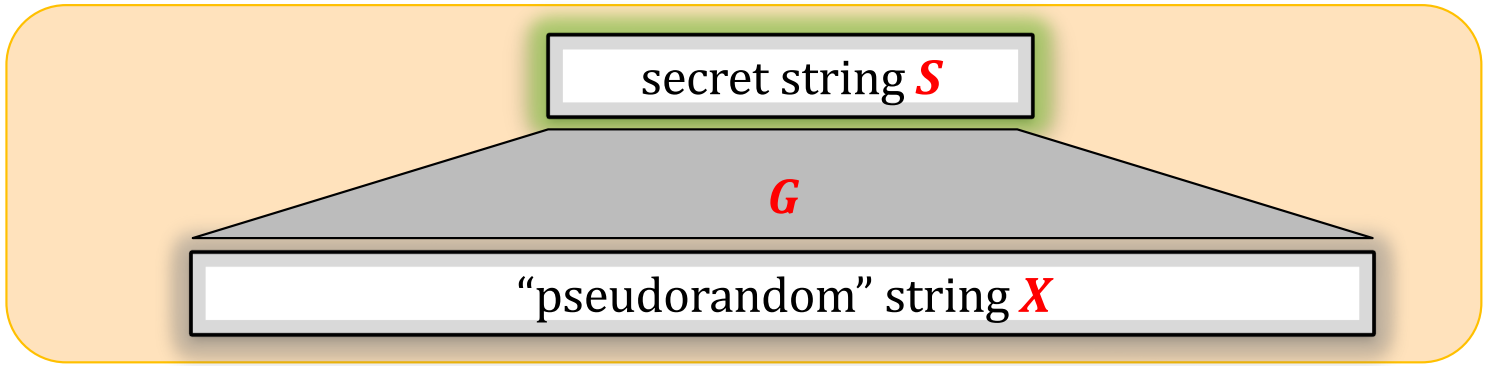
General rule

Take a secure system that uses some long secret string X.



Then, you can construct a system that uses a shorter string S, and expands it using a PRG:

$$X = G(S)$$



Constructions of PRGs

A theoretical result

a PRG can be constructed from any **one-way function**
(**very elegant**, **impractical, inefficient**)

Based on hardness of some
particular computational problems

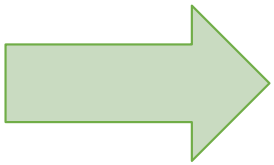
For example
[Blum, Blum, Shub. *A Simple Unpredictable Pseudo-Random Number Generator*]
(**elegant**, **more efficient**, still **rather impractical**)

"Stream ciphers"

ugly, **very efficient**, **widely used in practice**

Examples: RC4, Trivium, SOSEMANUK,...

Plan



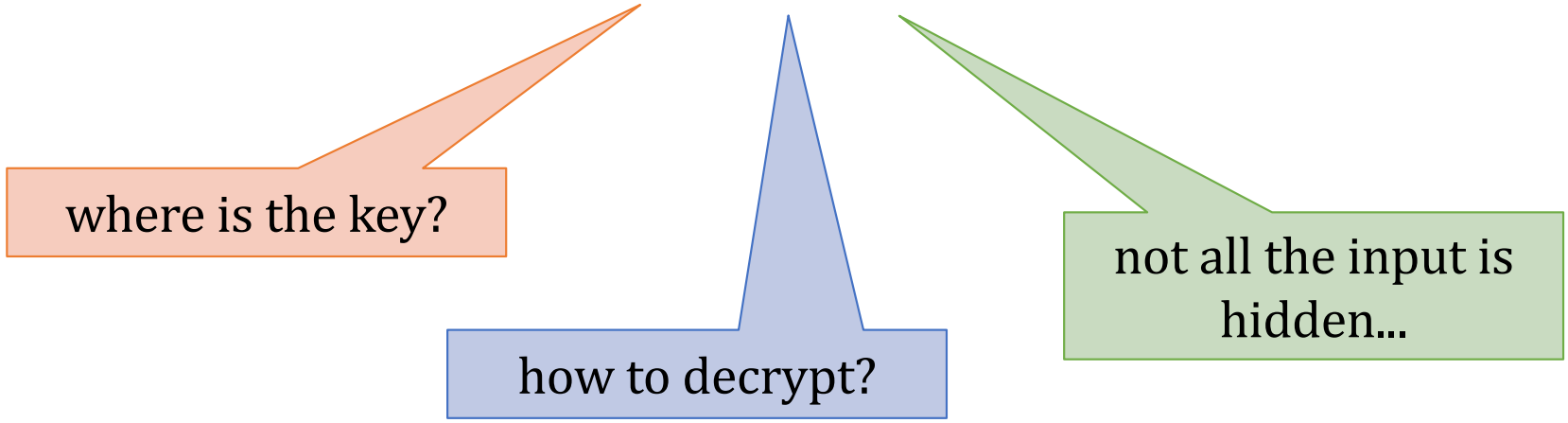
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How to encrypt with one-way functions?

Naive (and wrong) idea:

1. Take a one-way function $\boldsymbol{f}$,
2. Let a ciphertext of a message $\boldsymbol{M}$ be equal to

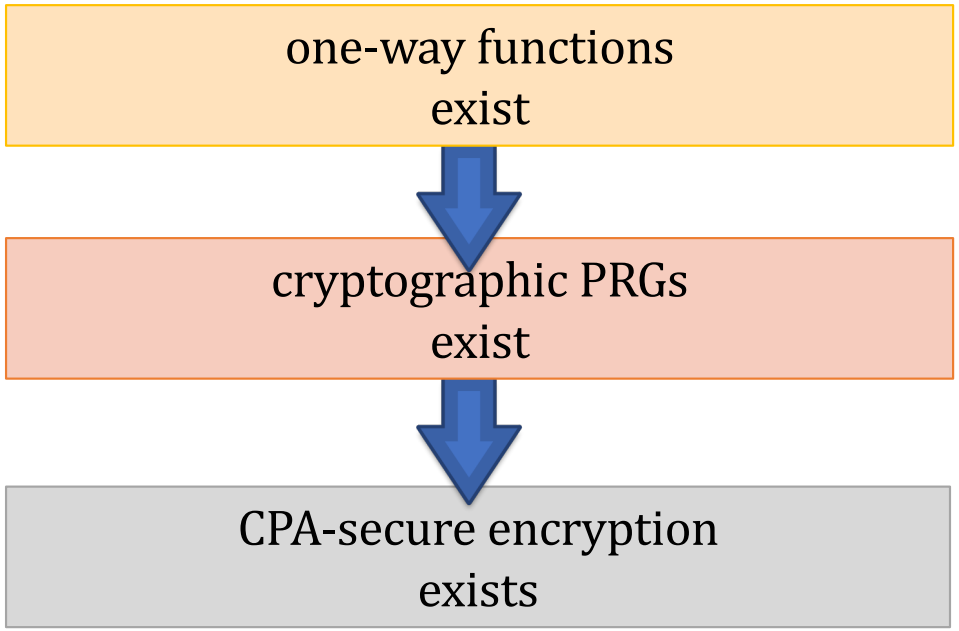
$$\boldsymbol{C := f(M)}$$



One of the most fundamental results in symmetric cryptography

[Håstad, Impagliazzo, Levin, Luby *A Pseudorandom Generator from any One-way Function*]:

"a PRG can be constructed from any **one-way function**"



Why is this result so important?

It connects **two types of hardness**:

- hardness of **guessing** some value



K **?**

- hardness of **distinguishing** two experiments

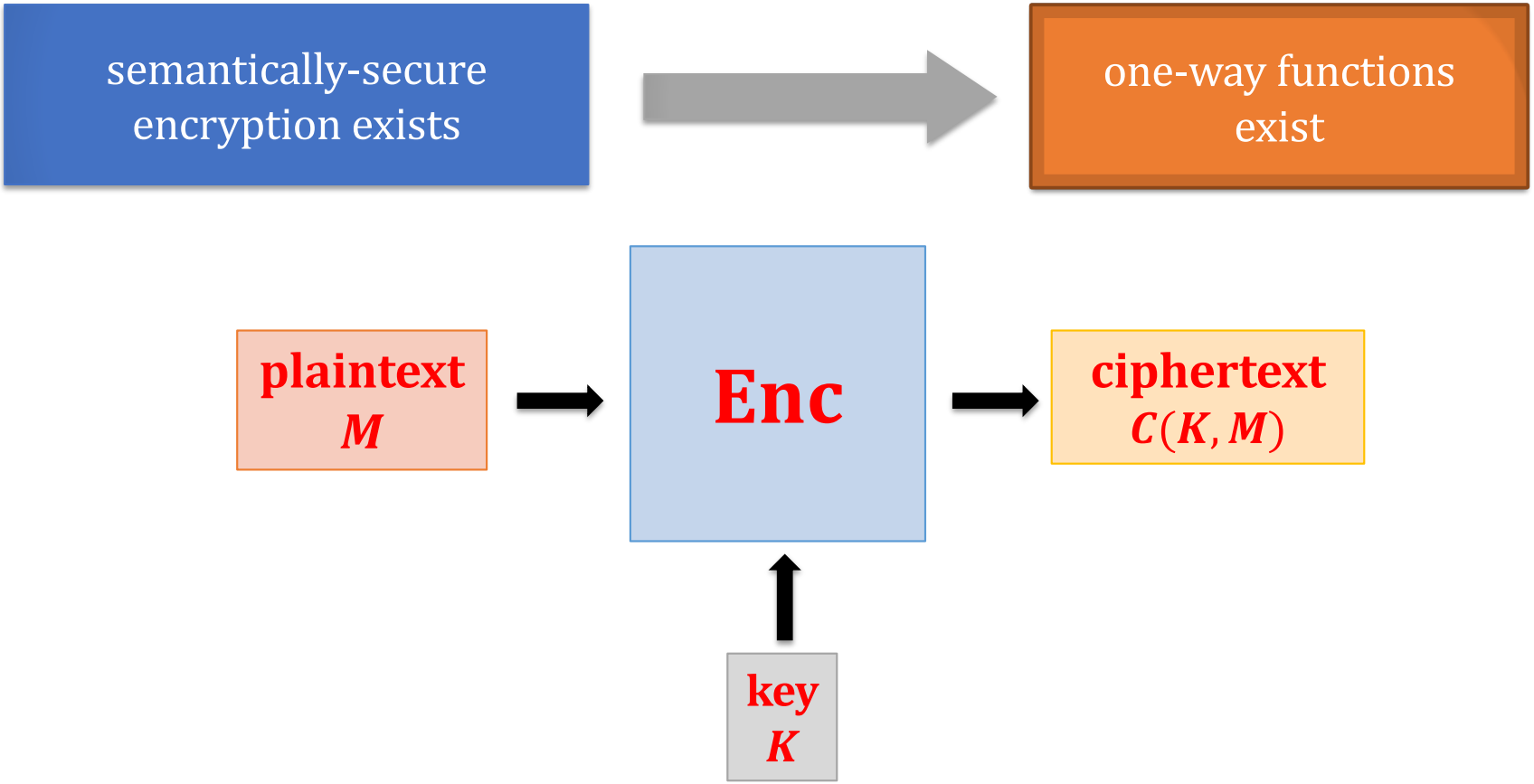
experiment **0**



?

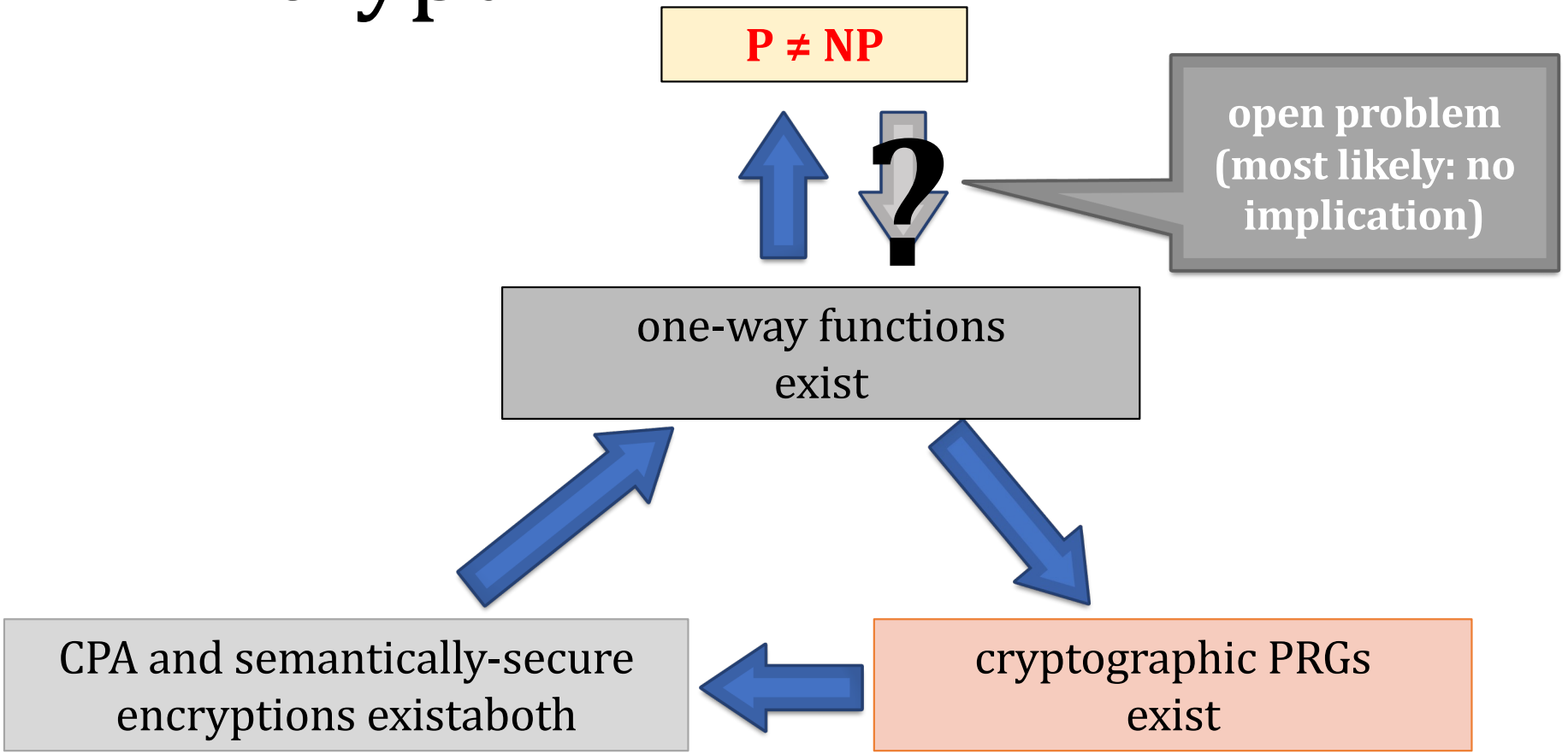
experiment **1**

The implication also holds in the other direction



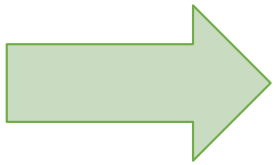
$f(K) = \mathbf{Enc}(K, (0, \ldots, 0))$ is a one-way function

“Minicrypt”



The “world” where the one-way functions exist is called “**minicrypt**”.

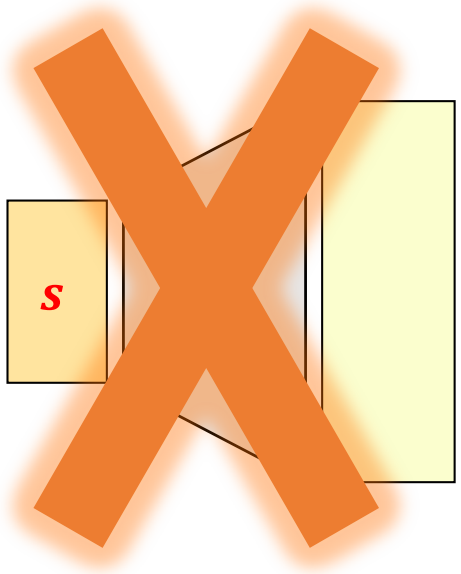
Plan

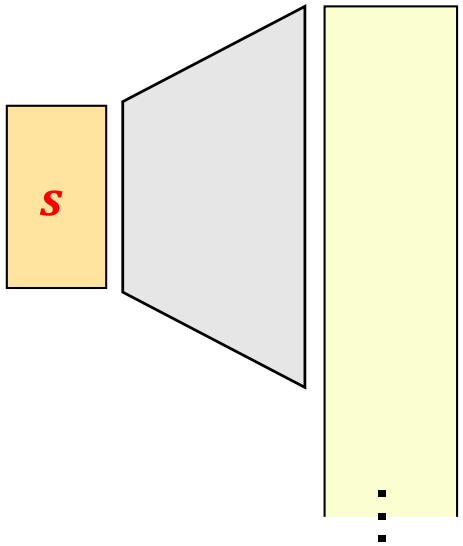


1. Computational definitions of security
2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
6. Pseudorandom functions and permutations
7. Block ciphers
8. Practical considerations

Stream ciphers

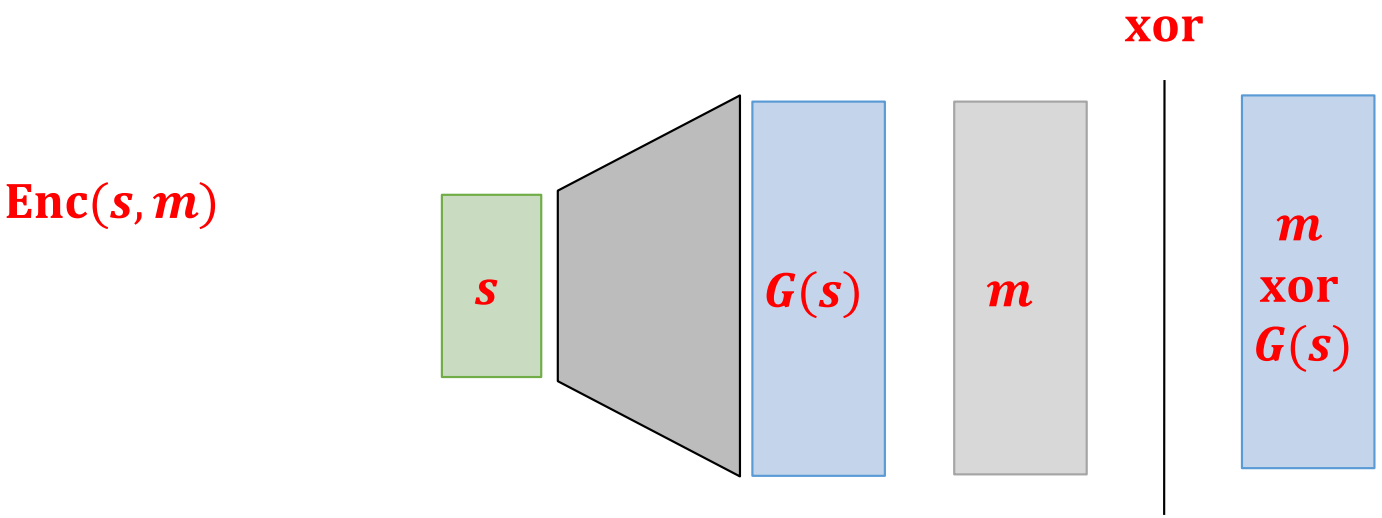
The pseudorandom generators used in practice are called **stream ciphers**





They are called like this because their output is an “infinite” **stream** of bits.

How to encrypt multiple messages using pseudorandom generators?



Of course we **cannot** just reuse the same seed
(remember the problem with the one-time pad?)

It is not just a theoretical problem!

Misuse of RC4 in Microsoft Office
[Hongjun Wu 2005]

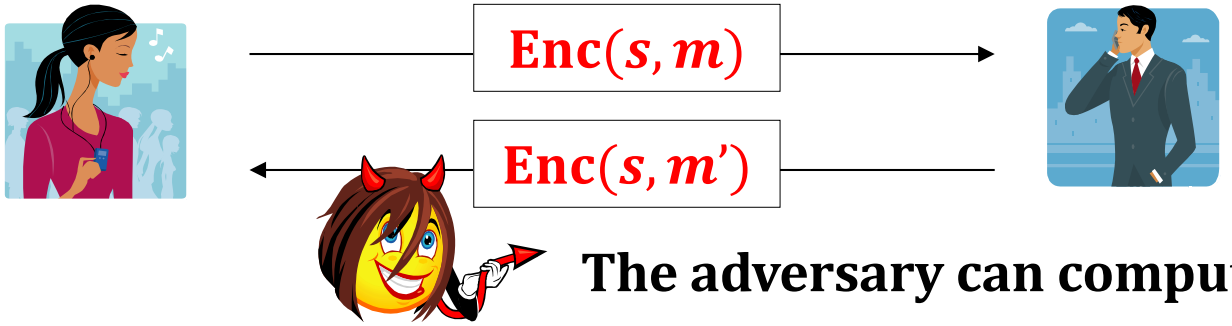
RC4 – a popular PRG/"stream cipher" (now broken)

> **"Microsoft Strong Cryptographic Provider"**
> (encryption in Word and Excel, Office 2003)

The key ***s*** is a function of a **password** and an **initialization vector**.

These values **do not change between the different versions** of the document!

Suppose **Alice** and **Bob** work together on some document:



The adversary can compute *m* xor *m'*

What to do?

There are two solutions:

1. The **synchronized mode**

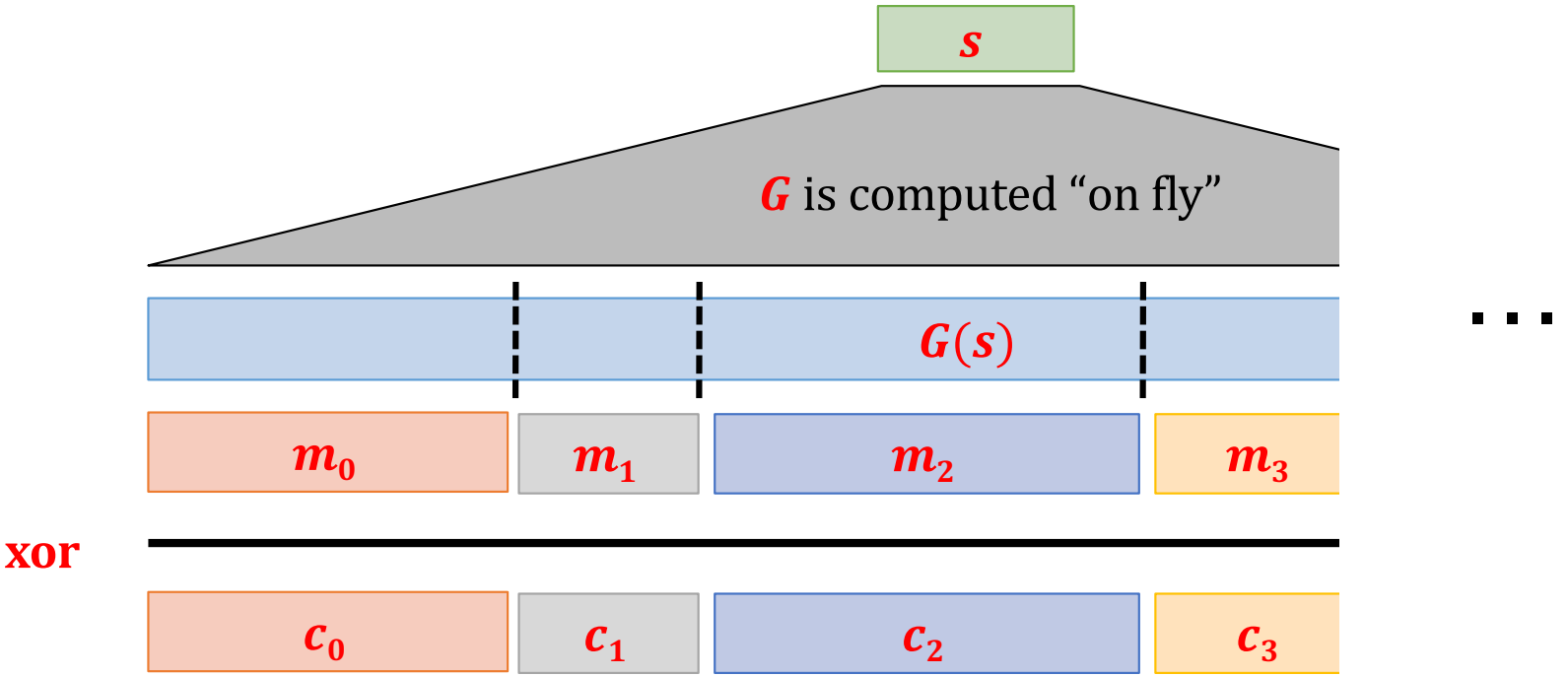
2. The **unsynchronized mode**

How to encrypt several messages

$G: \{0,1\}^n \rightarrow \{0,1\}^{\text{very large}}$ – a PRG.

this can be proven to be CPA-secure

divide $G(s)$ in blocks:



...

Unsynchronized mode

Idea

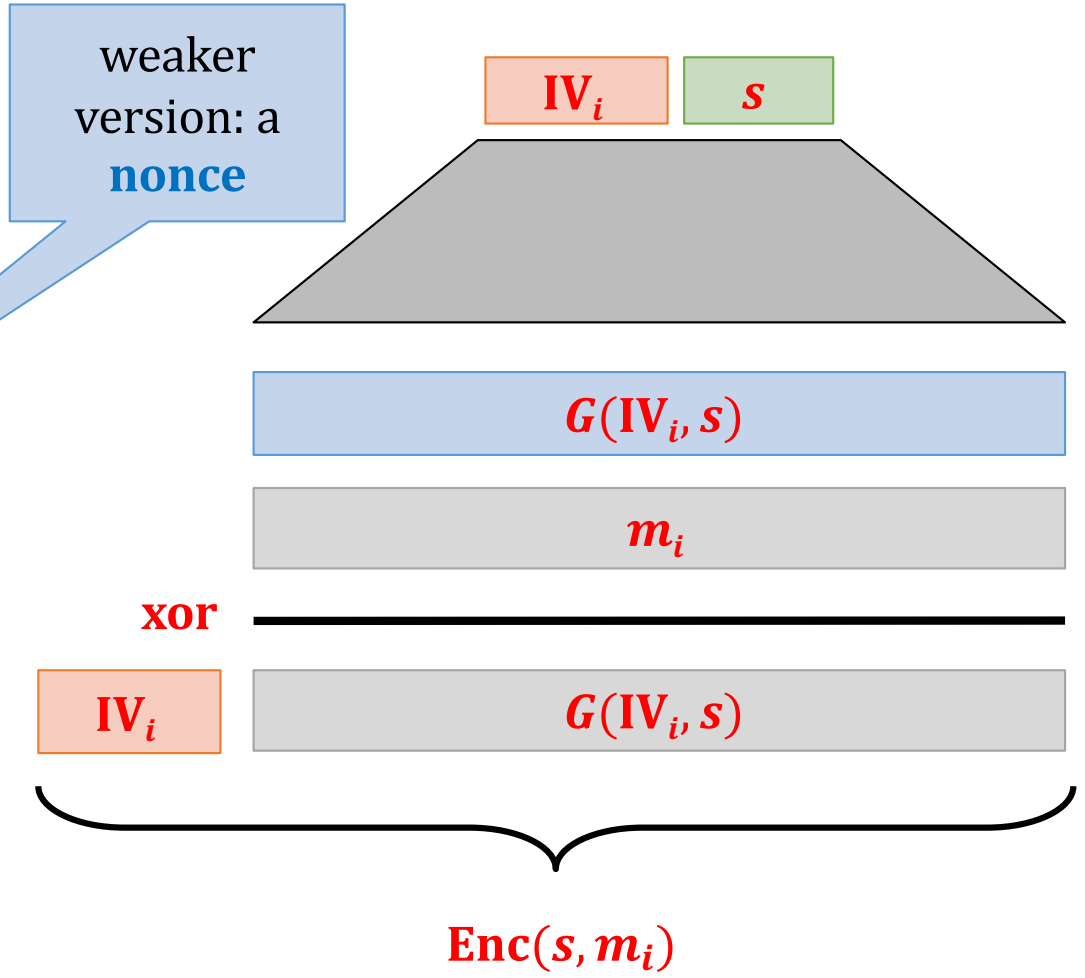
Randomize the encryption procedure.

Assume that G takes as an additional input

an **initialization vector** (**IV**).

The **Enc** algorithm selects a fresh random $\mathbf{IV}_i$ for each message $\mathbf{m}_i$.

Later $\mathbf{IV}_i$ is included in the **ciphertext**



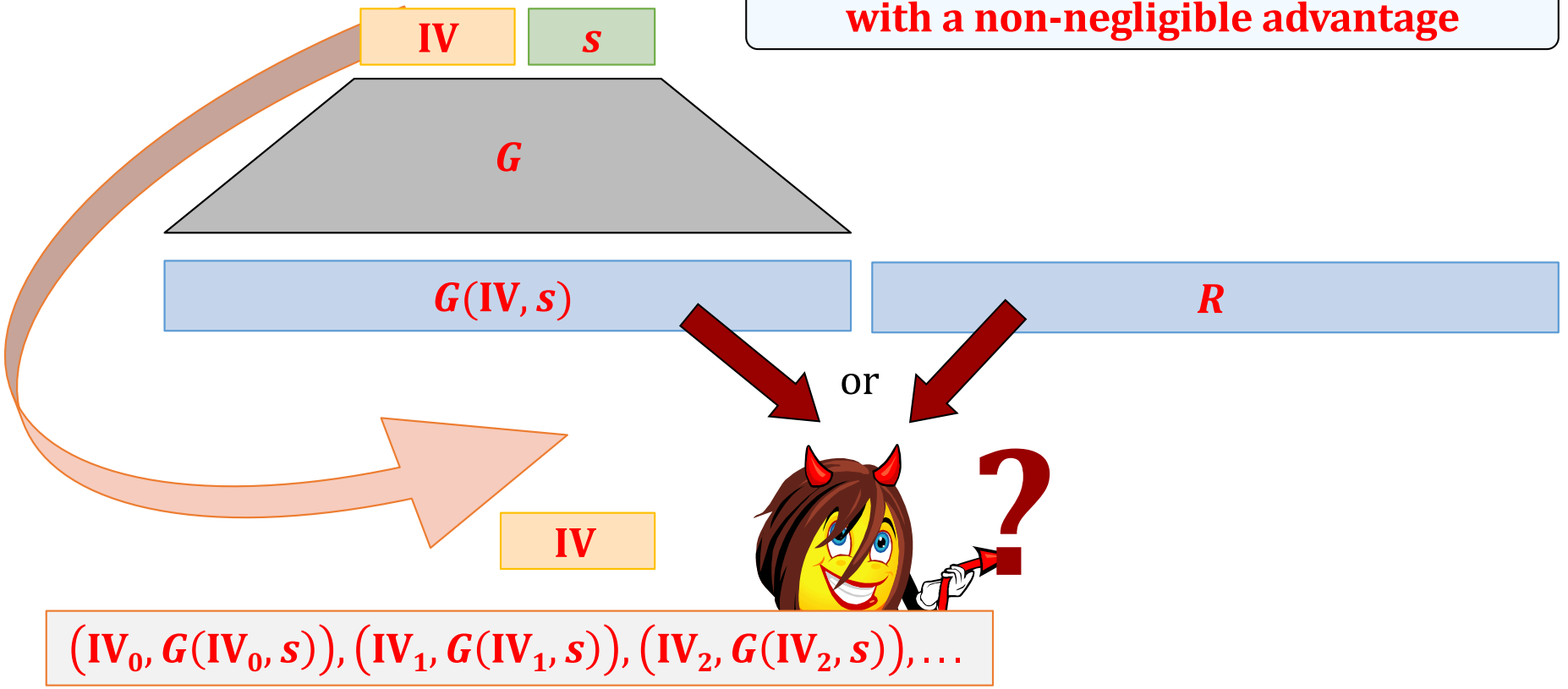
We need an "augmented" PRG

We need a **PRG** such that the adversary cannot distinguish $G(\mathbf{IV}, s)$ from a random string even if she knows **IV** and some pairs

$$(\mathbf{IV}_0, G(\mathbf{IV}_0, s)), (\mathbf{IV}_1, G(\mathbf{IV}_1, s)), (\mathbf{IV}_2, G(\mathbf{IV}_2, s)), \ldots$$

where $s, \mathbf{IV}, \mathbf{IV}_0, \mathbf{IV}_1, \mathbf{IV}_2 \ldots$ are random.

with a non-negligible advantage



How to construct such a PRG?

An **old-fashioned approach**:

1. take a standard **PRG** G
2. set $G'(\mathrm{IV}, s) := G(H(\mathrm{IV}, S))$

often:
just concatenate
IV and S

where H is a "hash-function" (we will define cryptographic hash functions later)

A more **modern approach**:

design such a G from scratch.

Popular historical stream ciphers

Based on the **linear feedback shift registers**:

- **A5/1** and **A5/2** (used in **GSM**)
  Ross Anderson:

  "there was a terrific row between the NATO signal intelligence agencies in the mid 1980s over whether GSM encryption should be strong or not. The Germans said it should be, as they shared a long border with the Warsaw Pact; but the other countries didn't feel this way, and the algorithm as now fielded is a French design."

  completely broken

- **Content Scramble System (CSS)** encryption

  completely broken

Other:

- **RC4**

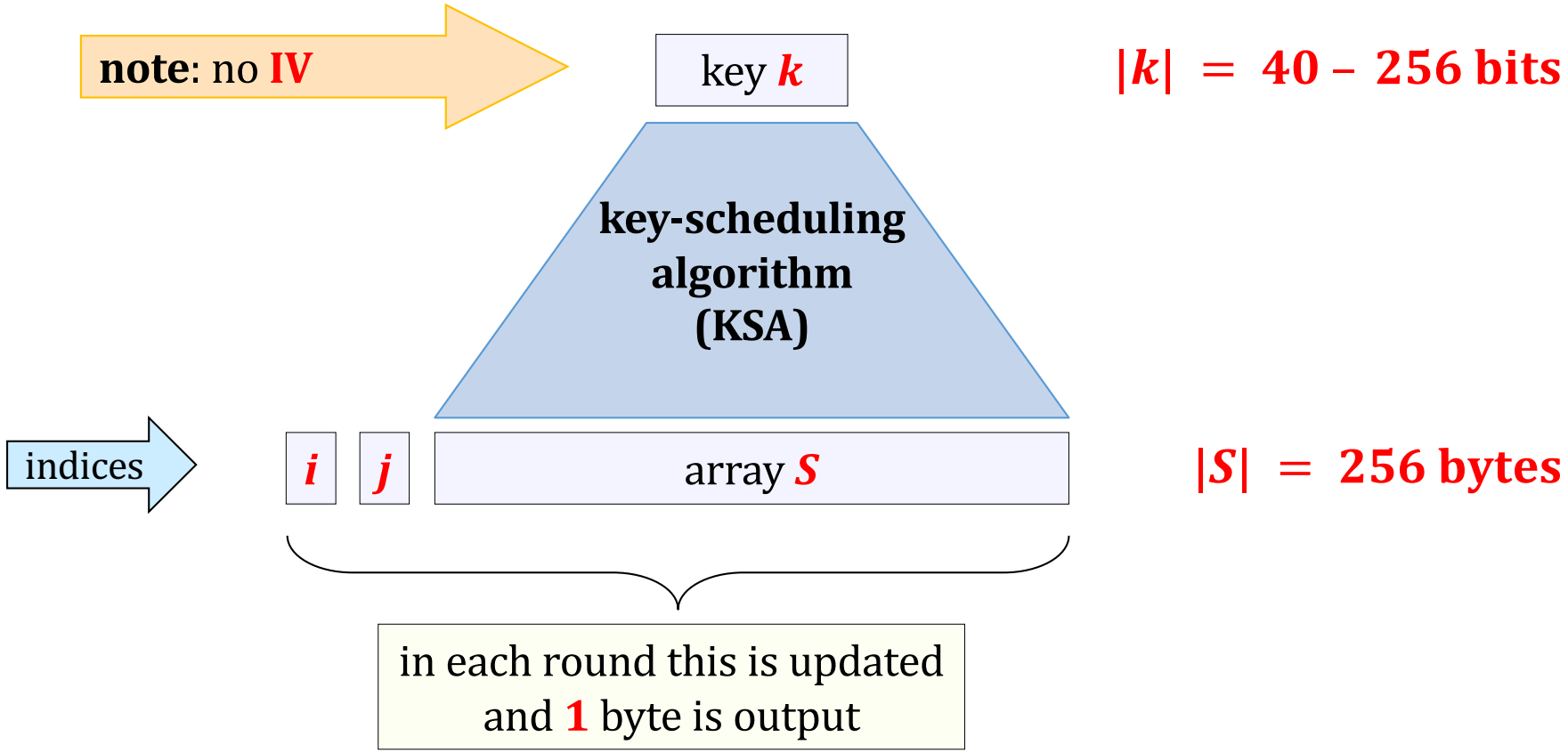
  very popular in the past, now broken

RC4



- Designed by **Ron Rivest** (**RSA Security**) in 1987.
  RC4 = "**Rivest Cipher 4**", or "**Ron's Code 4**".
- Trade secret, but in **September 1994** its description leaked to the internet.
- For legal reasons sometimes it is called: "**ARCFOUR**" or "**ARC4**".
- Used in **WEP** and **WPA** and **TLS**.
- **Very efficient and simple**, but has serious **security flaws**

RC4 – an overview



(this is called a "**pseudo-random generation algorithm (PRGA)**")

RC4

KSA

```
for i from 0 to 255
        S[i] := i
end
for j := 0 for i from 0 to 255
        j := (j + S[i] + key[i mod 256]) mod 256
        swap(S[i],S[j])
endfor
```

PRGA

```
i := 0
j := 0
while GeneratingOutput:
        i := (i + 1) mod 256
        j := (j + S[i]) mod 256
        swap(S[i],S[j])
        output S[(S[i] + S[j]) mod 256]
endwhile
```

don't read it!

Problems with RC4

1. Doesn't have a separate **IV**.

2. It was discovered that some bytes of the output are **biased**.
   [Mantin, Shamir, 2001]

3. First few bytes of output sometimes **leak some information about the key**
   [Fluhrer, Mantin and Shamir, 2001]
   Recommendation: discard the first **768-3072** bytes.

4. Other weaknesses are also known...

Use of RC4 in WEP

- **WEP** = "Wired Equivalent Privacy"

- Introduced in **1999**, for a long time widely used to protect **WiFi** communication.

- How **RC4** is used:

  to get the **seed**, the key ***k*** is **concatenated** with the **IV**

  - old versions: **|*k*| = 40 bits, |IV| = 24 bits** (artificially weak because of the **US export restrictions**)
  - new versions: **|*k*| = 104 bits, |IV| = 24 bits.**

RC4 in WEP – problems with the key length

- **|k| = 40 bits** is **not enough**:
  can be cracked using a **brute-force attack**
- **IV** is changed for each packet.
  Hence **|IV| = 24 bits** is also not enough:
  - assume that each packet has length **1500 bytes**,
  - with **5Mbps** bandwidth the set of all possible **IVs** will be exhausted in half a day
- Some implementations reset **IV := 0** after each restart – this makes things even worse.

see **Nikita Borisov, Ian Goldberg, David Wagner (2001). "*Intercepting Mobile Communications: The Insecurity of 802.11*"**

RC4 in WEP – the weak IVs

[Fluhrer, Mantin and Shamir, 2001]
(we mentioned this attack already)

For so-called **"weak IVs"** the key stream **reveals some information about the key**.

In response the vendors started to "filter" the **weak IVs**.

But then **new weak IVs were discovered**.

[see e.g. Bittau, Handley, Lackey *The final nail in WEP's coffin.*]

Another attack

"RC4 NOMORE" (www.rc4nomore.com)
[Vanhoef and Piessens, USENIX Security 2015]

Conclusion: RC4 should not be used in real life.

RC4 forbidden in TLS

Internet Engineering Task Force (IETF)
Request for Comments: 7465
Updates: 5246, 4346, 2246
Category: Standards Track
ISSN: 2070-1721

A. Popov
Microsoft Corp.

February 2015

Prohibiting RC4 Cipher Suites

Abstract

This document requires that Transport Layer Security (TLS) clients and servers never negotiate the use of RC4 cipher suites when they establish connections. This applies to all TLS versions. This document updates RFCs 5246, 4346, and 2246.

Competitions for new stream ciphers

- **NESSIE** (New European Schemes for Signatures, Integrity and Encryption, 2000 – 2003) project **failed to select** a new stream cipher (all 6 candidates were broken)

  (where "*broken*" can mean e.g. that one can distinguish the output from random after seeing 2^{36} bytes of output)

- **eStream** project (November 2004 – May 2008) chosen a portfolio of ciphers: **HC-128, Grain v1, Rabbit, MICKEY v2, Salsa20/12, Trivium, SOSEMANUK**.

Salsa 20

One of the winners of the **eStream** competition.
Author: Dan Bernstein.
Very efficient both in **hardware** and in **software**.

key k
(size: **256** bits)

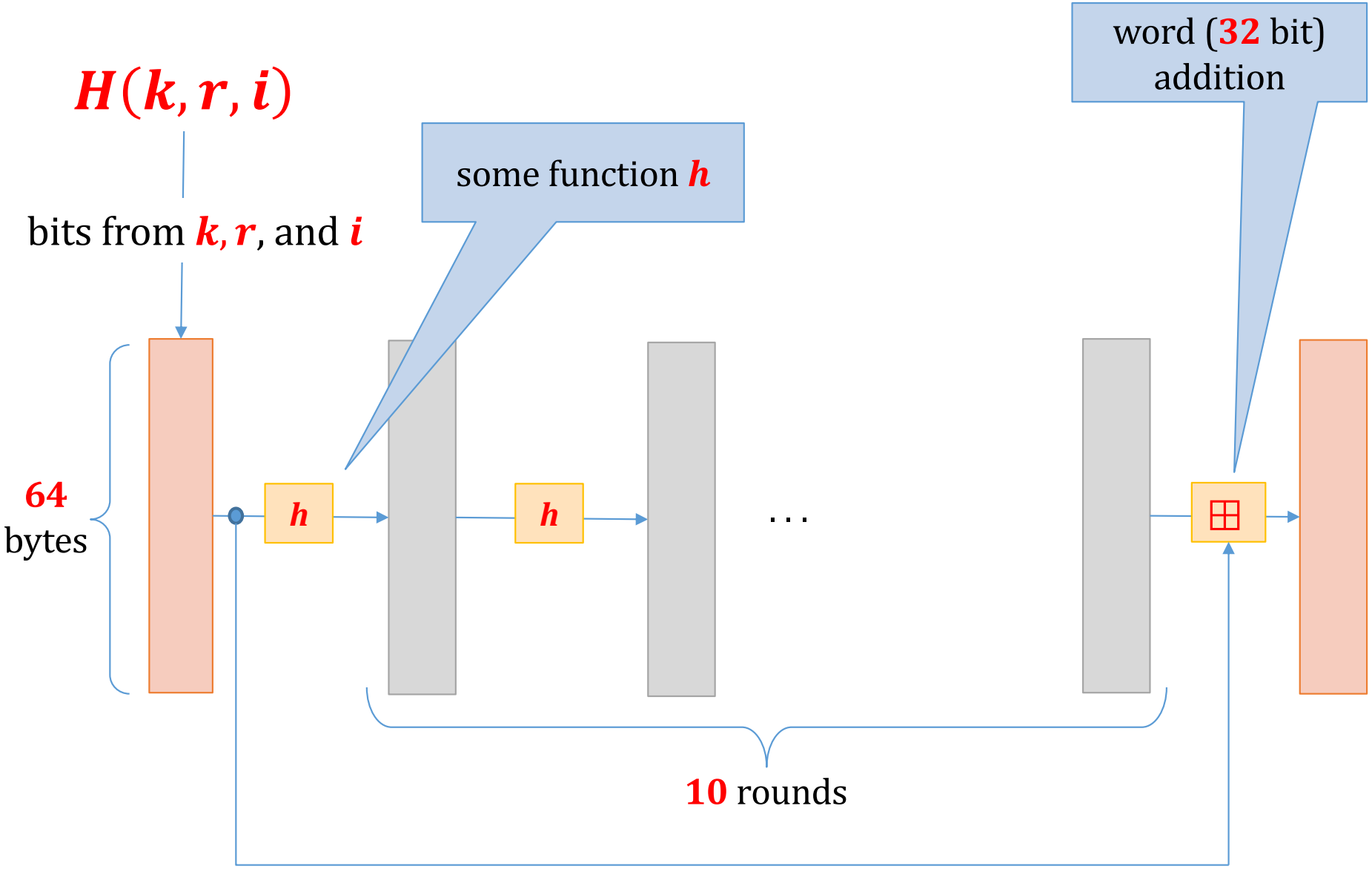
$$\mathbf{Salsa20}(k, r) := H(k, r, 0) || H(k, r, 1) || \cdots$$

nonce r
(size: **64** bits)

Note:

- **parallelizable**
- allows "**random access**"

How is H defined?

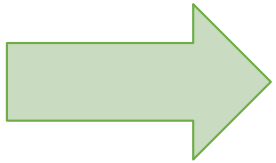


Is there an alternative to the stream ciphers?

Yes!

the **block ciphers**

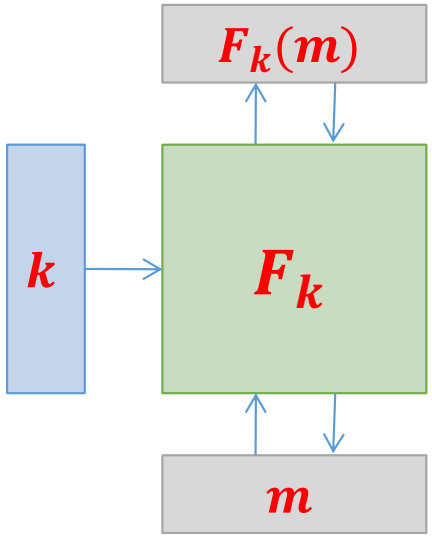
Plan



1. Computational definitions of security
2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
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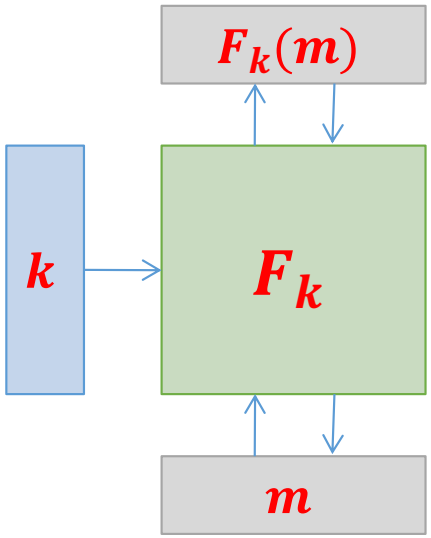
A common way of constructing "ciphers"

A cipher is just "a **deterministic box that encrypts**".



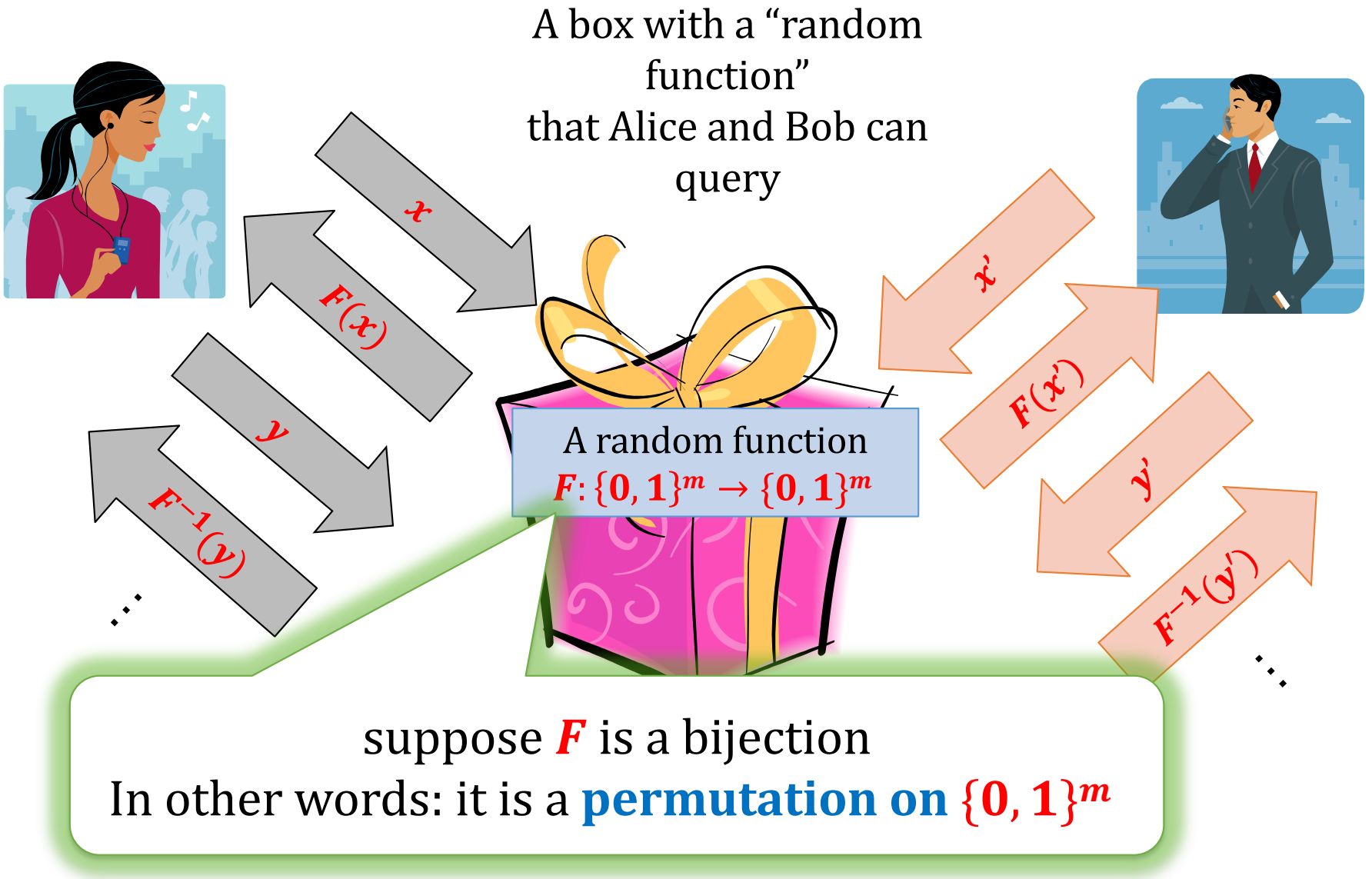
Name: block ciphers
Examples: DES, AES,…
Warning: not a good way to encrypt "directly"

How to model them formally?



block cipher ≈ "a **deterministic box that behaves randomly**".
≈ **a pseudorandom permutation**

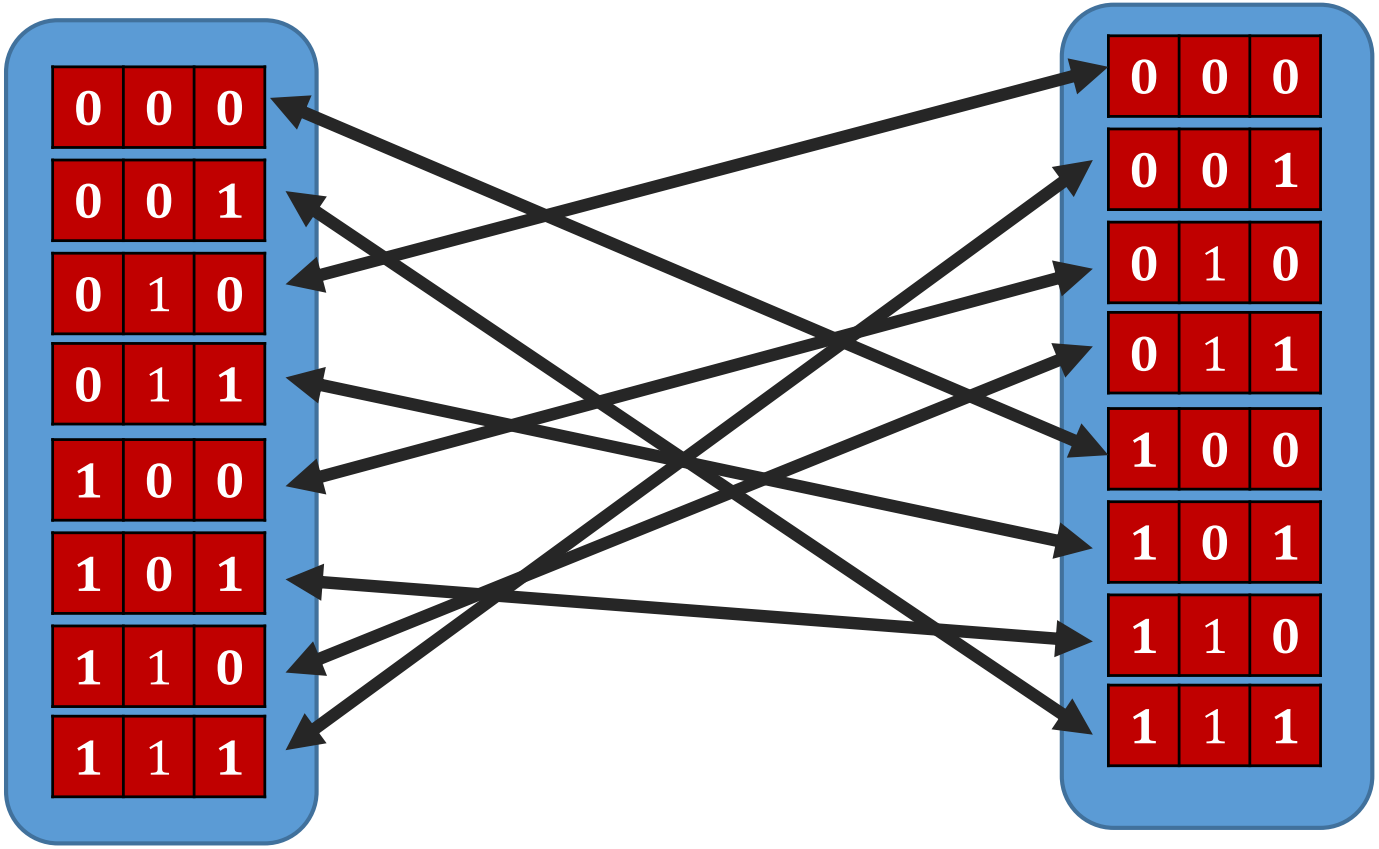
Random permutations



Note

We consider permutations on $\{0, 1\}^m$, **not** on $\{1, \dots, m\}$

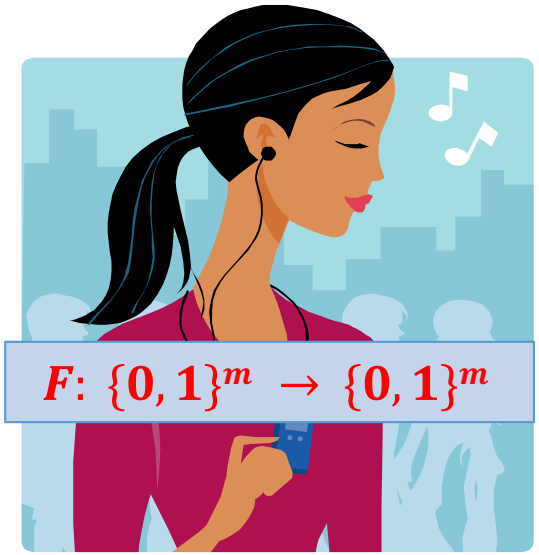
Example:

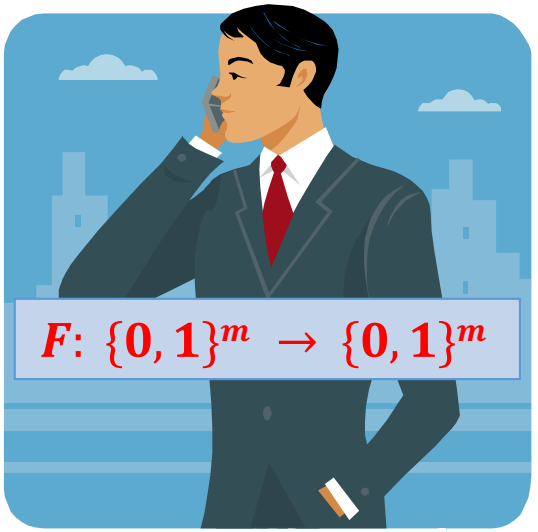


Can this box be simulated in real life?

Naive solution:

Select a random permutation $F: \{0,1\}^m \rightarrow \{0,1\}^m$ and give it to both parties.





Problem:

The number of possible permutations is $(2^m)!$

That's why **pseudo**random

One **cannot** describe a random permutation

$$F: \{0,1\}^m \rightarrow \{0,1\}^m$$

in a short space.

But we can describe a permutation that

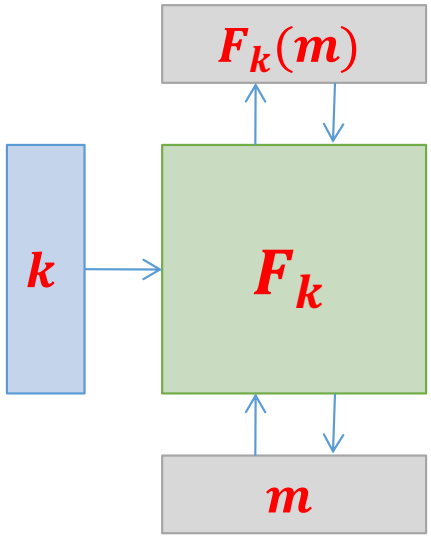
"behaves almost random"

Terminology

- **pseudorandom permutations (in theory)**
- **block ciphers (in practice)**

Keyed permutations

For a partial function
$F: \{0,1\}^* \times \{0,1\}^* \rightarrow \{0,1\}^*$
let $F_k(m)$ denote $F(k,m)$.



A **keyed-permutation** is a function
$F: \{0,1\}^* \times \{0,1\}^* \rightarrow \{0,1\}^*$ such that

1. for every k function F_k is a permutation on some $\{0,1\}^n$

2. for every k functions F_k and F_k^{-1} are poly-time computable.

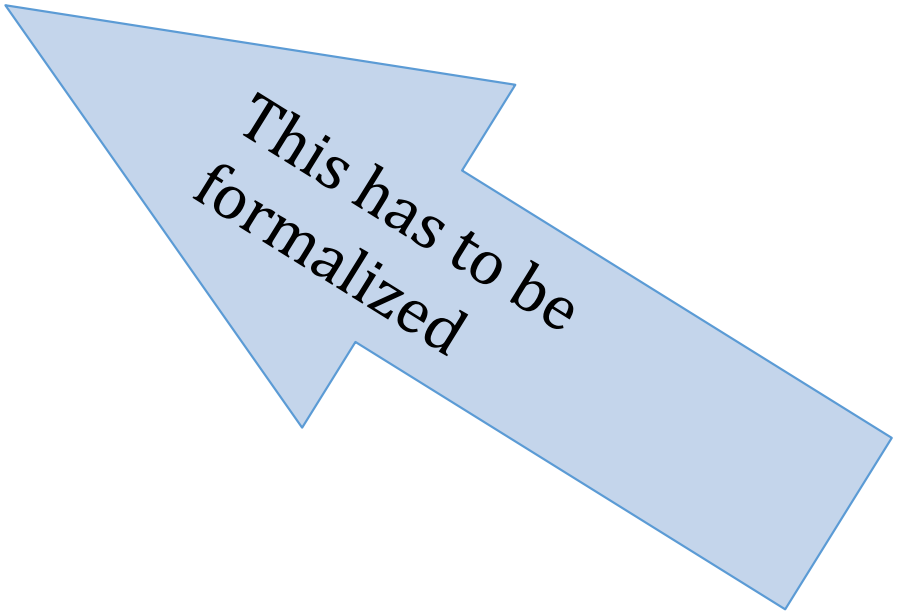
n is a function of $|k|$

for simplicity assume: $n = |k|$

Pseudorandom permutations

Intuition:

A keyed permutation ***F*** is **pseudorandom** if it cannot be distinguished from a completely random permutation.



Scenario **0**

security parameter
1^n

distinguisher D



oracle
chooses a random $k \in \{0,1\}^n$.



$m_1 \in \{0,1\}^n$

$F_k(m_1)$

$m_2 \in \{0,1\}^n$

$F_k(m_2)$

$\vdots$

$m_t \in \{0,1\}^n$

$F_k(mt)$

outputs $b \in \{0,1\}$

Scenario 1

security parameter
1^n

distinguisher D



$m_1 \in \{0,1\}^n$

$F(m_1)$

$m_2 \in \{0,1\}^n$

$F(m_2)$

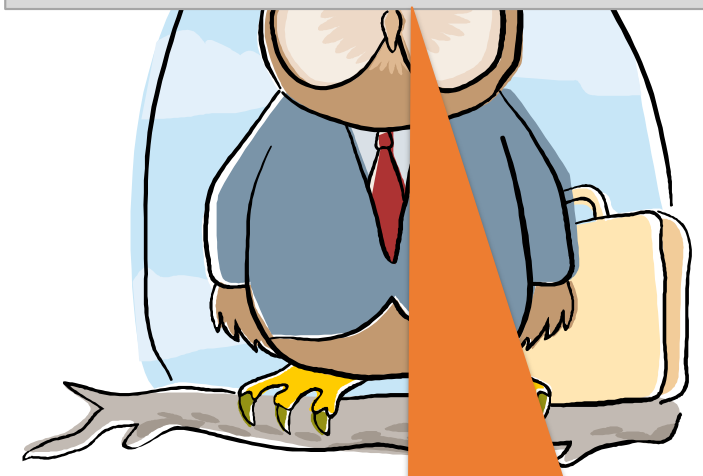
$\vdots$

$m_t \in \{0,1\}^n$

$F(mt)$

outputs $b \in \{0,1\}$

oracle
chooses a random permutation
$F : \{0,1\}^n \rightarrow \{0,1\}^n$



This of course **cannot** be done efficiently, but it doesn't matter

Pseudorandom permutations – the definition

We say that a **keyed-permutation** $F: \{0,1\}^* \times \{0,1\}^* \to \{0,1\}^*$ is a **pseudorandom permutation (PRP)** if

any **polynomial-time randomized distinguisher** D cannot distinguish **scenario 0** from **scenario 1** with a **non-negligible advantage**.

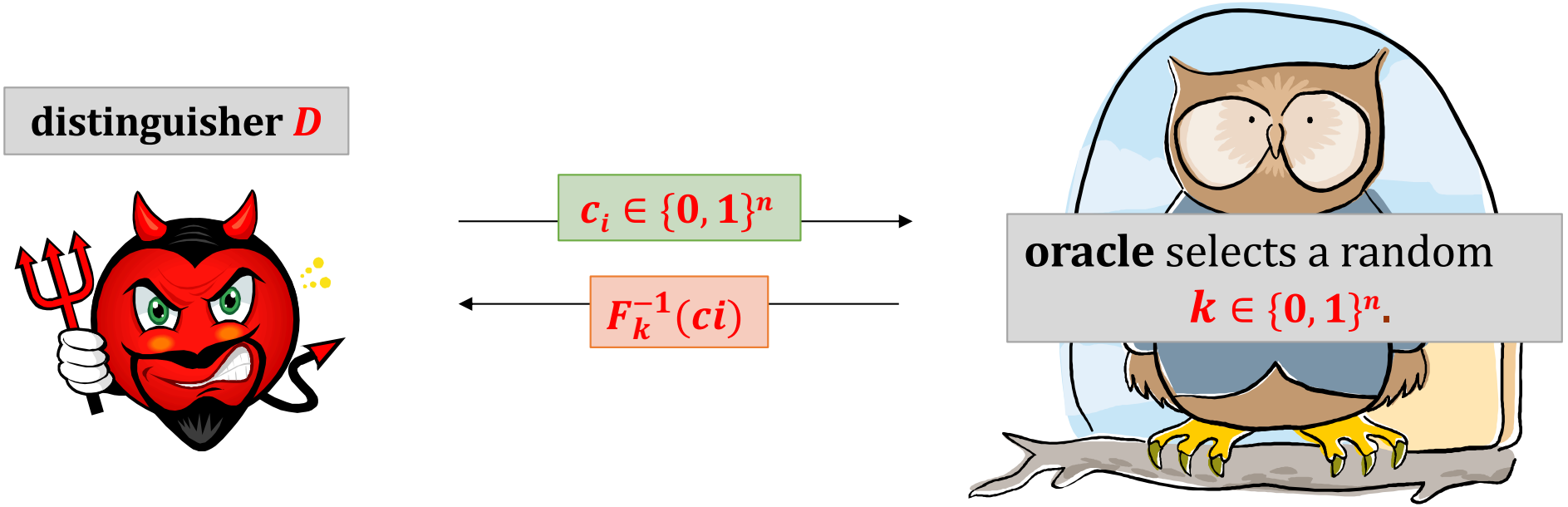
That is:

$$|P(D \text{ outputs "1" in scenario 0}) - P(D \text{ outputs "1" in scenario 1})|$$

is negligible in n.

Strong pseudorandom permutations

Suppose we allow the distinguisher to **additionally** ask the oracle for inverting F:



Then we get a definition of a **strong pseudorandom permutation.**

PRFs vs PRP

If we drop the assumption that

F_k **has to be a permutation**

we obtain an object called

a "pseudorandom **function (PRF)**".

The security definition doesn't change.

In fact those two objects are **indistinguishable** for a polynomial-time adversary.

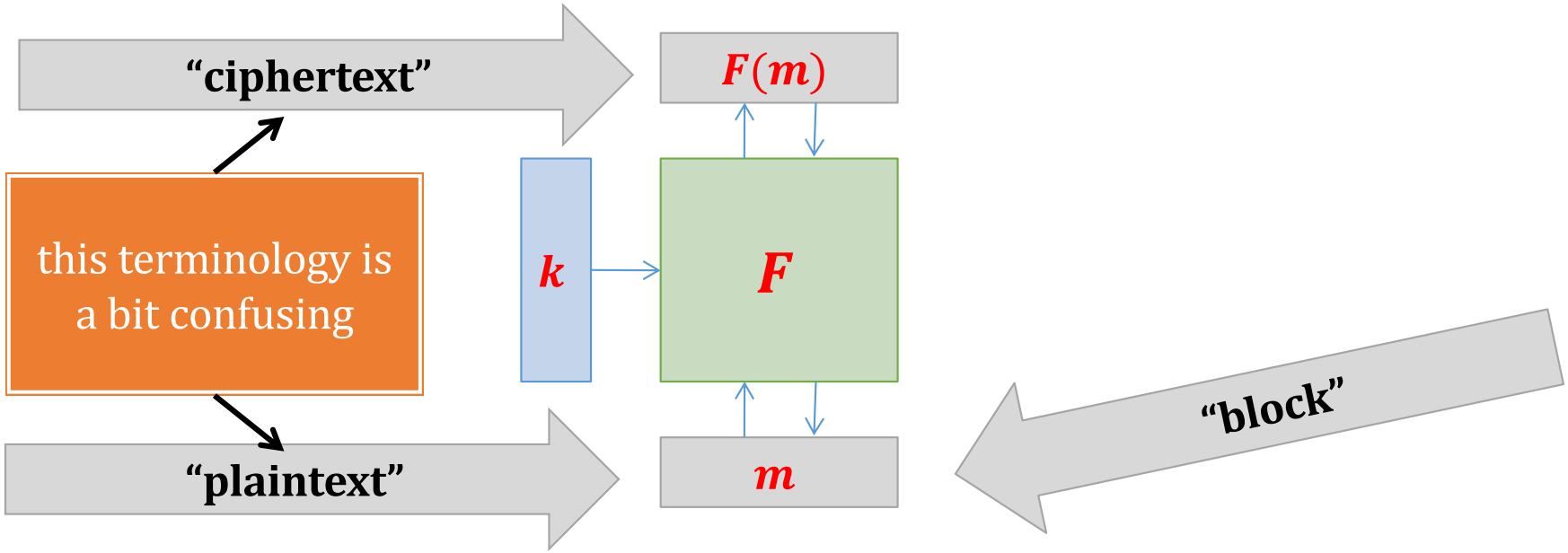
Terminology

Before we had:

stream ciphers ≈ pseudorandom generators

Similarly:

block ciphers ≈ pseudorandom permutations



Another way to look at the stream ciphers :

m is a parameter



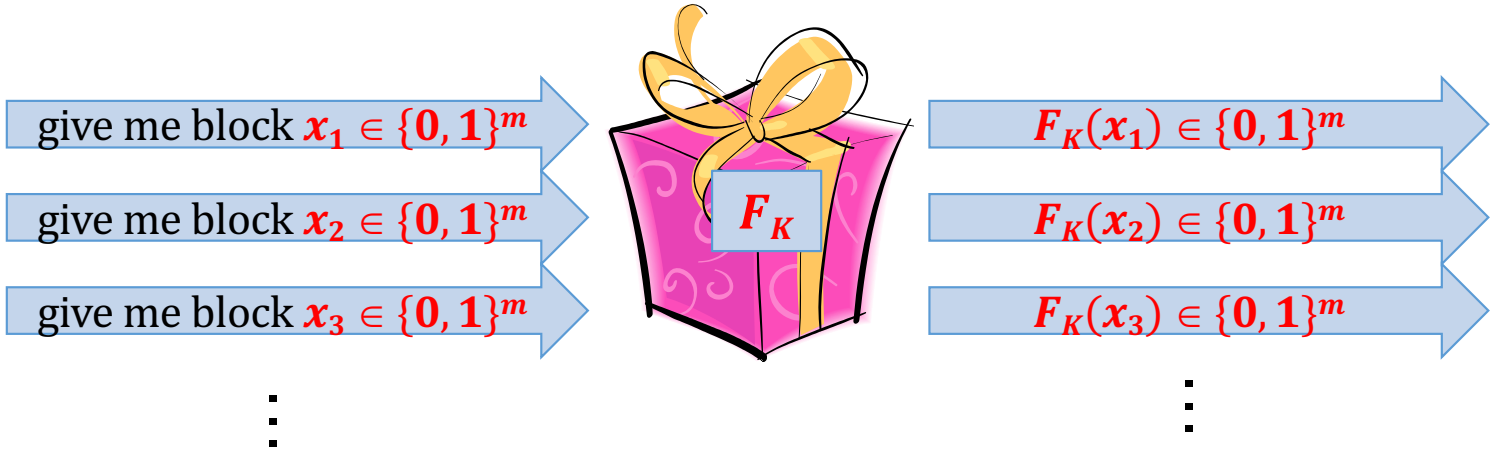
Requiremenent:

$$G_K(1), G_K(2), G_K(3), \ldots$$

has to "look random" if is K random and secret.

Block ciphers:

m is a parameter



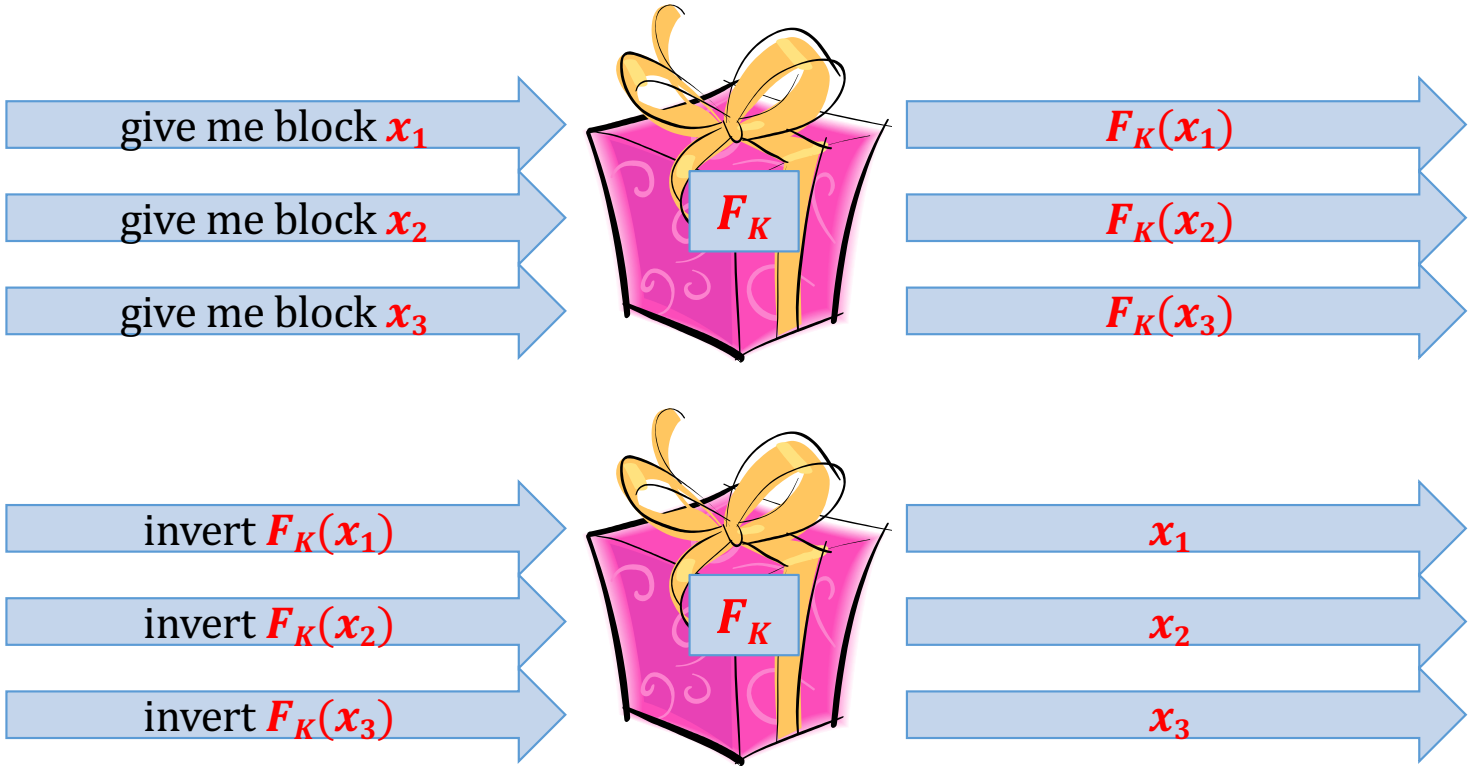
for x_1, x_2, x_3 … chosen adversarily

Requirement:

$$F_K(x_1), F_K(x_2), F_K(x_3), \ldots$$

has to "look random" if is K random and secret.

An additional property of the block ciphers



Popular block ciphers

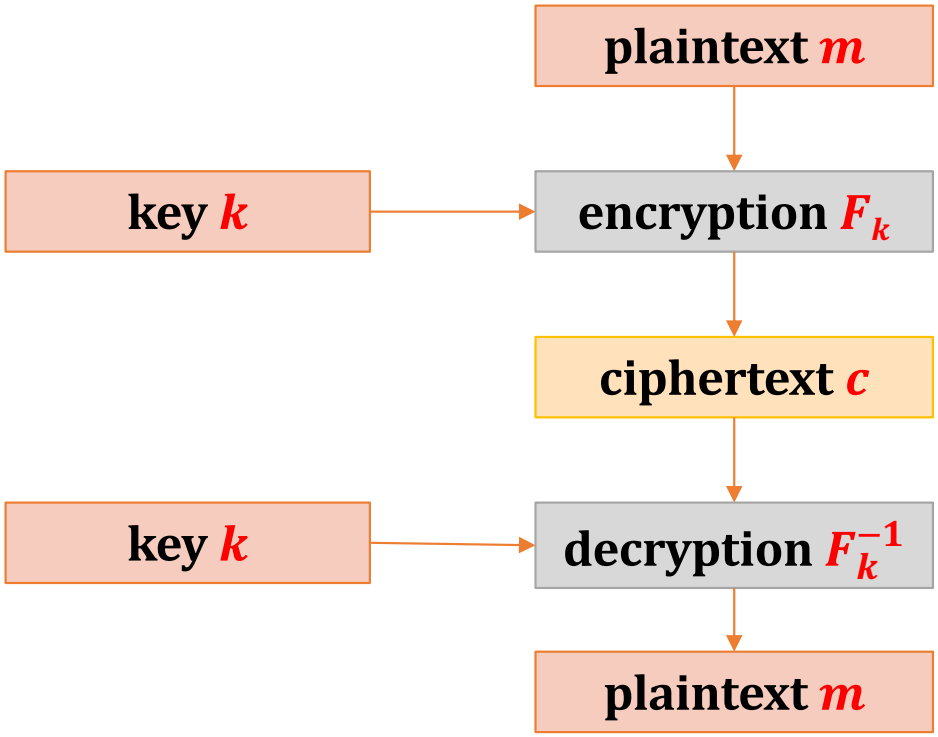
A great design.
The only practical weakness: **short key**.
Can be broken by a **brute-force attack**.

	key length	block length
DES (1976) (**Data Encryption Standard**)	**56**	**64**
IDEA (1991) (**International Data Encryption Algorithm**)	**128**	**64**
AES (1998) (**Advanced Encryption Standard**)	**128**, **192** or **256**	**128**

Other: **Blowfish, Twofish, Serpent,...**

How to encrypt using the block ciphers?

A naive (wrong) idea: Encrypt short blocks:



Problems:

1. the messages have to be short
2. it is **deterministic** and **has no state**, so it **cannot be CPA-secure**.

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2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
6. Pseudorandom functions and permutations
7. Block ciphers
    1. Modes of operation
    2. Popular construction paradigms
    3. Cascade ciphers
8. Practical considerations



Block cipher modes of operation

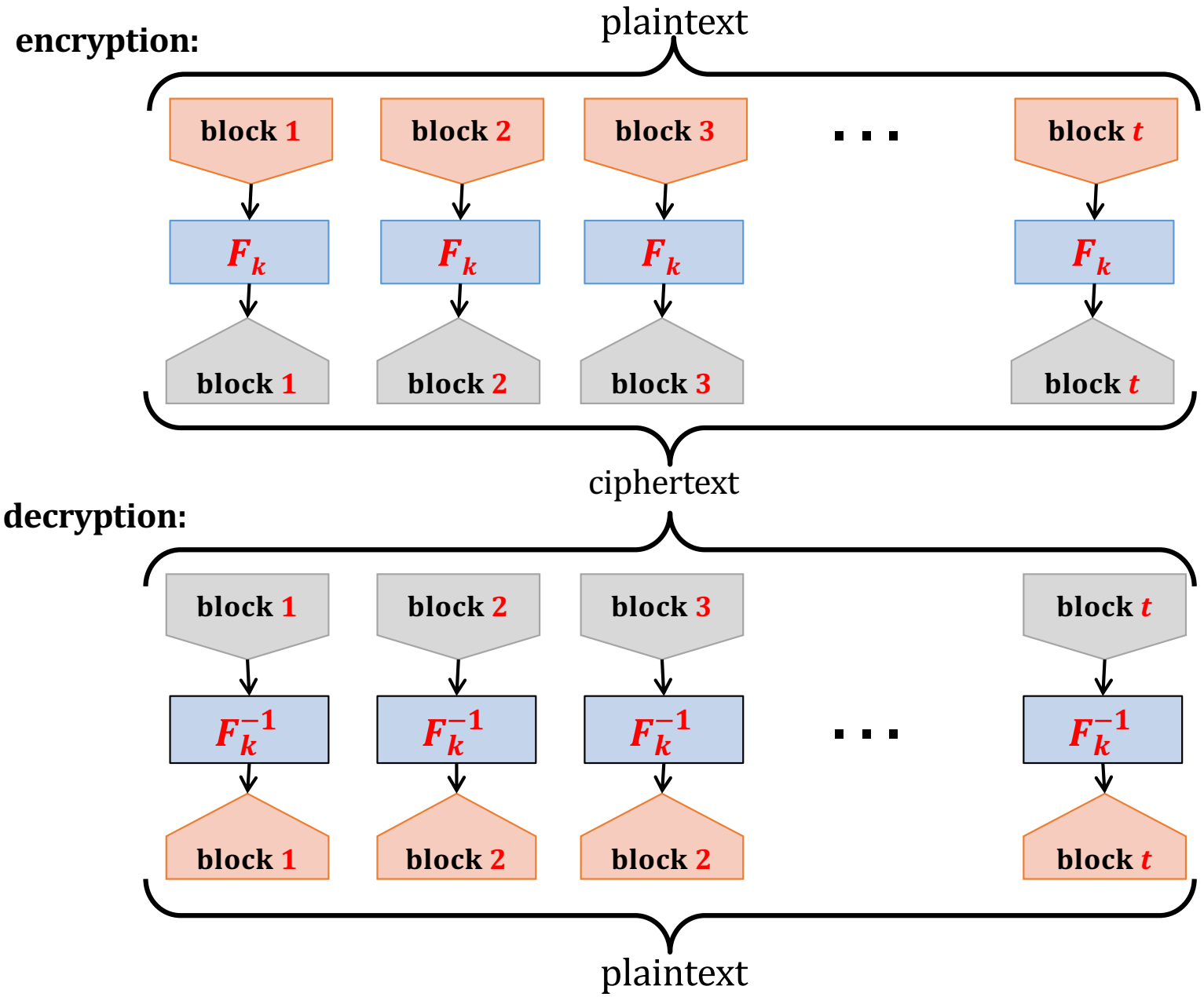
Block ciphers **cannot be used directly for encryption**.

They are always used in some "**modes of operation**"

1. **Electronic Codebook (ECB)** mode ← **not secure,**
2. **Cipher-Block Chaining (CBC)** mode,
3. **Output Feedback (OFB)** mode,
4. **Counter (CTR)** mode,

. . .

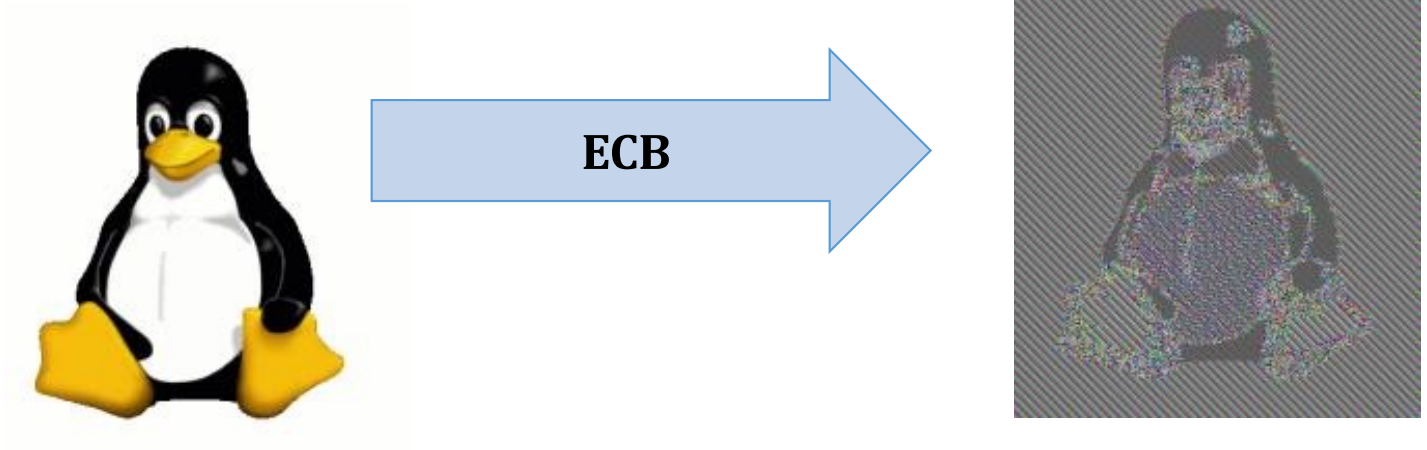
Electronic Codebook mode



This mode was used in the past.

It is not secure, and should not be used.

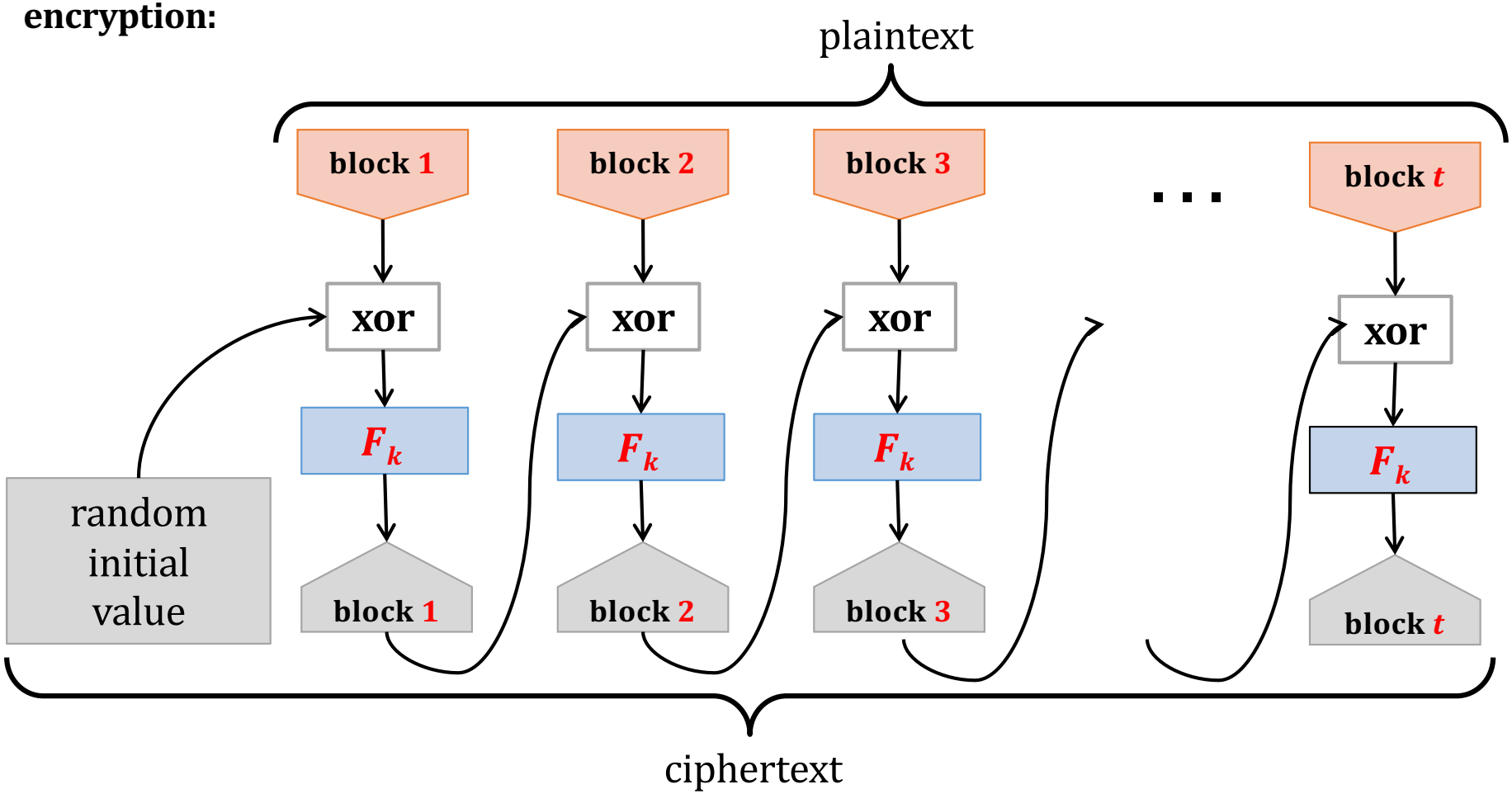
Example:



© *wikipedia*

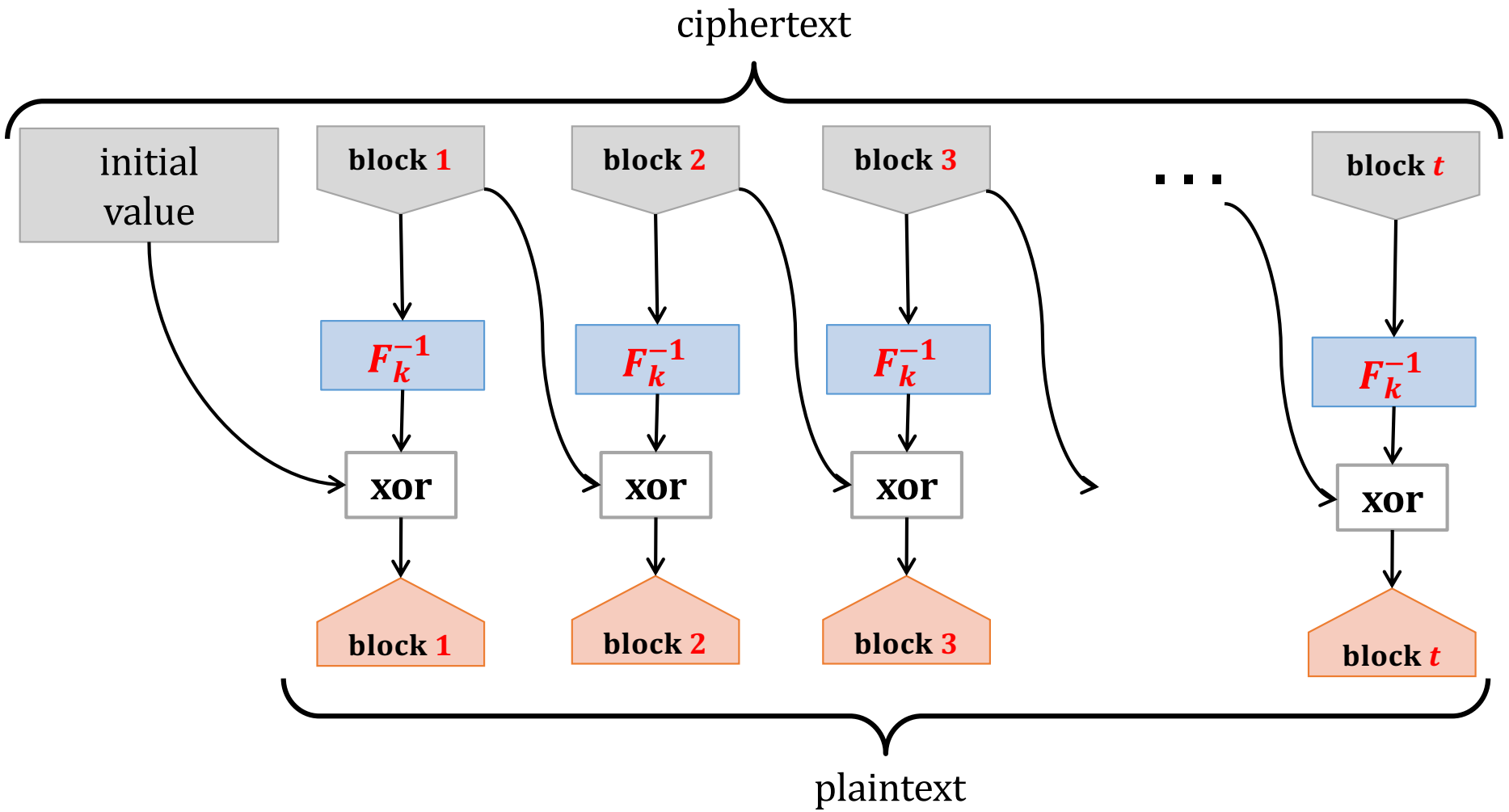
Cipher-Block Chaining (CBC)

encryption:



Cipher-Block Chaining (CBC)

decryption:



CBC mode – properties

Error propagation?

Error in block c_i affects only c_i and c_{i+1}.
So, errors don't propagate (This mode is **self-synchronizing**)



Can encryption be parallelized?

No



Can decryption be parallelized?

Yes



What if one bit of plaintext is changed?

Everything needs to be recomputed
(not so good e.g. for disc encryption)

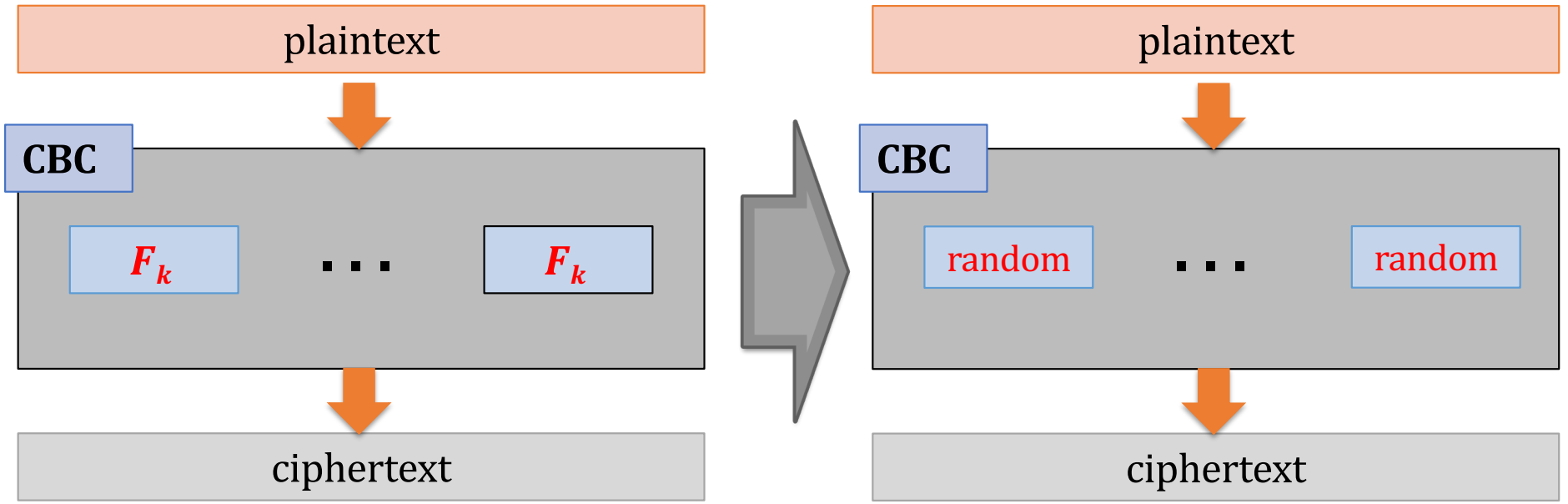


CBC mode is secure

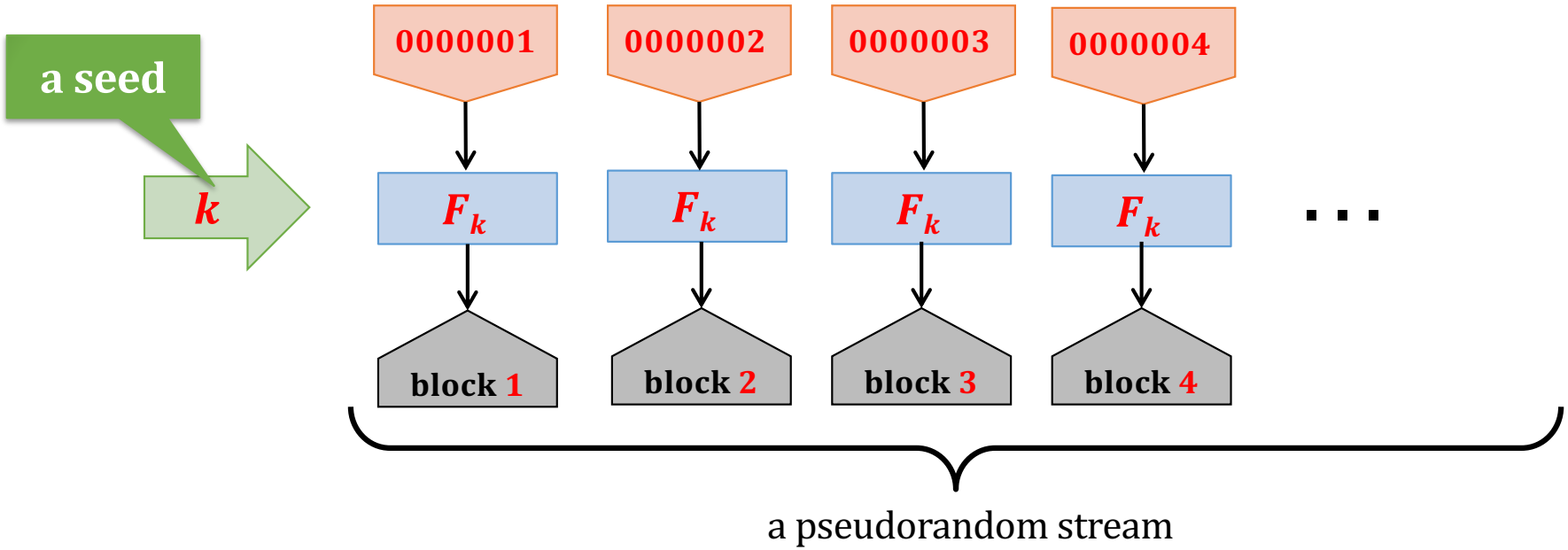
Theorem. If F is a **PRP** then F**-CBC** is secure.

[M. Bellare, A. Desai, E. Jokipii and P. Rogaway 1997]

In the proof one can assume that F_k is a completely random function.
(If **CBC** behaves differently on a pseudorandom function, then one could construct a distiguisher.)



How to convert a pseudorandom **permutation** into a pseudorandom **generator**?

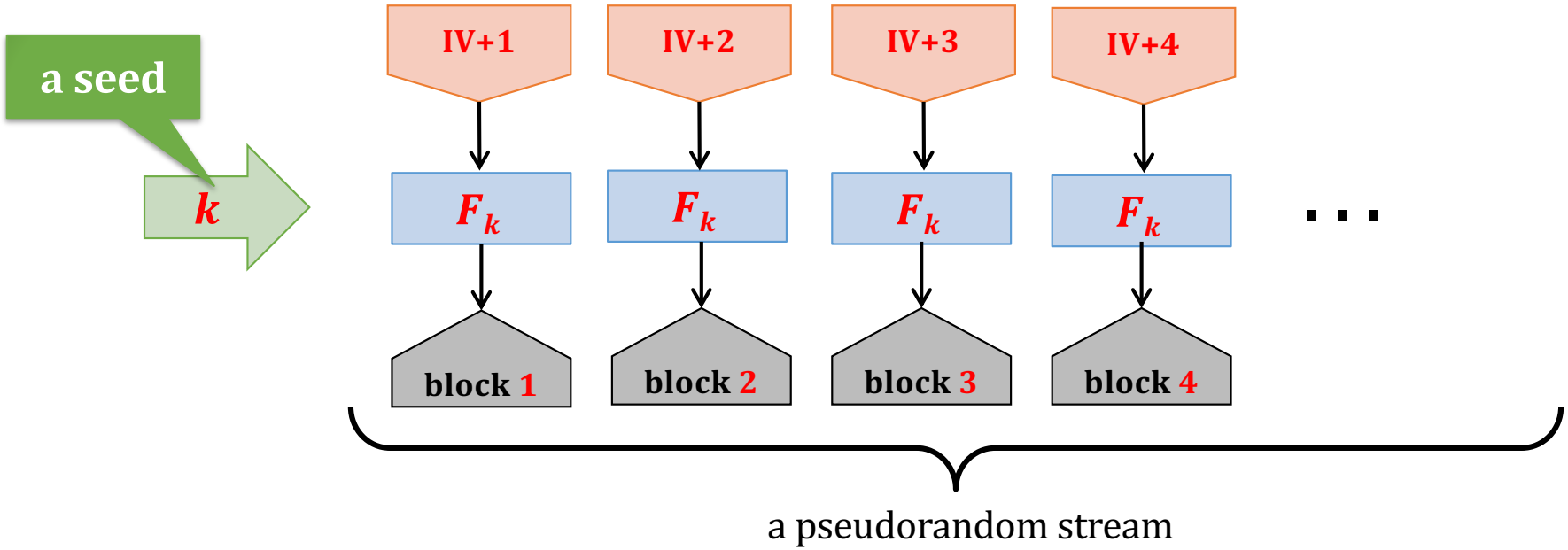


$G(k)$:
$= Fk(1) \,||\, Fk(2) \,||\, Fk(3) \,||\, \cdots$

Essentially, this is called a "**counter mode**" (**CTR**).

How to “randomize” this?

take some random **IV**



$$G(k, \mathrm{IV}) := Fk(\mathrm{IV} + 1) \,||\, Fk(\mathrm{IV} + 2) \,||\, Fk(\mathrm{IV} + 3) \,||\, \cdots$$

Note:

We have to be sure that **IV + *i*** never repeats.

This is why it is bad if the block length is too small (like in **DES**).

CTR mode – properties

Error propagation?

Error in block c_i affects only c_i.



(But this mode is **not** **self-synchronizing**)



Can encryption be parallelized?

Yes



Can decryption be parallelized?

Yes

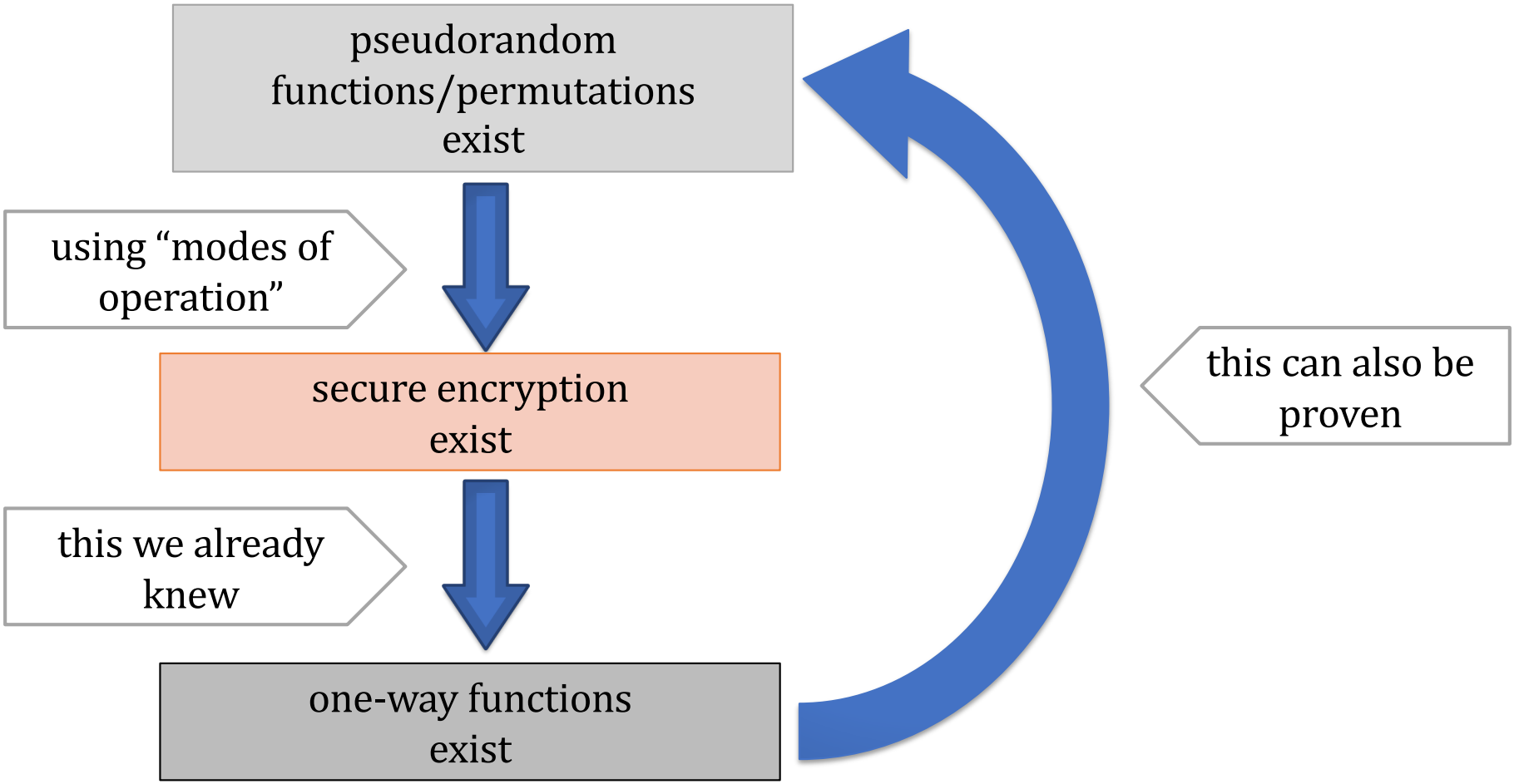


What if one bit of plaintext is changed?

Only one block needs to be recomputed



One more member of minicrypt!



There are many constructions of block ciphers that are **believed** to be secure

Why do we believe it?





- Someone important say “it is secure”.

(But is he honest?)

- Many people tried to break it and they failed...

Plan

1. Computational definitions of security
2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
6. Pseudorandom functions and permutations
7. Block ciphers
    1. Modes of operation
    2. Popular construction paradigms
    3. Cascade ciphers
8. Practical considerations



Block ciphers – typical requirements

- **security**: ideally the best attack should be the **brute force key search**.
- **efficiency** when implemented on:
  - **8 bit microcontrollers** and **smart cards** with limited memory
  - **tablets**, **phones**, **palmtops**,
  - **PCs**, **workstations**, **servers**,
  - dedicated hardware (**ASICs**, **FPGAs**) – here we might require speeds up to **gigabits/second**
- **key agility** – **changing the key** can be done very efficiently

Block ciphers – more “informal” requirements

- **simplicity** – advantages:
  - **easier to implement**
  - more confidence that there is **no backdoor**
- **symmetry** (repeating patterns):
  - **smaller circuits** (in hardware)
  - **easier to program** (in software).

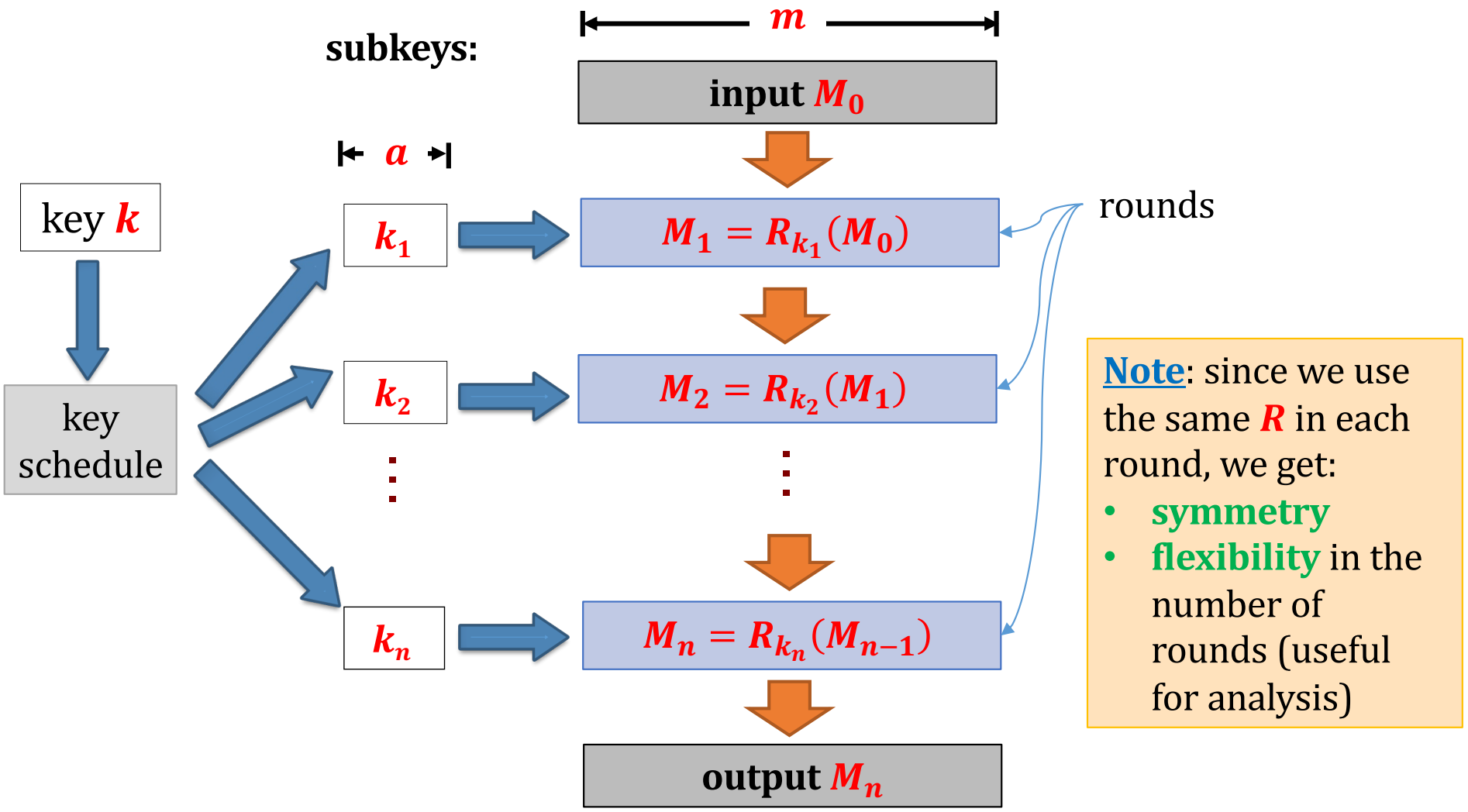
Block ciphers –advanced security requirements

- resistance to the **side-channel attacks**,
- resistance to the **key-related attacks**.

A very popular paradigm: iterated ciphers

$R: \{0,1\}^a \times \{0,1\}^m \rightarrow \{0,1\}^m$ – a **round function**

Typically we write the first argument in a subscript.



Note: since we use the same R in each round, we get:

- **symmetry**
- **flexibility** in the number of rounds (useful for analysis)

Popular types of iterated ciphers

1. **Feistel** ciphers
2. **Substitution-permutation networks**
3. **Lai-Massey** ciphers

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Feistel ciphers

Invented by **Horst Feistel** (1915-1990) in **1970s** while working at **IBM**.



First used in **Lucifer.** Most famous use: **Data Encryption Standard (DES)**.

Other ciphers that use it:

Blowfish, Camellia, CAST-128, FEAL, GOST 28147-89, ICE, KASUMI, LOKI97, MARS, MAGENTA, MISTY1, RC5, Simon, TEA, Twofish, XTEA,...

DES (Digital Encryption Standard)

- **Key length**:
  - effective: **56** bits
  - formally: **64** bits (**8** bits for checking parity).
- **Block length**: **64** bits



History of DES



- First version designed by **IBM** in **1973-74**, based on a **Lucifer** cipher (by **Horst Feistel**).
- **National Security Agency (NSA)** played some role in the design of **DES**.
- Made public in **1975**.
- Approved as a **US federal standard** in November **1976**.

Criticism of DES

- The key is to short (only **56** bits).
- Unclear role of **NSA** in the design
  - hidden **backdoor**?
  - 2^{56} : feasible for **NSA**, infeasible for the others (in the **1970s**)?

Security of DES

- The main weakness is the **short key** (**brute-force** attacks are possible).

- Also the **block length is too small**.

Apart from this – **a very secure design**:

after almost **5 decades** still the most practical attack is **brute-force**!

The only attacks so far:
- **differential cryptanalysis**
- **linear cryptanalysis**

are rather theoretical

The role of NSA

The **United States Senate Select Committee on Intelligence (1978)**:

"In the development of **DES**, **NSA** convinced **IBM** that a reduced key size was sufficient; indirectly assisted in the development of the **S-box** structures; and certified that the final **DES** algorithm was, to the best of their knowledge, free from any statistical or mathematical weakness."

"**NSA** did not tamper with the design of the algorithm in any way. **IBM** invented and designed the algorithm, made all pertinent decisions regarding it, and concurred that the agreed upon key size was **more than adequate** for all commercial applications for which the **DES** was intended."

Brute-force attacks on DES

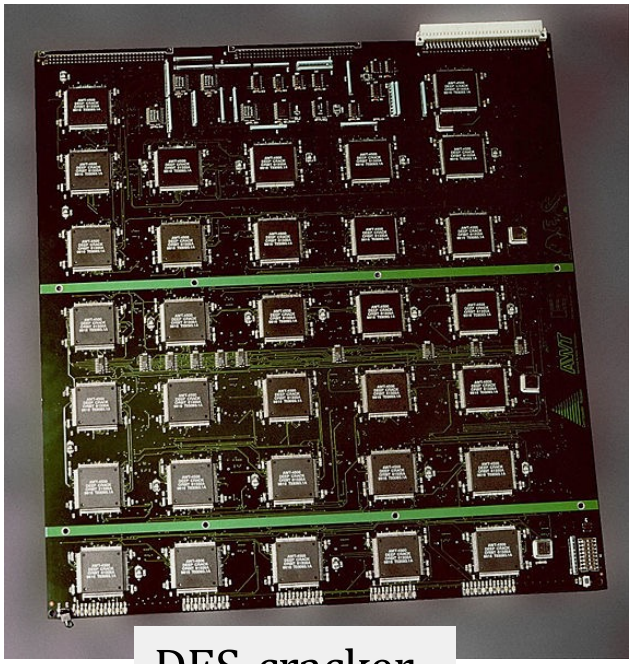
- **1977**
  Diffie and **Hellman** proposed a machine costing **20 million $** breaking DES in **1 day**.

- **1993**
  Wiener proposed a machine costing **1 million $** breaking DES in **7 hours**.

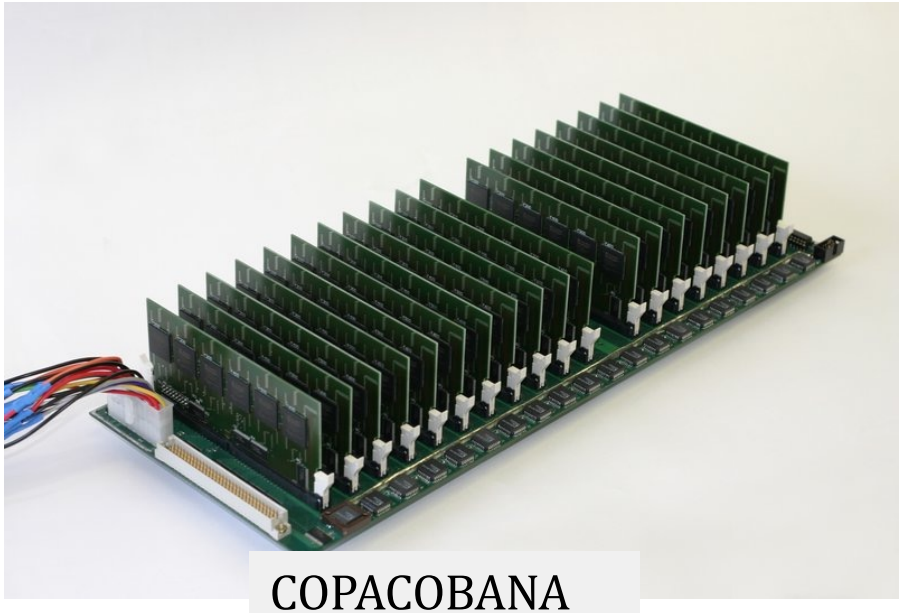
- **1997**
  DESCHALL Project broke a "**DES Challenge**" (published by **RSA**) in **96 days** using idle cycles of thousands of computers across the Internet.

- **1998**
  a **DES-cracker** was built by the **Electronic Frontier Foundation (EFF),** at the cost of approximately **250,000$**

- **2000s**
  COPACOBANA (the Cost-Optimized Parallel COde Breaker) breaks **DES** in **1 week** and costs **10,000$**



DES-cracker



COPACOBANA

Theoretical attacks on DES – differential cryptoanalysis

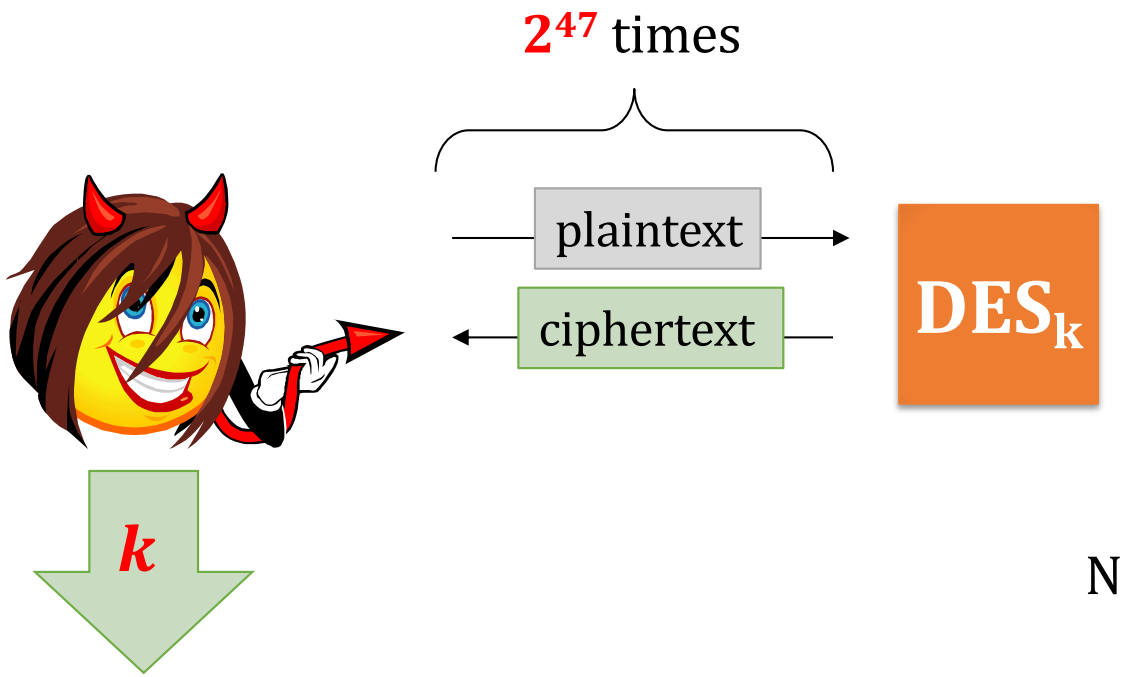
Biham and Shamir (late 1980s):

differential cryptoanalysis





They show how to break **DES** using a **chosen-plaintext attack**.



Not very practical...

Differential cryptoanalysis – an interesting observation

A **small change** in the design of **DES** would make the **differential cryptanalysis** much more sucesfull.

Moral

NSA and **IBM** knew it!





"After discussions with NSA, it was decided that **disclosure of the design considerations would reveal the technique of differential cryptanalysis**, a powerful technique that could be used against many ciphers. This in turn **would weaken the competitive advantage the United States enjoyed over other countries** in the field of cryptography."



Don Coppersmith, IBM

see: Coppersmith, Don (May 1994). "The Data Encryption Standard (DES) and its strength against attacks" (PDF). *IBM Journal of Research and Development* **38** (3): 243. http://www.research.ibm.com/journal/rd/383/coppersmith.pdf.

Theoretical attacks on DES – linear cryptoanalysis

Matsui (early **1990s**):

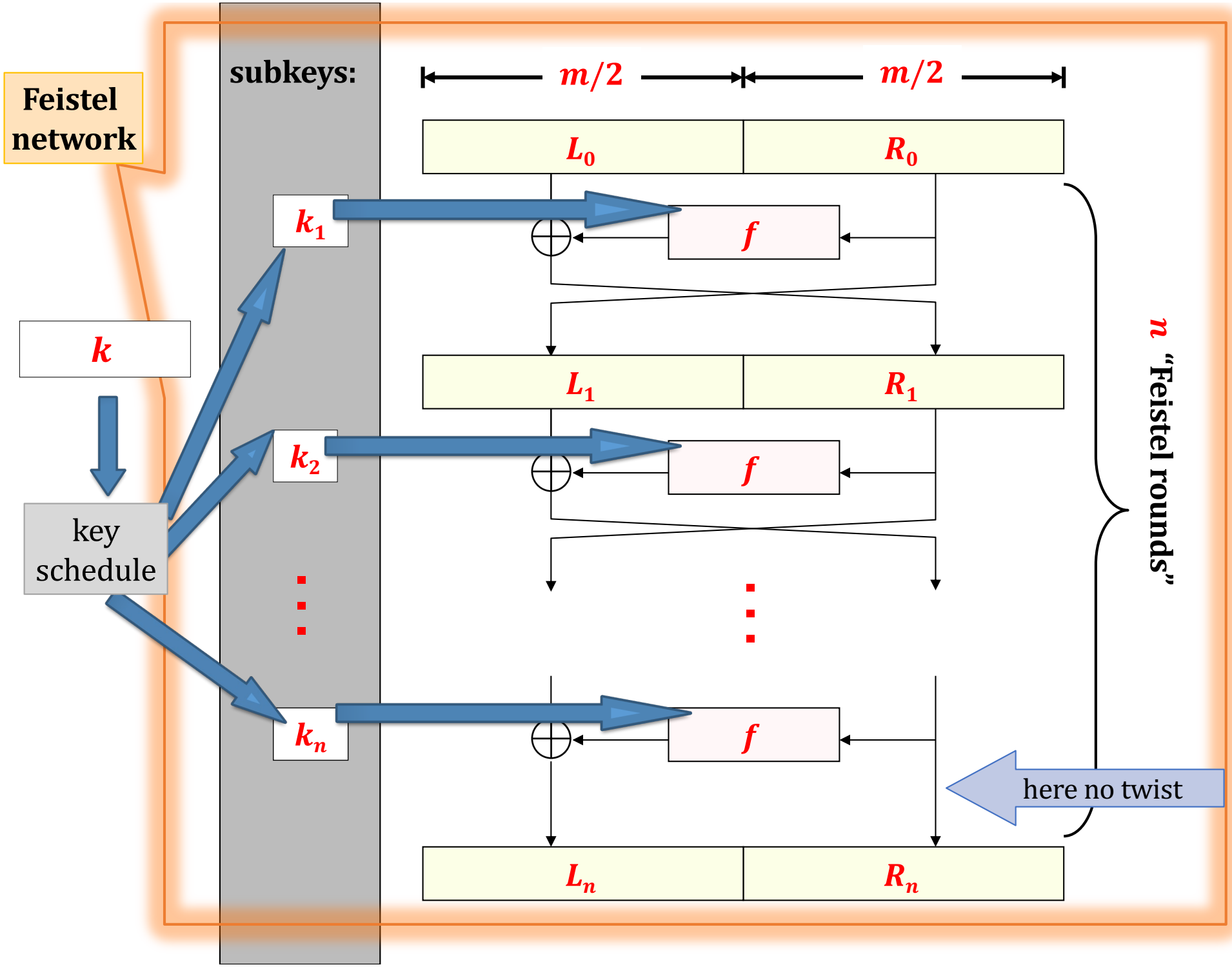
linear cryptoanalysis

uses a **known-plaintext attack**

2^{43} **(plaintext, ciphertext)** pairs

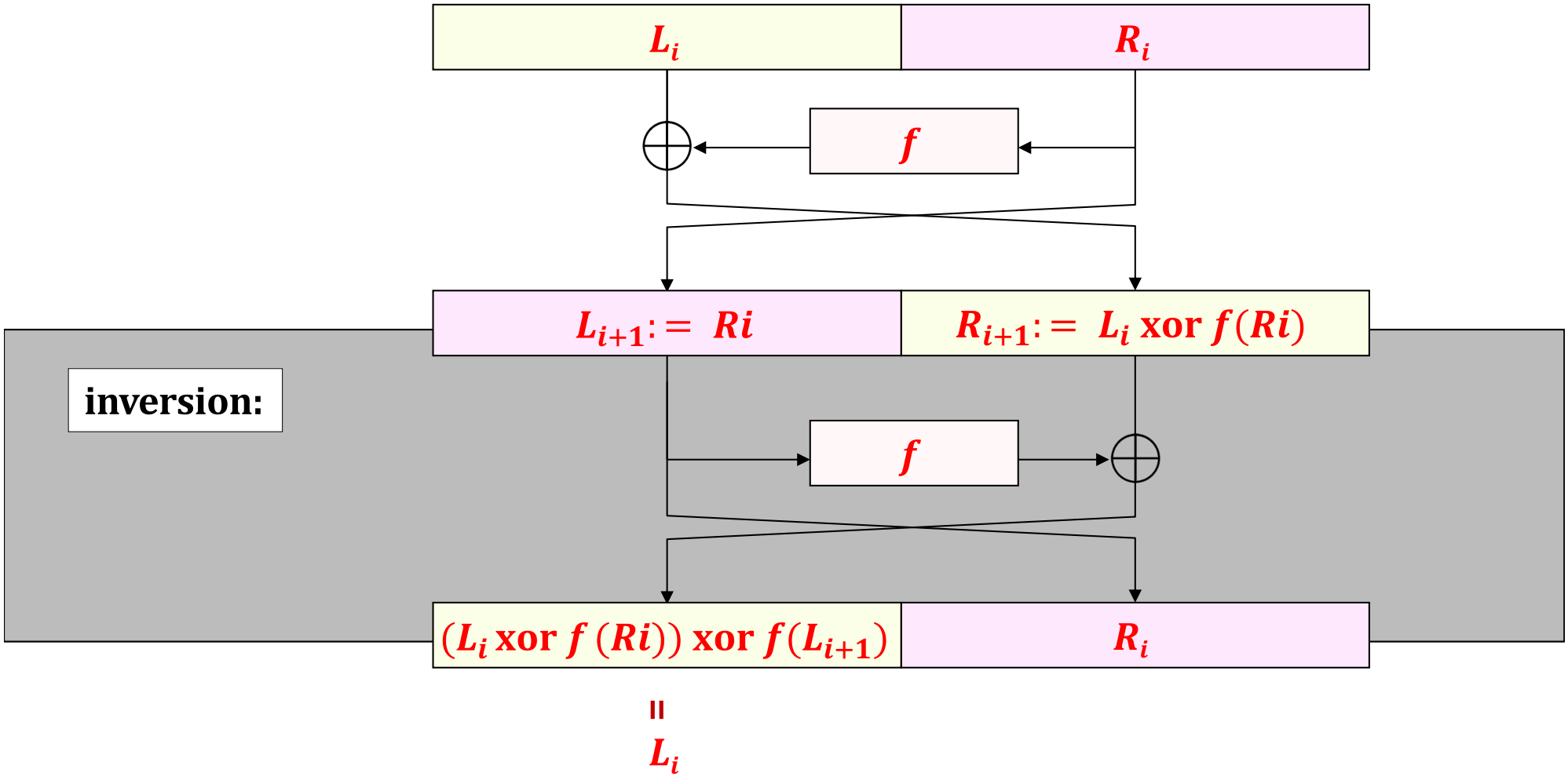
this means:
the adversary doesn't need to choose the plaintexts

Let's now discuss in detail how **DES** is built.

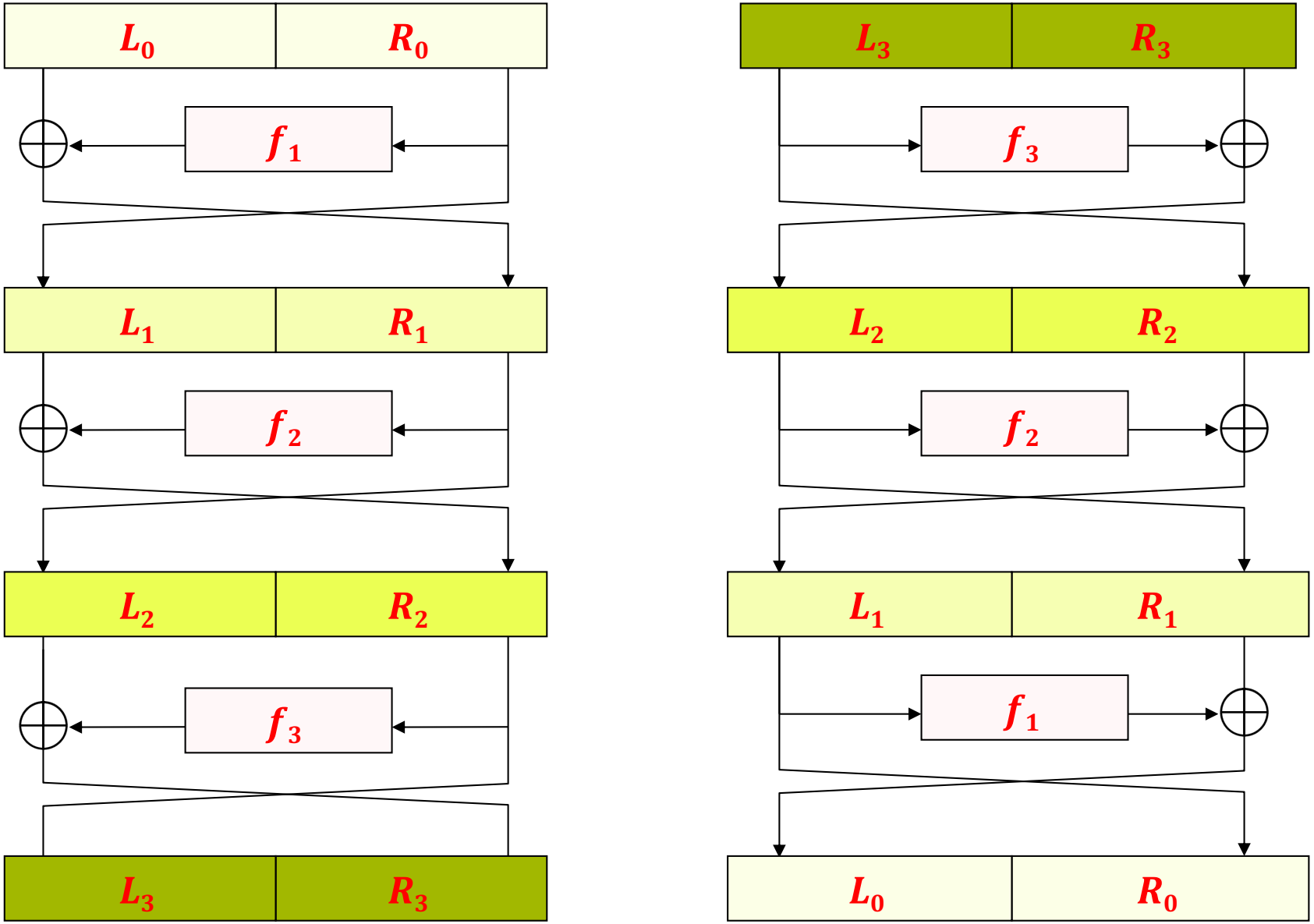

Feistel network
subkeys:
m/2
m/2
L0
R0
k1
f
k
L1
R1
k2
f
key schedule
n "Feistel rounds"
kn
f
here no twist
Ln
Rn

A nice propery of Feistel rounds

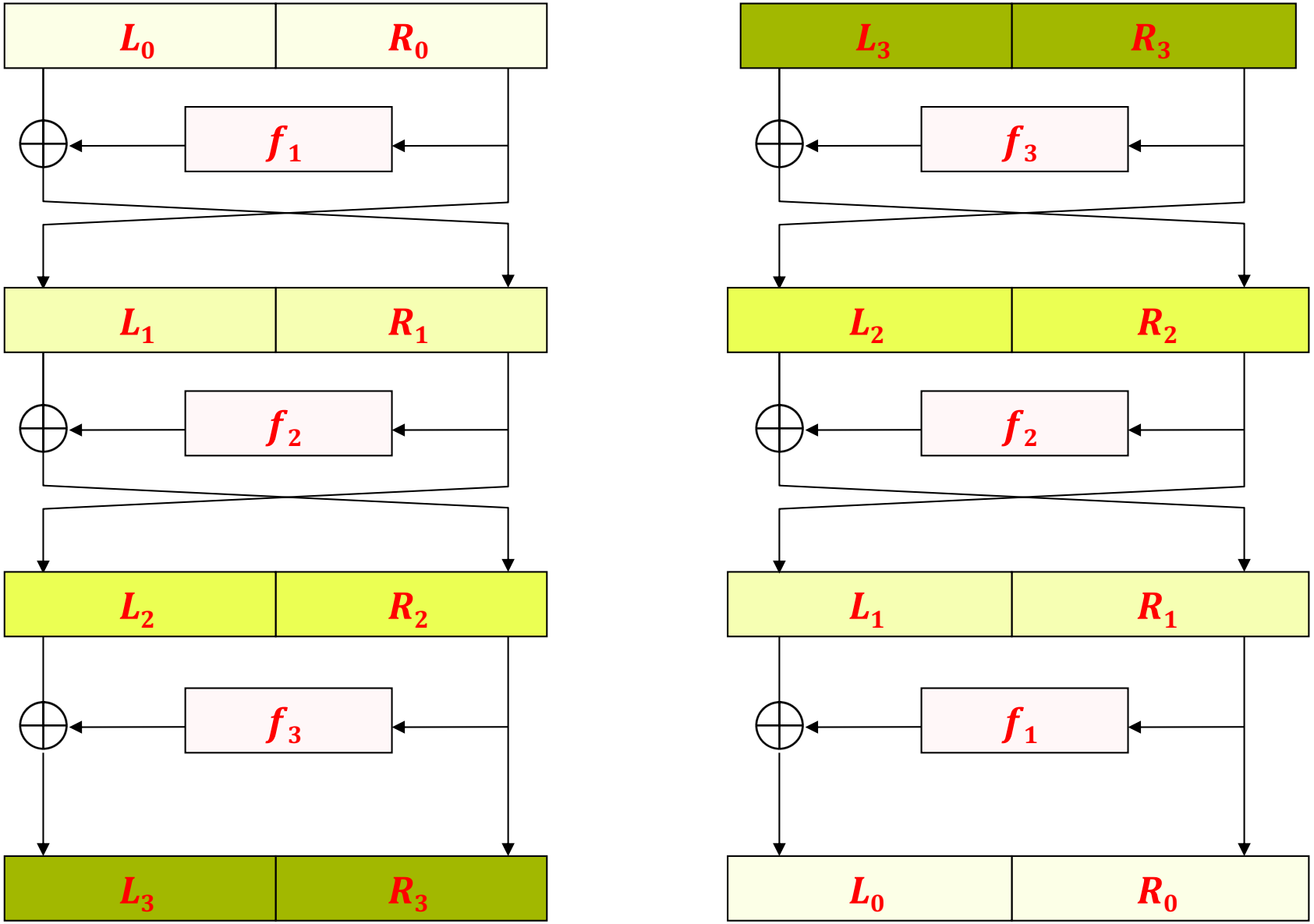
Even if f is **not** easily invertible, each round **can be easily inverted**!



Hence: the Feistel network can be “inverted”!
Example: 3 round Feistel network

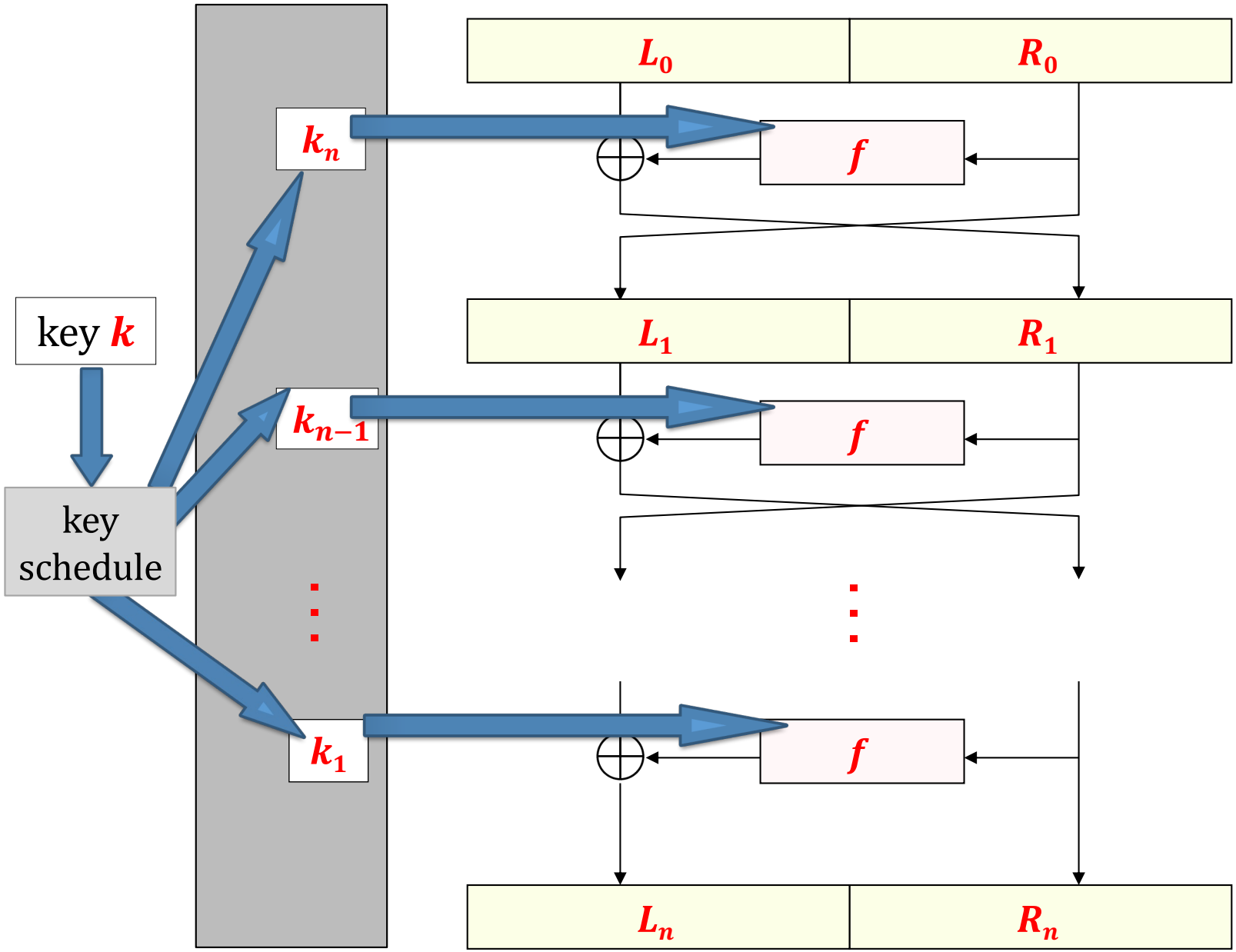


Without a “twist” in the last round:



How to decrypt?

Reverse the key schedule (**note**: **symmetry**)!



Feistel networks are also studied by the theoreticians

Suppose ***f*** is a pseudorandom **function**, and we use it to construct a Feistel network.

Then:

- the **3**-round Feistel network is a **pseudorandom permutation**,
- the **4**-round Feistel network is a **strong pseudorandom permutation**.

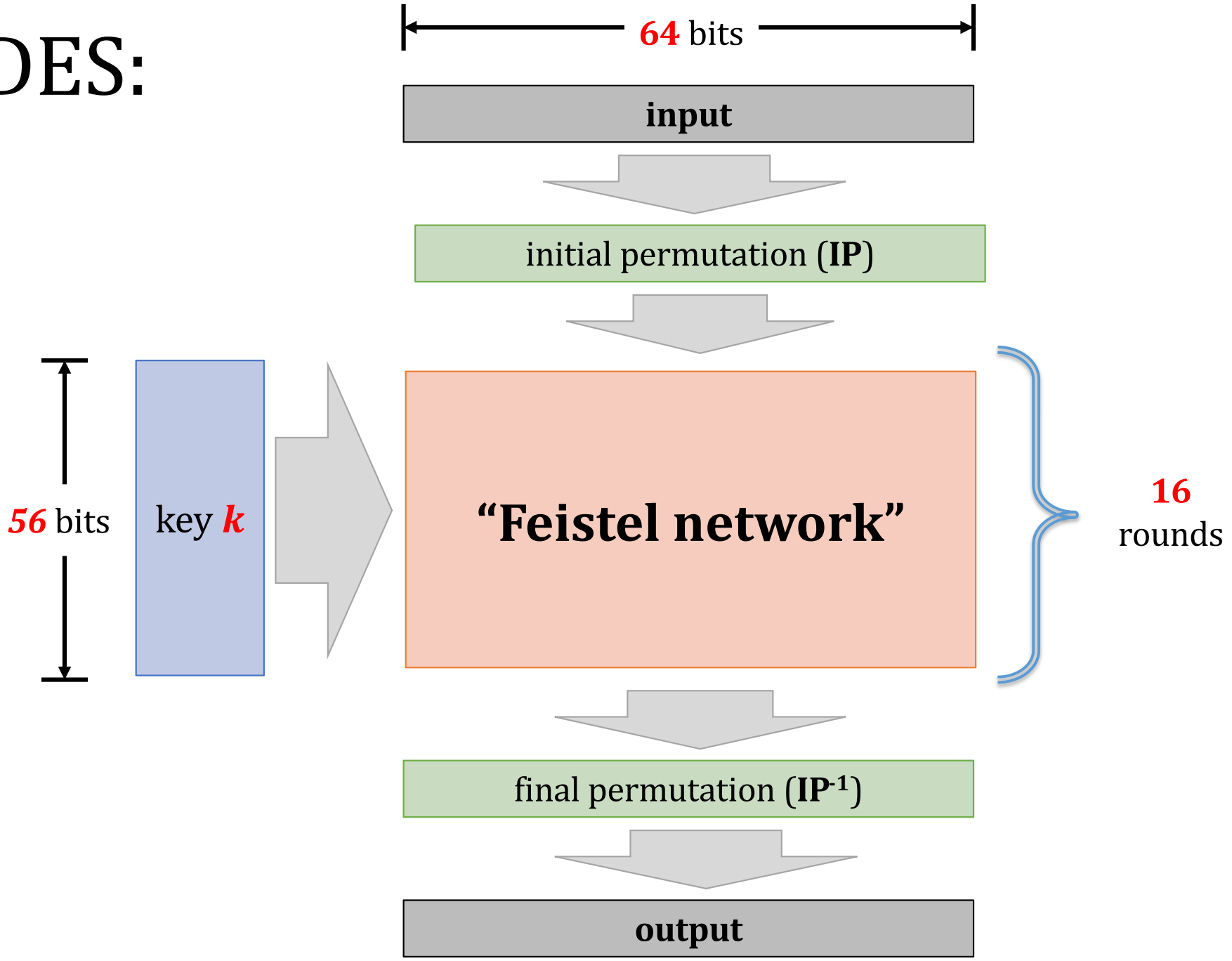
see **M. Luby and C. Rackoff. "How to Construct Pseudorandom Permutations and Pseudorandom Functions." In *SIAM J. Comput.*, vol. 17, 1988, pp. 373-386**.

How is the Feistel network used in DES?

The following needs to be described:

1. The concrete parameters
2. The key schedule algorithm.
3. The functions f.

DES:


64 bits
input
initial permutation (IP)
56 bits
key k
"Feistel network"
16
rounds
final permutation (IP-1)
output

DES – the conclusion

- The design of **DES** is extremally good.
- The only weaknesses: **short key** and **block**.
- Enormous impact on research in cryptography!

One practical weakness of Feistel networks

Only **half of the message** is processed at one time.

Question: Is there any alternative construction that does not have this problem?

Yes! (**substitution-permutation networks**)

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Substitution-permutation networks

Based on the ideas of **Claude Shannon** (1916–2001) from **1949**.



Used in **AES (Rijndael), 3-Way, SAFER, SHARK, Square**...

Advanced Encryption Standard (AES)

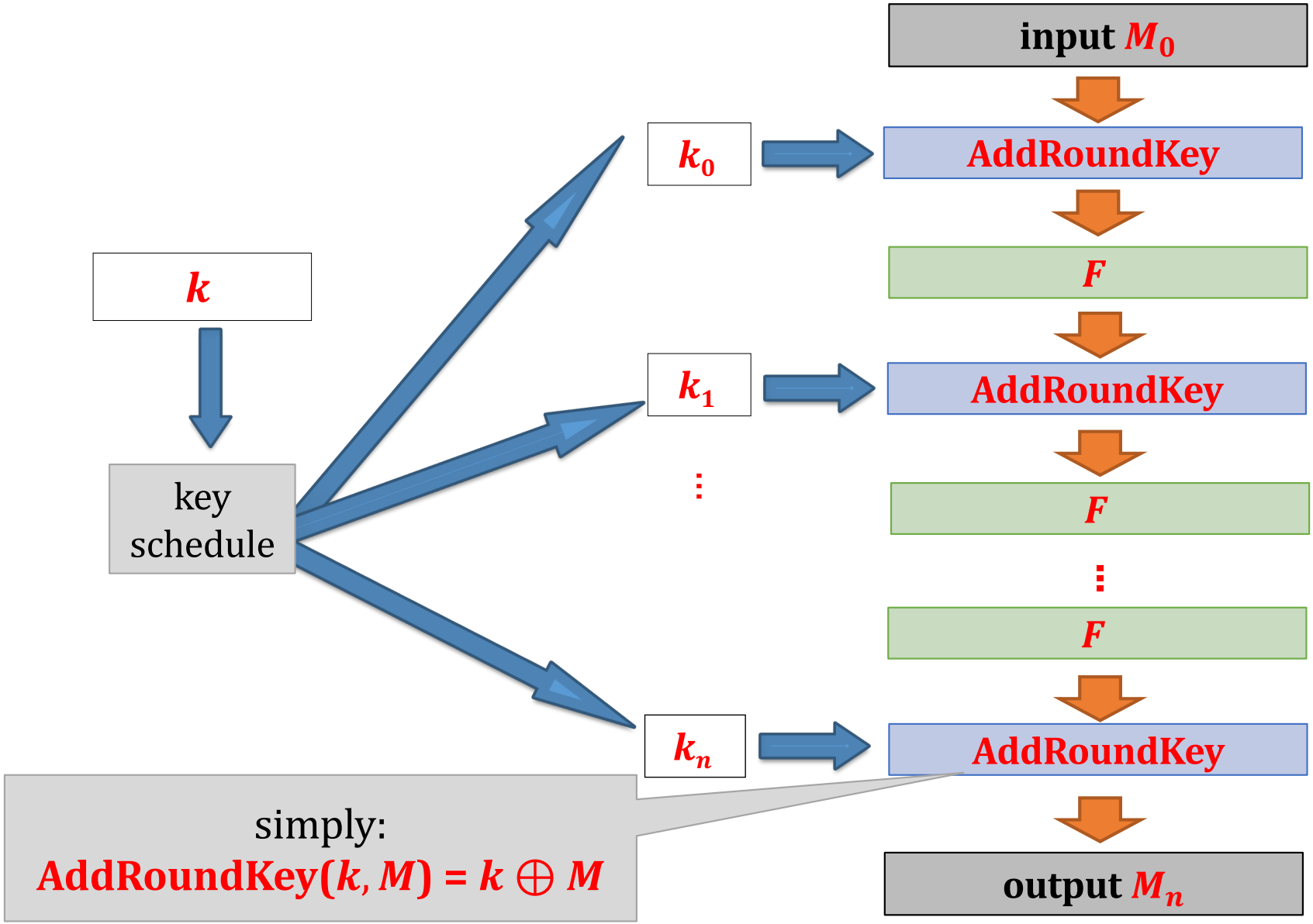
- Competition for **AES** announced in **January 1997** by the US **National Institute of Standards and Technology** (NIST)
- **15** ciphers submitted
- **5** finalists: **MARS, RC6, Rijndael, Serpent**, and **Twofish**
- **October 2, 2000**: **Rijandel** selected as the winner.
- **November 26, 2001**: **AES** becomes an official standard.

- Authors : **Vincent Rijmen**, **Joan Daemen** (from Belgium)
- **Key sizes**: **128**, **192** or **256** bit, **block size**: **128** bits

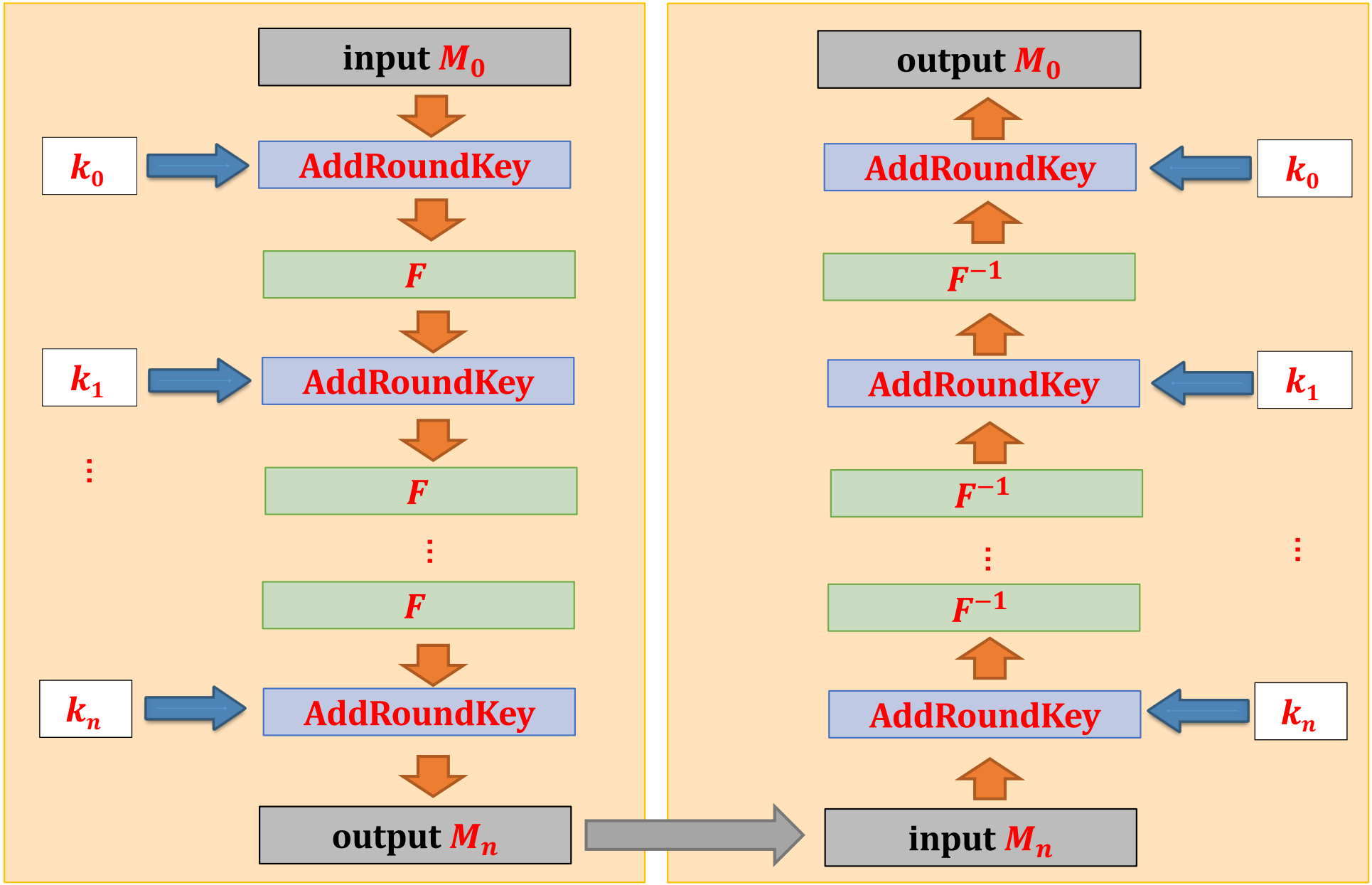




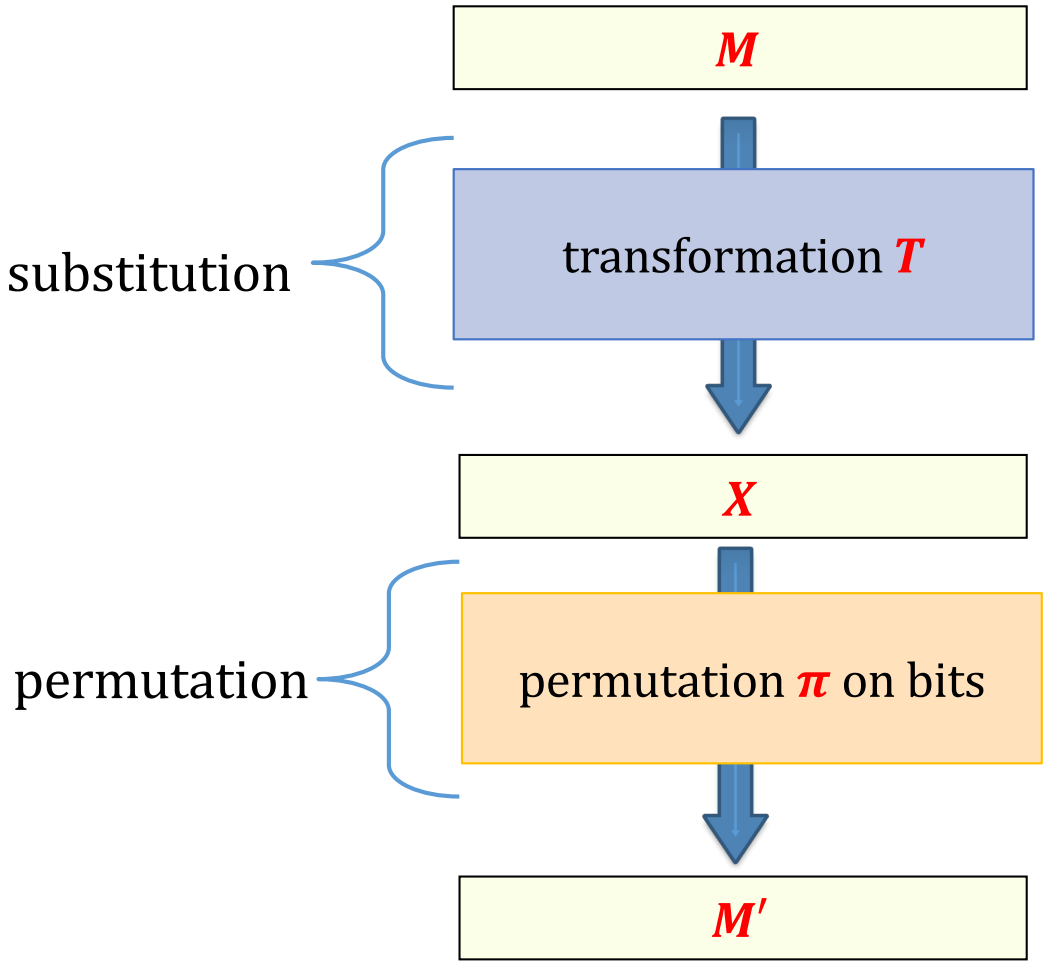
Substitution-permutation networks



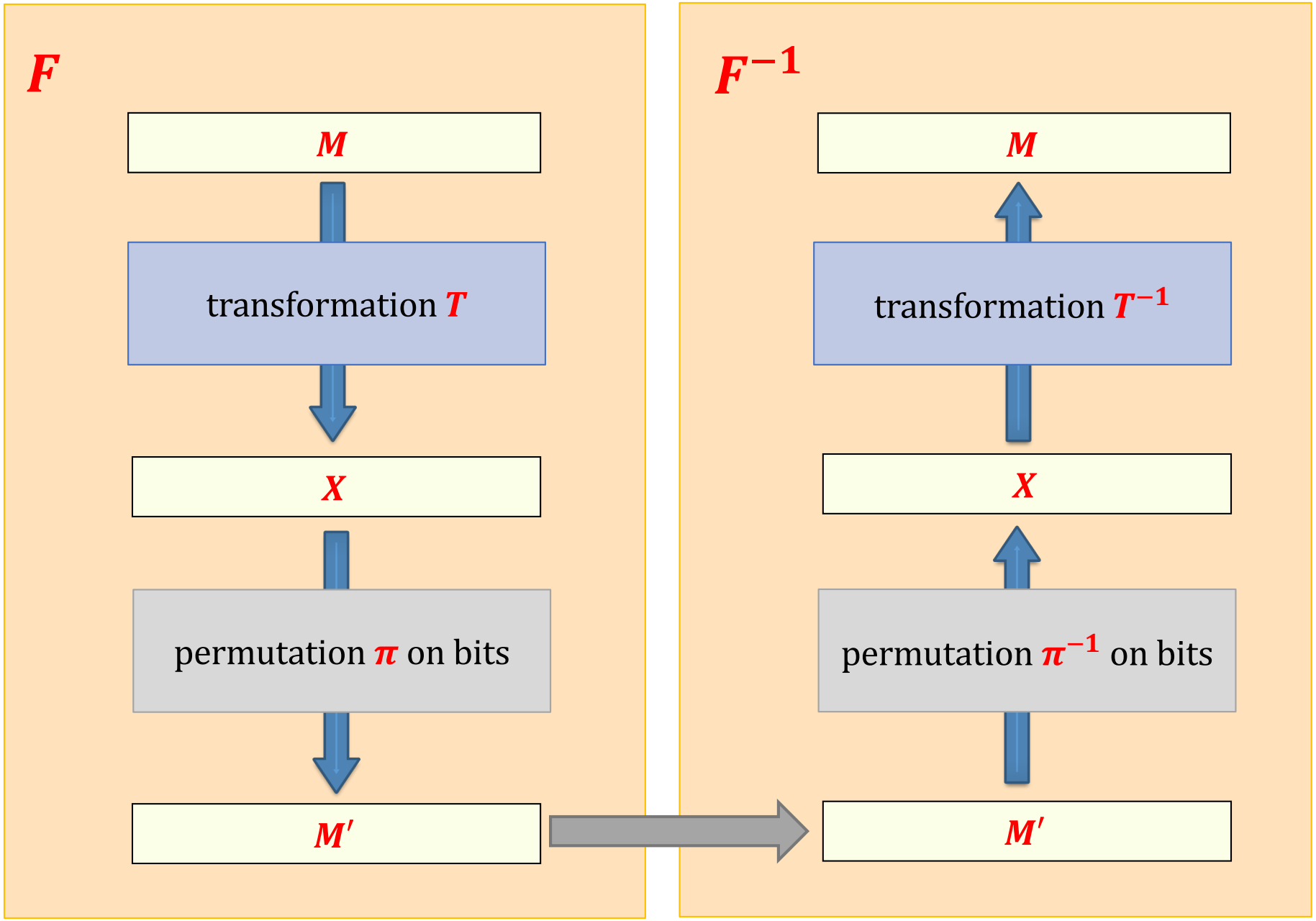
To invert: invert the order and apply F^{-1} instead of F.



A construction of F

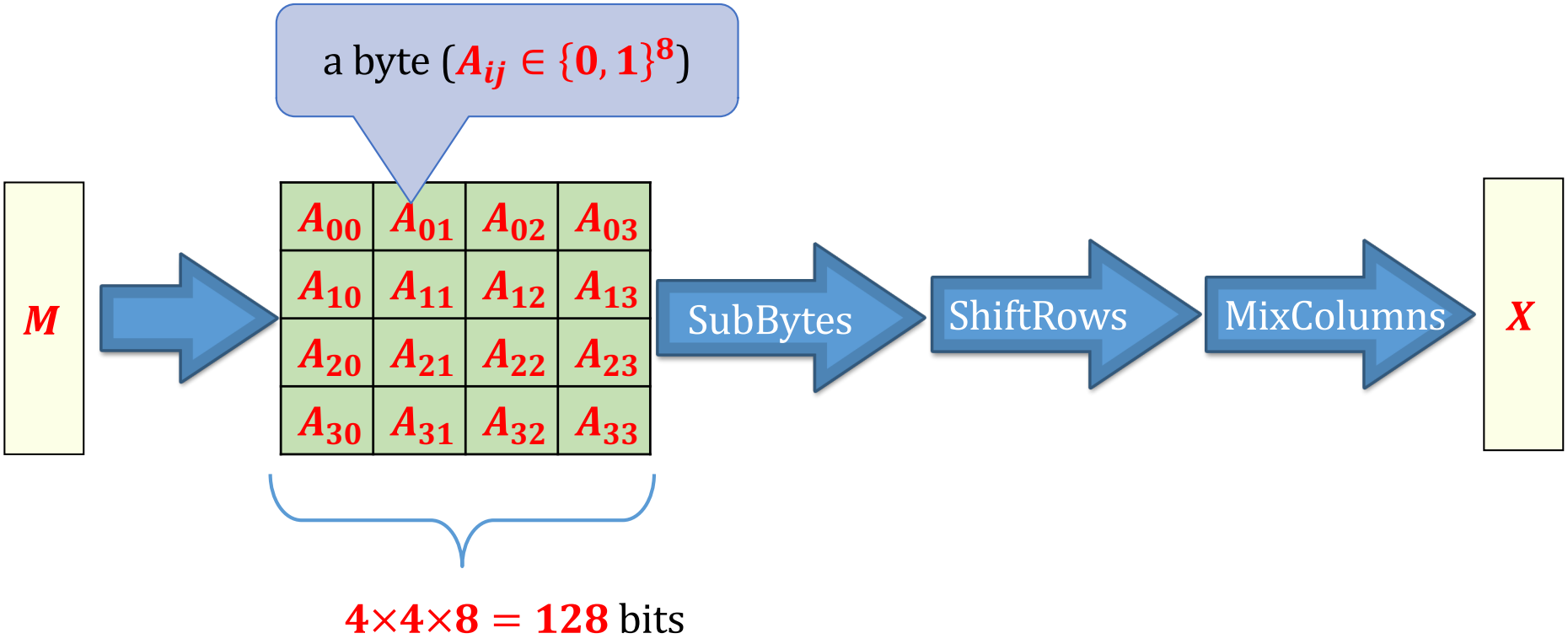


How to construct F^{-1}?

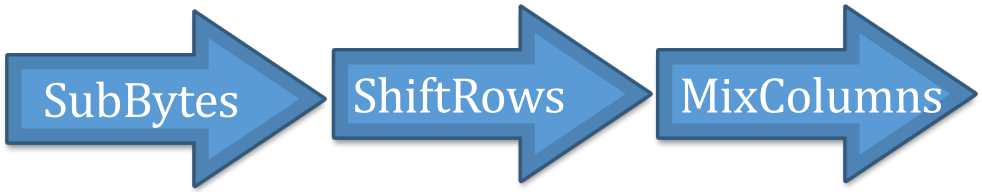


Transformation T in AES

In AES M is represented as a 4×4-matrix of bytes.



Main challenge



This transformation needs to be **invertible**...

On the other hand: it cannot be “too simple”.

AES idea: use **finite field algebra**.

Groups

A **group** is a set G along with a binary operation $\circ$ such that:

- [**closure**] for all $g, h \in G$ we have $g \circ h \in G$,
- there exists an **identity** $e \in G$ such that for all $g \in G$ we have
$$e \circ g = g \circ e = g,$$
- for every $g \in G$ there exists an **inverse of**, that is an element h such that
$$g \circ h = h \circ g = e,$$
- [**associativity**] for all $g, h, k \in G$ we have
$$g \circ (h \circ k) = (g \circ h) \circ k$$
- [**commutativity**] for all $g, h \in G$ we have
$$g \circ h = h \circ g$$

if this holds, the group is called **abelian**

Additive/multiplicative notation

Convention:

[**additive notation**]
If the groups operation is denoted with $+$, then:
the inverse of g is denoted with $-g$,
the neutral element is denoted with $\mathbf{0}$,
$g + \cdots + g$ (n times) is denoted with ng.

[**multiplicative notation**]
If the groups operation is denoted "$\times$" or "$\cdot$", then:
sometimes we write gh instead of $g \cdot h$,
the inverse of g is denoted g^{-1} or $1/g$.
the neutral element is denoted with $\mathbf{1}$,
$g \cdot \cdots \cdot g$ (n times) is denoted with g^n
$\left(g^{-1}\right)^n$ is denoted with g^{-1}.

Fields

$(F, +, \times)$ is a **field** if

- $(F, +)$ is an **additive group** with **neutral element 0**
- $(F \setminus \{0\}, \times)$ is a **multiplicative group**
- **Distributivity of multiplication over addition**:
  for all $a, b, c \in F$, we have

$$a \times (b + c) = a \times b + a \times c$$

How to define a “field over bytes”?

A very simple additive group over $\{0,1\}^n$:

$$(\{0,1\}^n, +)$$

where

$$(a_1, \ldots, a_n) + (b_1, \ldots, b_n) = (a_1 \oplus b_1, \ldots, a_n \oplus b_n)$$

xor

Extremely efficient to implement.

How to extend $(\{0,1\}^n, +)$ to a field?

"**Galois fields**" $\mathbf{GF}(2^n)$:

Represent each $(a_0, \ldots, a_{n-1}) \in \{0,1\}^n$ as a polynomial A over Z_2 of degree $n-1$.

$$A(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}$$

Note: if $(b_0, \ldots, b_{n-1})$ is represented as a polynomial $B(x) = b_0 + b_1 x + \cdots + b_{n-1} x^{n-1}$, then

$$(A+B)(x) = \sum_{i=0}^{n-1} (a_i + b_i) \cdot x^i \pmod 2$$

Observe: this is the same as the **xor** operation on bits.

How to multiply?

Suppose:

$$A(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1}$$
$$B(x) = b_0 + b_1 x + \cdots + b_{n-1} x^{n-1}.$$

Then $A \times B$ is a polynomial of max degree $2n - 2$.

How to reduce this degree?

do not exits non-constant polynomials q_0, q_1 such that $p = q_0 \cdot q_1$

Take an **irreducible polynomial** p of degree n, and compute

$$C = A \cdot B \pmod{p}$$

Then C is a polynomial of degree $n - 1$.

Write $C(x) = c_0 + c_1 x + \cdots + c_{n-1} x^{n-1}$.

Define: $(a_0, \ldots, a_{n-1}) \times (b_0, \ldots, b_{n-1}) = (c_0, \ldots, c_{n-1})$

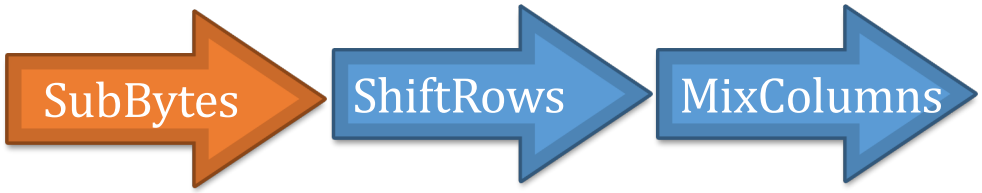
Fact from algebra: this defines a field.

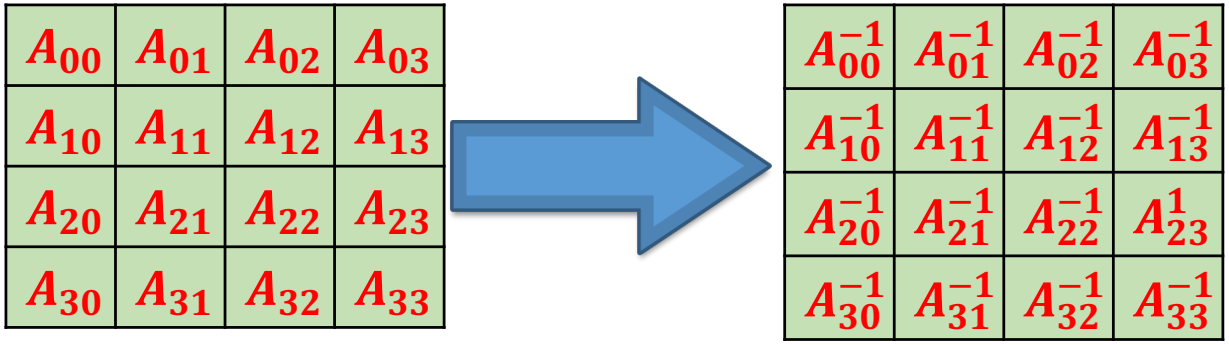
AES field

AES uses $\mathbf{GF(2^8)}$, where the polynomial $\boldsymbol{p}$ is defined as

$$\boldsymbol{p(x) = 1 + x + x^3 + x^4 + x^8}$$

First step of SubBytes





Invert every $\boldsymbol{A_{ij}}$ (in the multiplicative group of $\mathbf{GF}(\boldsymbol{2^n})$).

Convention: $\boldsymbol{0^{-1} = 0}$.

Another observation

We can look at Z_2^n as a **linear space**.

AES defines the following affine transformation:

$$\varphi(x_1, \dots, x_8) :=$$

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} * \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \\ x_8 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}$$

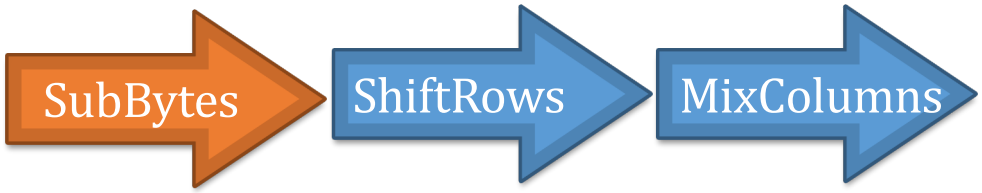
Advantage: **SubBytes** is not an operation only in $\mathbf{GF}(2^n)$.

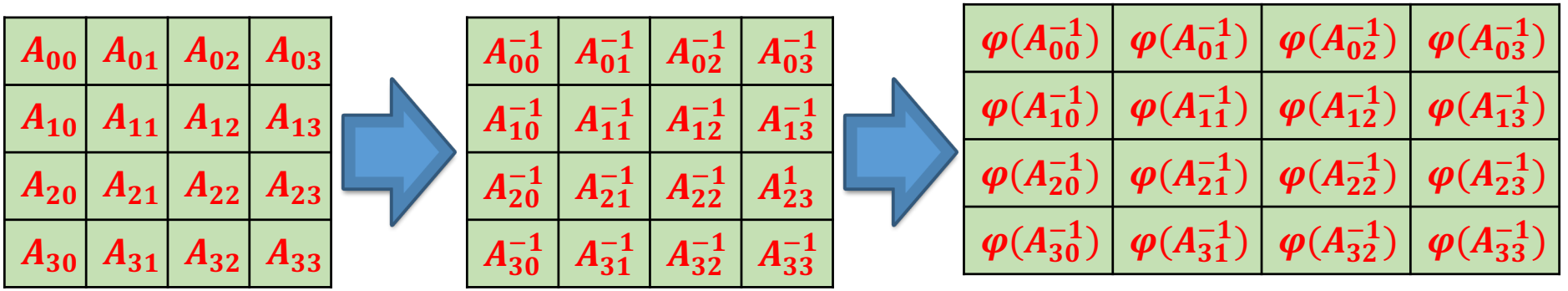
The constant vector is chosen in such a way that there are no

- **fixpoints** $\varphi(X) = X$
- **anti-fixpoints** $\varphi(X) = \overline{X}$

φ is invertible.

Complete SubBytes





Observe that

$$A_{ij} \mapsto A_{ij}^{-1} \mapsto \varphi(A_{ij}^{-1})$$

is invertible (since $A_{ij} \mapsto A_{ij}^{-1}$ and φ are invertible)

ShiftRows



Cyclic shifts of rows:

B_{00}	B_{01}	B_{02}	B_{03}
B_{10}	B_{11}	B_{12}	B_{13}
B_{20}	B_{21}	B_{22}	B_{23}
B_{30}	B_{31}	B_{32}	B_{33}



B_{00}	B_{01}	B_{02}	B_{03}
B_{11}	B_{12}	B_{13}	B_{10}
B_{22}	B_{23}	B_{20}	B_{21}
B_{33}	B_{30}	B_{31}	B_{32}

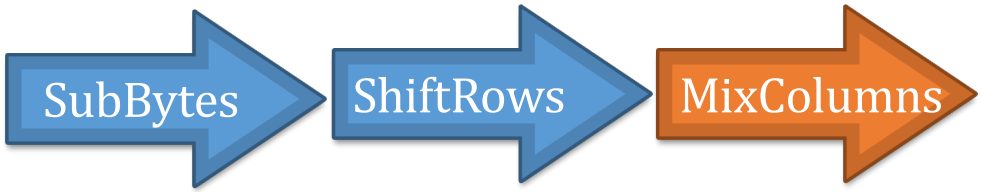
shift **1** cell left

shift **2** cells left

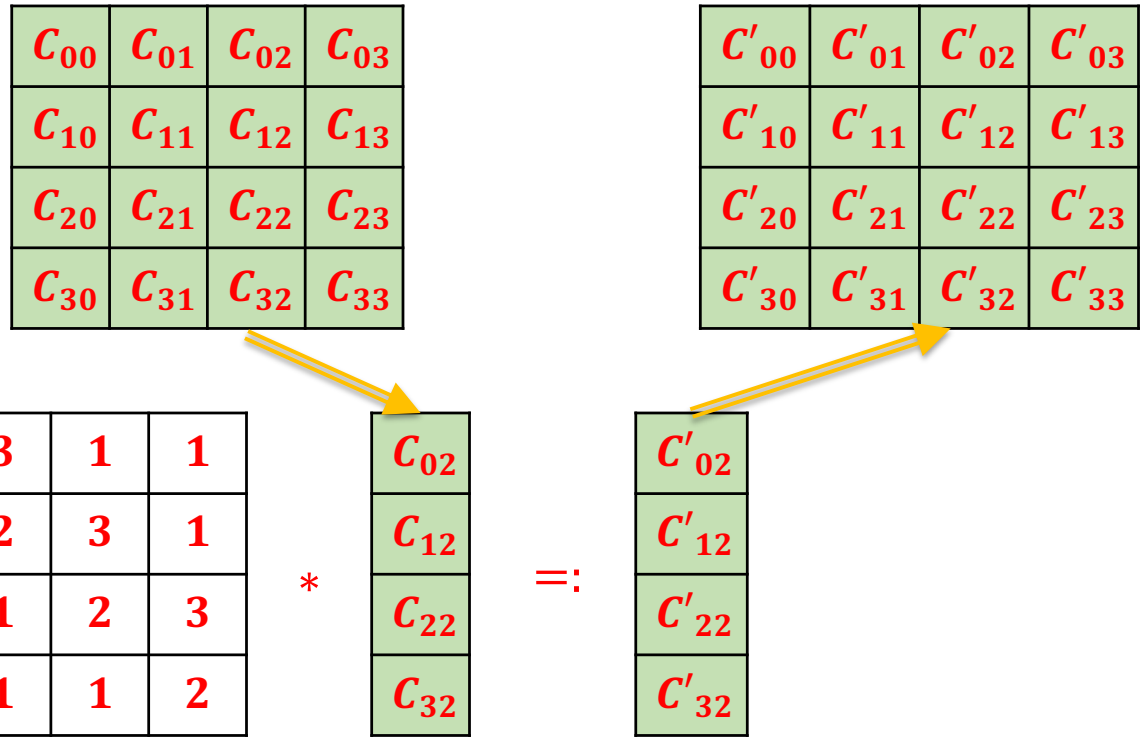
shift **3** cells left

Clearly: **ShiftRows** is invertible.

MixColumns



Multiply every column by a matrix $\boldsymbol{M}$ (in $\mathbf{GF}(\mathbf{2^8})$):



bytes treated as polynomial coefficients

Clearly $\boldsymbol{M}$ is invertible, so the whole operation also is.

AES construction – more details

Concrete parameters:

key size: 128, **192** or **256** bit,
block size: **128** bits

We omit the description of the key schedule.

Security:

best known attack: **biclique attacks [Bogdanov, Khovratovich, and Rechberger, 2013]**:

- **AES-128** complexity $2^{126.1}$,
- **AES-192** complexity $2^{189.7}$,
- **AES-256** complexity $2^{254.4}$.

Plan

1. Computational definitions of security
2. If semantically-secure encryption exists, then **P ≠ NP**
3. A proof that "the PRGs imply secure encryption"
4. Theoretical constructions of PRGs
5. Stream ciphers
6. Pseudorandom functions and permutations
7. Block ciphers
    1. Modes of operation
    2. Popular construction paradigms
    3. Cascade ciphers
8. Practical considerations



An idea

The main problem of **DES** is the short key!

Maybe we could increase the length of the key?

But how to do it?

Idea: cascade the ciphers!

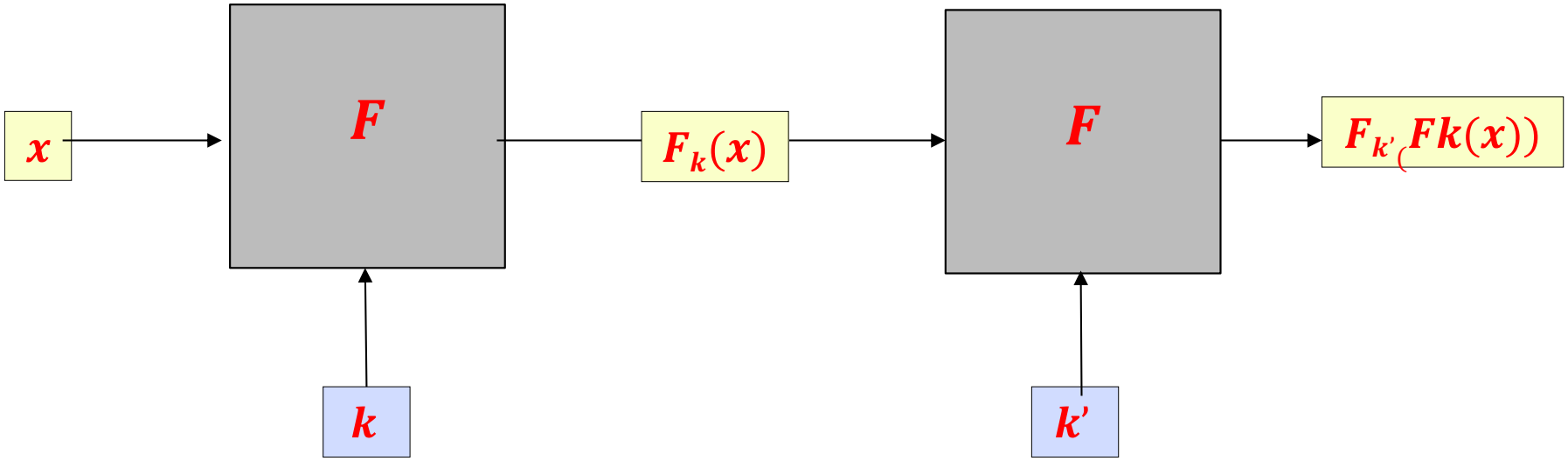
We now describe it in an abstract way (for any block cipher F)

How to increase the key size?

Cascade encryption.

For example, **double encryption** is defined as:

$$F'_{(k,k')}(x) := Fk'(Fk(x))$$



Does it work?

- Double encryption – not really...
- Triple encryption is much better!

Double encryption

$\boldsymbol{n}$ = block length = key length

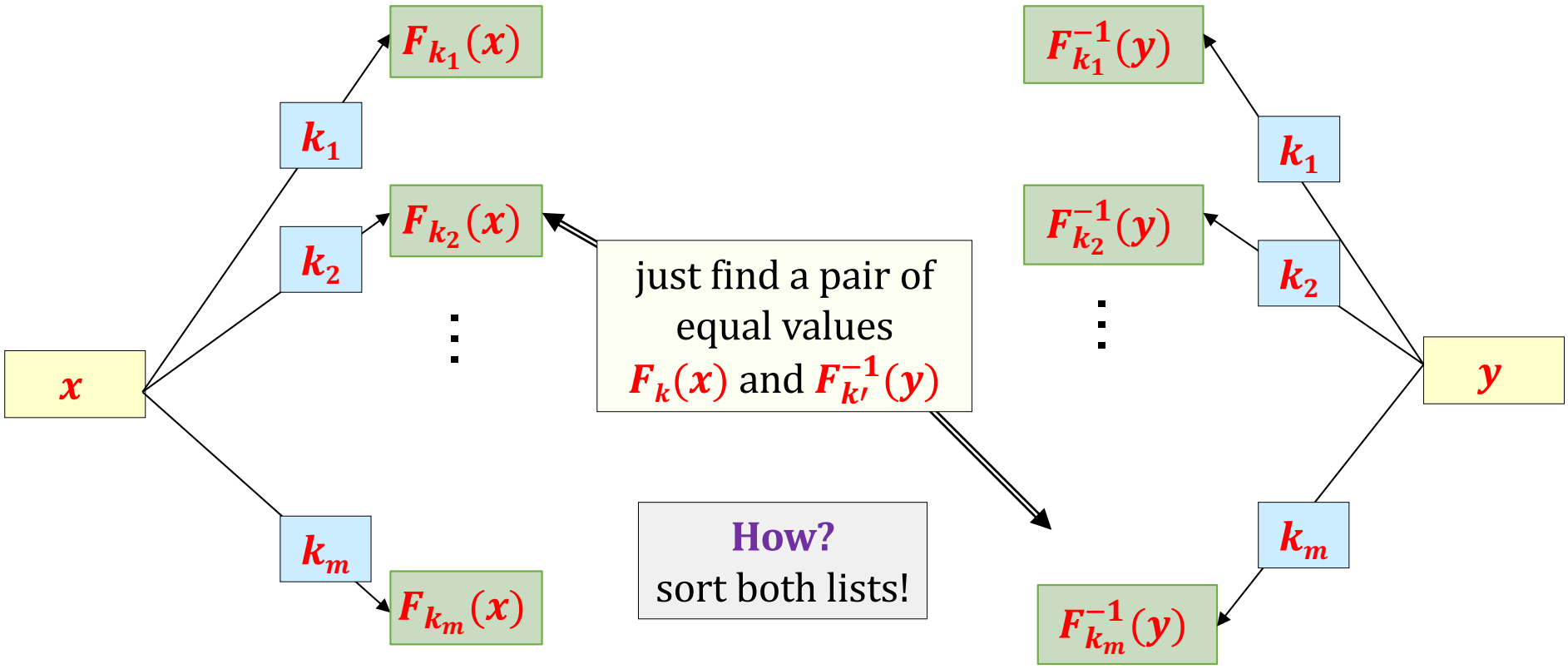
Double encryption can be broken using

- time $\boldsymbol{O(2^n)}$,
- space $\boldsymbol{O(2^n)}$,
- and **3 (plaintext,ciphertext)** pairs.

The attack is called "**meet in the middle**".

Meet-in-the middle attack – the idea

Goal: Given (x, y) find (k, k') such that $y = F_{k'}(F_k(x))$ $\quad m = 2^n$



Meet-in-the middle attack – the algorithm

Goal: Given (x, y) find (k, k') such that $y = F_{k'}(F_k(x))$.

Algorithm:

1. For each k compute $z = F_k(x)$ and store (z, k) in a list L.
2. For each k compute $z = F_k^{-1}(y)$ and store (z, k') in a list L'.
3. Sort L and L' by their first components.
4. Let S denote the list of all pairs all pairs (k, k') such that for some z we have
   $(z, k) \in L$ and $(z, k') \in L'$.
5. Output S.

Meet-in-the middle attack – analysis [1/2]

Suppose: n = block length = key length, x and y are fixed

P (**a random pair** (k, k') **satisfies** $y = F_{k'}(Fk(x))) \approx 2^{-n}$

why? because $F_{k'}(Fk(x))$ can take 2^n values

The number of all pairs (k, k') is equal to 2^{2n}. Therefore

$$E(|S|) \approx 2^{2^n} \cdot 2^{-n} = 2^n.$$

So we have around 2^n "candidates" for the correct pair (k, k').

How to eliminate the "false positives"?

For each "positive" check it against another pair (x', y').

Meet-in-the middle attack – analysis [2/2]

The probability that (k, k') is a false positive for (x, y) and for (x', y') is around

$$2^{-n} \cdot 2^{-n} = 2^{-2n}.$$

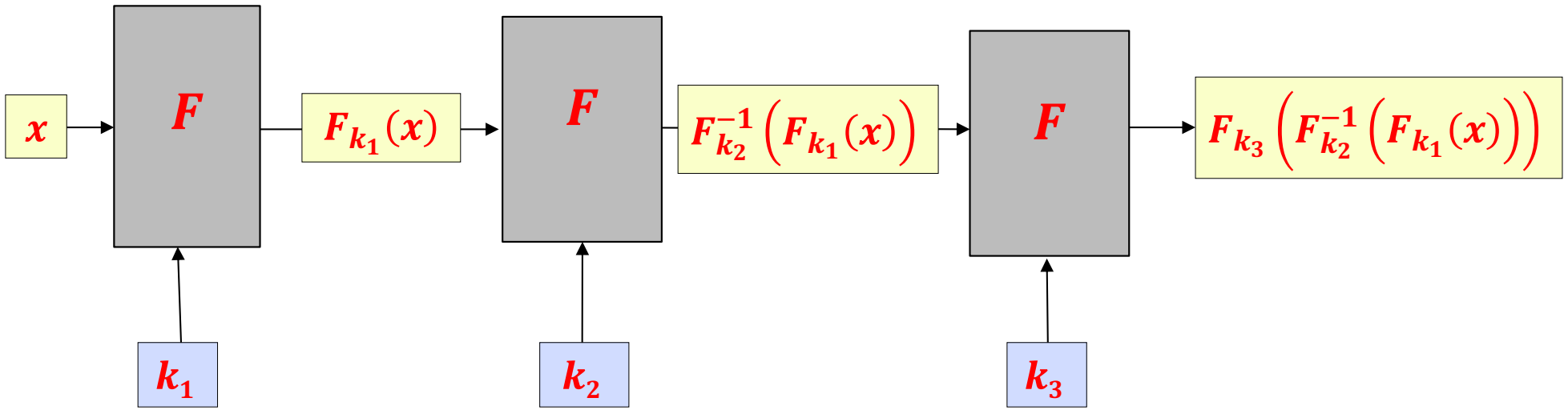
Hence, the expected number of "false positives" is around

$$2^{2n} \cdot 2^{-2n} = 1$$

An additional pair (x'', y'') allows to eliminate the false positive.

A much better idea: triple encryption

$$F_{(k_1,k_2,k_3)}(x) := F_{k_3}\left(F_{k_2}^{-1}\left(F_{k_1}(x)\right)\right)$$



Sometimes $k_1 = k_3$.

Triple DES (3DES) is a standard cipher.

Disadvantages:

- rather slow,
- small block size.

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Benchmarks

	Algorithm	MiB/Second	Cycles Per Byte
stream	Salsa20	643	2.7
	Sosemanuk	727	2.4
block	AES/CTR (128-bit key)	139	12.6
	AES/CTR (192-bit key)	113	15.4
	AES/CTR (256-bit key)	96	18.2
	AES/CBC (128-bit key)	109	16
	AES/CBC (192-bit key)	92	18.9
	AES/CBC (256-bit key)	80	21.7
	DES/CTR	32	54.7
	DES-EDE3/CTR	13	134.5

Source: www.cryptopp.com/benchmarks.html
All algorithms coded in C++, compiled with Microsoft Visual C++ 2005 SP1 (whole program optimization, optimize for speed), and ran on an Intel Core 2 1.83 GHz processor under Windows Vista in 32-bit mode.

Hardware implementations of AES

(taken from J Daemen, V Rijmen The design of Rijndael, 2001):

Example of a hardware record:

ASIC. H. Kuo and I. Verbauwhede report a throughput of 6.1 Gbit/s, using 0.18 μm standard cell technology [55] for an implementation without pipelining. Their design uses 19 000 gates. B. Weeks et al. report a throughput of 5 Gbit/s [91] for a fully pipelined version. They use a 0.5 μm standard cell library that is not available outside NSA.

Stream ciphers vs. block ciphers

- Stream ciphers are a **bit more efficient**.
- But they appear to be "**less secure**".
- It is easier to misuse them (use the same stream twice).
- If you encrypt a stream of data, you can always use a block cipher in a **CTR** mode.
- Probably at the moment **block ciphers are a better choice** for most of the applications.

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